Towards 3-Minutes: Application of Holding and Crew Interventions to Improve Service Regularity on a High Frequency Rail Transit Line

by

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Abstract

Transit service regularity is an important factor in achieving reliable high frequency operations. This thesis explores aspects of headway and dwell time regularity and their impact on service provision on the MBTA Red Line, with specific reference to the agency's objective of operating a future 3-minute trunk headway, and to issues of service irregularity faced today. Current operating practices are examined through analysis of historical train tracking and passenger fare card data. Headway regularity is explored in the areas of initial headway regularity, stability of line headways, and passenger impacts. A metric of effective utilization is proposed which quantifies the effect of headway irregularity on passenger capacity on an individual train basis. Dwell times are explored through a statistical approach without passenger regard, and through a regression approach with passenger regard, both applied to automated data. The proposed methods of dwell time investigation are applicable for other agencies with track occupancy and passenger fare card data for continuous monitoring and evaluation of dwell times.

Following the identification of capacity consequences of service irregularity on the MBTA Red Line, three interventions were proposed and trialed in order to demonstrate operating practices consistent with the need to improve service regularity for future conditions. The first intervention trialed a drop-back crewing strategy at the trunk terminal, finding that the strategy is applicable to achieve necessary future reductions in operator layover time. Significant short-term benefits were also found, including reduction in terminal queuing delays, and improvements to schedule recovery following disruptions. The second intervention trialed a target headway-based holding strategy to mitigate initial headway irregularity caused by differences in allowable speeds for the two departure movements at the terminal. The intervention successfully improved mean journey times for the majority of passengers, and reduced extreme downstream bottleneck occupancy times, consistent with future capacity objectives. The third intervention trialed a variable target headway-based holding strategy following the branch merge to mitigate headway irregularity caused by stochastic branch travel times and simple schedule based coordination. The intervention successfully improved trunk headway regularity, contributing to reduced journey times and overcrowding at the peak load point. These interventions represent a starting point for a program of incremental service regularity improvements necessary to achieve future capacity objectives.

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Chapter 1. Introduction

1.1 Background and Motivation

In response to ridership demands and backlogged state-of-good-repair (SGR) projects, the Massachusetts Bay Transportation Authority (MBTA) has entered a phase of intensive capital investment. Between 2019 and 2023, a combined $1.98B are budgeted for the MBTA Red and Orange Line Modernization Program, including signaling system modernization, complete rolling stock replacement, and yard expansion (MBTA, 2018b). On the Red Line, already the highest frequency line in the network, the MBTA aims to achieve a 3-minute trunk headway by 2025 (MBTA, 2018a), a substantial 50% increase in service frequency compared to today. With parts of the system operating at capacity during peak hours today, the capacity increase is likely to spur suppressed ridership growth, and continue to fuel transit-oriented development in the region.

However, there are questions as to whether the current state of operating practices on the Red Line can accommodate such dramatic levels of frequency improvements, and to what extent the investments will provide relief to riders in the interim. Today, the MBTA reports a Red Line reliability of approximately 91% (MBTA, 2019a), however this number masks many sources of unreliability as experienced by customers, such as passengers left-behind at stations when unable to board overly crowded trains, cascading delays, and increased peak period travel time due to train congestion. Many of these reliability issues can be linked to issues of service regularity.

To that end, this research aims to investigate current operating practices on the MBTA Red Line relevant to the feasibility of the future 3-minute headway objective, while simultaneously linking them to the reliability issues faced by passengers today. With such large investments, there is a responsibility to ensure the full extent of the planned benefits are realized. Interim improvements in service regularity and the culture surrounding it are essential to ensure that when the new fleet of vehicles is in place, it will be possible to deliver the promised 3-minute headway.

1.2 Research Objectives

This research aims to quantify current operating practices on the MBTA Red Line relevant to the feasibility and reliability of the future 3-minute headway objective including station dwell time, terminal operations, merge control, and headway control. This work does not aim to represent a comprehensive engineering assessment of the physical constraints necessary to achieve the 3-minute headway, but rather a review of the operating practices requiring particular attention. Future operating practices receive considerably less attention relative to capital-heavy areas, but will be equally important to future service objectives.

The objectives of this research can be summarized as follows:

(1) Quantify current state of operational practices on the MBTA Red Line and their contribution to line capacity and passenger delay.
(2) Test operational interventions to improve operational bottleneck performance and determine magnitude of improvements

1.3 Research Approach

The primary method which will be applied is an investigative approach, drawing on the large historical, automatically collected records of individual train and passenger movements. This research will also propose and test service interventions, developed using an understanding of baseline system performance, in order to confirm hypotheses about future operational practices. The results of the interventions will be explored using post-hoc analysis.

1.4 Data Sources

A number of data sources are used repeatedly throughout this research, including train movements obtained from track circuit occupancy records, and passenger movements obtained from fare card transactions. More generally, these data sources are referred to as Automated Data Collection Systems (ADCS). This section will give a brief overview of the assumptions behind each data source with reference to further documentation. Generally, historical data is taken from October 2018 for the initial headway analysis, and for January-February 2019 for the analysis of service interventions in Chapters 6-8.

Historical train movements are obtained from a database of track circuit occupancy records. Train identification is performed by wayside automatic vehicle identification (AVI) devices and matched with track occupancy records. For more information on the track circuit occupancy data, the reader is referred to Dixon (2006) who provides an in-depth discussion of the MBTA operations control systems (OCS), which houses and processes the data, and the underlying assumptions and data issues.

Headways, dwell times, and running times are fundamental metrics used in analyzing any transit operation. Each can be extracted from OCS data as follows. Headways are determined by measuring the elapsed time between successive occupations of the same track circuit by different trains. Running times and dwell times are determined by matching track circuit occupancy times for a given train and two points. In case of dwell time, the nearest bounding track circuits are selected for each platform in order to provide the closest available estimate of dwell time. This means dwell time as measured from OCS data will always be greater than the actual dwell time measured by an observer at the platform. Issues with pre- and post-shunting, whereby track blocks incorrectly register as occupied before or after the actual occupation, create some inaccuracies with these forms of measurement, but their occurrence is assumed to be a random variable and not affecting the overall distribution (Dixon, 2006). In this research, the first registered occupation of

a given track circuit is taken as actual for headway and dwell time determination and others are discarded.

Passenger movements are obtained by applying an origin-destination inference (ODX) model to disaggregate fare transaction, with assumptions originally developed at the New York MTA (Barry et al., 2007) and later extended by the MIT Transit Lab for use in multi-modal systems (Gordon et al., 2013) and adopted for use internally at the MBTA. The key assumption is that of symmetrical journeys, with destination, and transfer locations inferred from sequential taps. Transactions without inferred origins or destinations are distributed in probability with inferred rides sharing the same origin tap-location and 15-minute tap window within the given month. Passengers are assumed to board the first train compatible with their destination, with train capacity assumed to be sufficient in all cases.

1.5 Literature Review

This section will briefly explore the relevant literature pertaining to the topics explored in this research. Section 1.5.1 begins with an exploration of metro bottleneck identification and calculations, which is relevant for identifying pertinent issues on the MBTA Red Line. Section 1.5.2 reviews studies on dwell time estimation and contributing factors, which is important for determining the contribution of dwell time bottlenecks to future capacity on the Red Line. Section 1.5.3 reviews studies on transit reliability and headway regularity, important concepts which form the complement to transit frequency, and reviews strategies to improve reliability.

1.5.1 High-Frequency Metro Bottlenecks

The *Transit Capacity and Quality of Service Manual (TCQSM*) (2013) provides a valuable starting point for the review of rail transit bottlenecks, describing simple analytical models to determine the bottleneck capacity of station dwell time, turnbacks, and junctions. Each of these bottlenecks will be discussed individually in Section 1.6.3 - Line Bottlenecks.

Canavan et. al (2019) provide a useful inventory of best practices in operating high-frequency metro lines from a worldwide survey of 17 high-frequency lines (defined as at least 25 trains per hour (tph)). Factors constraining frequency were grouped into five main categories; signaling and train control, station and train crowding, fleet, terminal turnarounds, service complexity. In terms of signaling and train control, the survey finds that although the highest frequency lines (Taipei and Paris, both over 40 tph) both use moving block signaling with fully automated train control, nearly two-thirds of high-frequency systems are still using fixed block signaling with manual control. The benefits of full automation for reliability are emphasized in an analysis of coefficient of variation (CV—the standard deviation divided by the mean of any collection of distributed data) of headways; fully automated systems exhibit mean headway CVs of 0.14, compared to 0.36 for systems with no automation. In terms of crowding, the main constraining factor was realized in dwell times. It is emphasized that among high-frequency lines,

a key practice to reducing dwell times is achieving marginal reductions (1-3s) at every stage of station stop times, including: wheel start, door open/close, passenger boarding/alighting, signal clear, and dispatch delay. Automation is one solution to achieving such marginal gains. In terms of fleet, it is noted that although longer trains require some additional separation time, the additional capacity outweighs the reduction in throughput, and the highest capacity (passengers/hour) metros typically operate with 8-10 car trains. For terminal turnarounds, it is noted that examples of high-frequency terminals exist both with trains turning before or after the terminal, although examples with trains turning before the terminal typically are designed with higher speed crossovers, around 50-60kph, compared to 30kph for turnarounds beyond the terminal. Reducing layover times through operator drop-backs and speeding up passenger movements are listed as other methods for improving terminal capacity. Finally, service complexity such as branches and short turns are noted as a constraint to higher frequencies, although most surveyed lines contain some degree of complexity. Overall, a key takeaway from the report is that even between the surveyed high-frequency metros, significant variations exist in the key line attributes.

Typically, the identification of rail transit bottlenecks is calculated from fundamental system attributes, including a detailed representation of the signaling system and track layout (C. Gill & Goodman, 1992; Delaware River Port Authority, 1973; Jong et al., 2012; Toronto Transit Commission, 1988), which is beyond the scope of this research. Ultimately, this process is necessary, since alleviating one bottleneck may cause others to come to the forefront, meaning an empirical analysis of current operations may be insufficient for complete enumeration of future service constraints. However, the empirical methods presented in this research can still be invaluable as a starting point for more detailed investigations into metro bottlenecks.

1.5.2 Dwell Time Measurement

This section presents a review of dwell time estimation models, relevant to the application of automated data for dwell time measurement in 0. Dwell time modelling is important for future capacity analysis, since dwell time is a major constraint on line capacity, and dwell times cannot be assumed to remain constant with changing conditions, such as frequency and changes in rolling stock. To date, all known studies on rail transit dwell which incorporate passenger demand have relied on manual surveys to collect passenger volumes, limiting the scale of the studies, and the applicability of the findings in a wide range of scenarios. Indeed, in their inventory of rail dwell time models, Li et al. (2018) show that most influential models with passenger regard have been validated at between one and three stations, likely the busiest.

A large number of previous research have estimated the effect of various factors on rail transit dwell time. San & Masirin (2016) provide a review of them most influential models, and cite the three most important factors as passenger volume, crowding effect, and mixed flow effect. However, existing dwell time models, no matter how detailed are difficult to apply to a new system due to large data collection requirements for validation (Harris & Anderson, 2007), and station

specific factors, such as passenger distribution by door, which may vary even within a single line or by time of day.

Unlike rail transit, several studies on bus dwell time have employed AFC and/or automatic passenger counting (APC) technology to estimate dwell time models across large spatial and temporal ranges, controlling for detailed factors such as bus type, fare payment type, stop placement, and passenger type (Milkovits, 2008) (Egge et al., 2017; Glick & Figliozzi, 2017; Sun et al., 2014). Grisé et. al (2017) compared manually and automatically collected bus dwell observations in Montreal to identify biases in dwell time models estimated solely using ADCS, finding that the passenger activity portion of dwell time is overestimated, and that failing to account for passengers with accessibility needs results in an underestimate of passenger activity time on certain routes. Comparatively, rail transit is a less data-rich environment, typically lacking in-vehicle AFC or APC to precisely gather boarding and alighting data on a large scale, resulting in a lack of large scale studies.

Researchers have attempted to overcome the lack of readily available passenger data in rail dwell time models by developing models without regard for passenger volumes. Li et. al (2016) developed non-passenger regarding linear regression models using only track occupancy data, including dwell time at previous station, dwell time of preceding trains, and train length, all of which are normally available in real-time, as opposed to passenger volumes. Similarly, Martínez et al. (2007) used a large collection of automatically collected dwell time observations without passenger regard from the Madrid Metro, finding a mixture of two lognormal distributions to provide the best fit for the observed distribution. While these methods are valuable for real-time prediction, their utility for service planning and forecasting is limited due to the lack of passenger regard and thus applicability to different scenarios.

In the closest related study on rail transit, Buchmueller et. al (2008) gathered over 3-million observations of boarding and alighting flow rates on the Swiss Federal Railways regional rail system using vehicles outfitted with APC systems. The availability of such detailed data is extremely rare in rail transit, especially within the North American context, necessitating the development of dwell time estimation models for more readily available data, such as AFC.

Finally, the *TCQSM (2013)* provides ranges of values observed for loading diversity, passenger flow rates in different contexts, effect of door size/channels, and dwell time variability, noting the spread of values observed even within the North American context. These values are appropriate to provide a range for use in planning new systems, but lack sufficient detail for many applications in established systems. With increasingly available sources of ADCS, there is an opportunity to apply more data-driven dwell time models with wider range of temporal and spatial context. The methods presented in this research employ ADCS to estimate dwell models system-wide, for use in wide scale applications such as simulation modeling and service planning.

1.5.3 Transit Service Reliability

Important aspects of transit reliability include: definition and measurement, customer perceptions of service reliability, operator perception of service reliability, and control strategies to improve reliability.

Definition and Measurement

In the seminal study on transit service reliability, Abkowitz et al. (1978) define reliability as the regularity of service attributes which influence both passenger and transit providers. From the customer perspective, reliability is of concern as to how it effects wait time, in-vehicle travel time, in-vehicle crowding, and transfer time. From the operators' perspective, reliability is of concern as to how it impacts the cost and quality of service provision; for example, schedule adherence, dropped trips, and operator overtime pay. They go on to list different metrics for communicating headway regularity, with particular emphasis on describing the skew; suggested metrics include mean, standard deviation, coefficient of variation, and percent of headways below a certain threshold (i.e., twice the scheduled headway).

Tribone et al. (2014) suggest principles for design of reporting metrics in transit, including: reflective of both customer and operator experience, based on automated data sources for reproducibility, sufficiently concise and understandable as to not be overlooked, and detailed enough to allow identification of underlying issues.

Customer Perceptions of Service Reliability

In terms of customer perceptions, service reliability is often related to wait time. Wait time is commonly cited as more onerous than other components of the travel experience, especially travel time (Ortúzar & Willumsen, 2011; Raveau et al., 2014). Wardman (1998) conducted a review of value of time studies, finding that the average reported wait time valuation as a multiple of in-vehicle travel time is 1.47. Since wait times are minimized through even (regular) headways, this lends value to the focus on service regularity.

Chan (2007) proposes that reliability can be taken into account in traveler departure time and mode choice decision making through the use of a buffer time metric. Buffer time is defined as the difference between budgeted and median travel times. This forms a basis for calculating the effects of improvements in reliability on traveler decisions; if passengers perceive an improvement in reliability, they will respond by reducing their buffer time, and the reverse for a perceived decrease in reliability.

In a review of the full cost of public transportation unreliability, Munez (2019) enumerates the key attributes of service from a passenger perspective (speed, wait times, comfort, reliability), and demonstrates the impact which headway regularity can have on each of these attributes. Of particular interest, he emphasizes the degrading effect headway regularity has on customer satisfaction relating to comfort. With a simple example, he shows that since service evaluation

must be weighted by passenger and not by vehicle, and many more passengers travel aboard crowded vehicles rather than uncrowded vehicles, small perturbations in headway regularity can cause precipitous drops in crowding satisfaction.

Effect on Operator Cost

From the operators perspective, headway regularity can be communicated in its effect on fleet size required. Maltzan (2015) derives an Additional Vehicles Required (AVR) metric to communicate the additional fleet size required to provide a level of service as indicated by the schedule. Assuming no changes in cycle time or improvements to headway regularity, Maltzan shows the AVR can be calculated simply as the square of the line weighted coefficient of variation of headway. This metric is a valuable contribution to the literature on reliability, because it bridges the gap between passenger and operator perceptions of reliability, where most regularity metrics focus solely on measuring the passenger experience.

Control Strategies for Improving Service Reliability

Abkowitz et al. (1978) classify reliability improvement strategies as preventative or corrective. Preventative measures are primarily associated with scheduling, planning, and route/environmental modifications, while corrective measures are associated with operations control.

Control point holding strategies are one common corrective strategy; they "reduce the magnification of deviations further downstream" (M. Abkowitz et al., 1978) and must balance the needs of in-vehicle and downstream passengers. Literature on headway-based holding strategies in high-frequency transit are abundant, including in simulation environments (Eberlein et al., 1999; Fabian, 2017; Maltzan, 2015), and in real-world demonstrations (M. D. Abkowitz & Lepofsky, 2007; Berrebi et al., 2018; Fabian, 2017; Maltzan, 2015). However, holding strategies still appear to be elusive in practice, perhaps because the benefits of improved headway regularity incurred by holding are not well understood by agencies in contrast to more traditional approaches such as increasing the number of scheduled trips, or because of the perceived complexity in planning such strategies.

Turnquist and Blume (1980) proposed a simple screening model based on headway variability and the proportion of passengers in-vehicle vs. downstream to selecting appropriate locations to apply holding strategies.

Preventative approaches to improving service regularity can involve reducing variability in the operating environment, such as all-door boarding to limit bus dwell time, transit signal priority to limit variability in signal stop time, or automatic speed regulation to reduce operator differences in speed. Strategies to ensure trains depart on schedule can also be considered preventative measures for improving service regularity. Carrel (2009) provides a useful inventory of crew and train management techniques from discussions with service controllers on the London

Underground. Such strategies, including “dropping-back”, “stock and crew”, “switching operators”, and “substituting a spare operator” can be used for returning a late train to its schedule, without the use of the more disruptive short-turning. However, aside from drop-backs, these techniques do not appear to be commonly used in practice due to the requirements to have extra trains or operators standing by.

1.6 MBTA Red Line Characteristics

1.6.1 Line and Service Characteristics

This section provides an overview of the MBTA Red Line, which will be used as a case study throughout this thesis. Although the analysis presented is specific to the MBTA Red Line, the methods and findings are applicable broadly to other agencies and rail transit lines operating under similar service plans.

The Red Line is the longest, highest frequency, and highest ridership heavy rail line in the MBTA network with an average weekday ridership of approximately 244,000 (MBTA Performance Dashboard, 2017). The Red Line is composed of a trunk section and two branches, as shown in Figure 1-1 along with key connecting rapid transit lines. The trunk line runs from the northern terminus of Alewife Station to JFK/UMass Station (“JFK”) and is 14.2 km (8.8 miles) with 13 stations. The two branches are named after their terminus stations; the Braintree branch is 14.2 km (8.8 miles) with 5 stations, while the Ashmont branch is 4.7 km (2.9 miles) with 4 stations. The imbalanced length and passenger demand between branches contributes to complexity in service planning and delivery. Here, as elsewhere, JFK is classified as a trunk station, since it is served by both branches, although the two branches serve the station on separate platforms, as depicted below.

Figure 1-1: MBTA Red Line Schematic

The published headways on the Red Line are shown in **Table 1-1**, according to the MBTA website. Peak period branch headways of 9-minutes (rounded) result in a scheduled trunk headway of approximately 4.5 minutes, although peak hour headways closer to 4-minutes are often observed. Service on the trunk section generally operates with alternating trains from the two branches; there are no scheduled short-turns and all branch trains operate from end to end. To deal with higher demand on the Braintree branch, two back-to-back pairs of Braintree trains are scheduled in the morning and afternoon peaks each. Maintaining the correct sequence of branch trains is a top priority for service controllers, especially in the PM, in order to control platform crowding on southbound platforms downtown.

Table 1-1: MBTA Red Line Weekday March 2019 Scheduled Headways (min)

Branch	Direction	AM Peak	Midday	PM Peak	Evening	Late Night
Braintree	Northbound	9	14	9	12	12
	Southbound	9	14	9	12	12
Ashmont	Northbound	9	14	9	12	12
	Southbound	9	14	9	12	12

The signaling system on the Red Line is fixed-block system, with Automatic Train Control and five speed commands ("speed codes"): 0, 10, 25, 40, and 50 mph. When a track block is occupied, it communicates speed codes to upstream blocks (typically up to five upstream blocks), indicating the maximum allowable speed for trains in that block. Operators can proceed at any speed at or under the speed code. Speed codes vary based on track geometry and grade, but invariably the block immediately upstream of an occupied block receives a zero speed code, meaning that the following train cannot proceed through the block.

1.6.2 Passenger Demand Patterns

Estimated passenger movements are obtained through origin-destination inference of passenger fare card transactions as described in Section 1.4. Land use along the branch lines is heavily residential, leading to a strong directional flow throughout the network, with passengers traveling predominately inbound (towards downtown) in the morning peak and outbound (away from downtown) in the evening peak. In contrast to the outlying stations, the downtown stations of Park Street, Downtown Crossing, and South Station, exhibit primarily bi-directional passenger movements due to a high volume of transfer movements. The trunk section contains a mix of residential and commercial hubs, with major trip generators including Harvard, MIT, Charles/MGH, the downtown financial district, and major transfer stations at Park Street, Downtown Crossing, and South Station. This means trunk demand is less directionally dominant compared to the branches, with about one third of trunk-only passengers traveling in the non-dominant direction during each peak.

Table 1-2 provides a summary of the demand by time period, direction and branch, as calculated from ODX records from October 2018, including scaling for passengers with uninferred destinations. In this context, a branch passenger is defined as one who can only take a train from a single branch; that is, at least one of their origin or destination are on the branched portion of the line. Peak hour passenger flow along the line is shown in **Figure 1-2**. The passenger flow diagrams help to communicate the peak load points, and demand patterns, which are important considerations for service control interventions discussed in Chapters 6 and 7. Notably, Braintree passengers outnumber Ashmont passengers nearly 2:1 during the peak hours and peak direction, although they receive roughly the same service frequency.

Table 1-2: Peak Hour Boardings by Branch Destination

Time Period	Direction	Branch	Demand (Thousands)	Fraction (%)
AM Peak (8:00-9:00 AM)	Northbound	Ashmont	3.1	19%
		Braintree	6.1	37%
		Trunk	7.3	44%
		Total	16.5	100%
	Southbound	Ashmont	0.4	2.7%
		Braintree	0.8	5.5%
		Trunk	13.4	92%
		Total	14.6	100%
PM Peak (5:00-6:00 PM)	Northbound	Ashmont	0.6	4.1%
		Braintree	1.1	7.9%
		Trunk	11.7	88%
		Total	13.4	100%
	Southbound	Ashmont	2.6	17%
		Braintree	4.9	32%
		Trunk	7.8	51%
		Total	15.3	100%

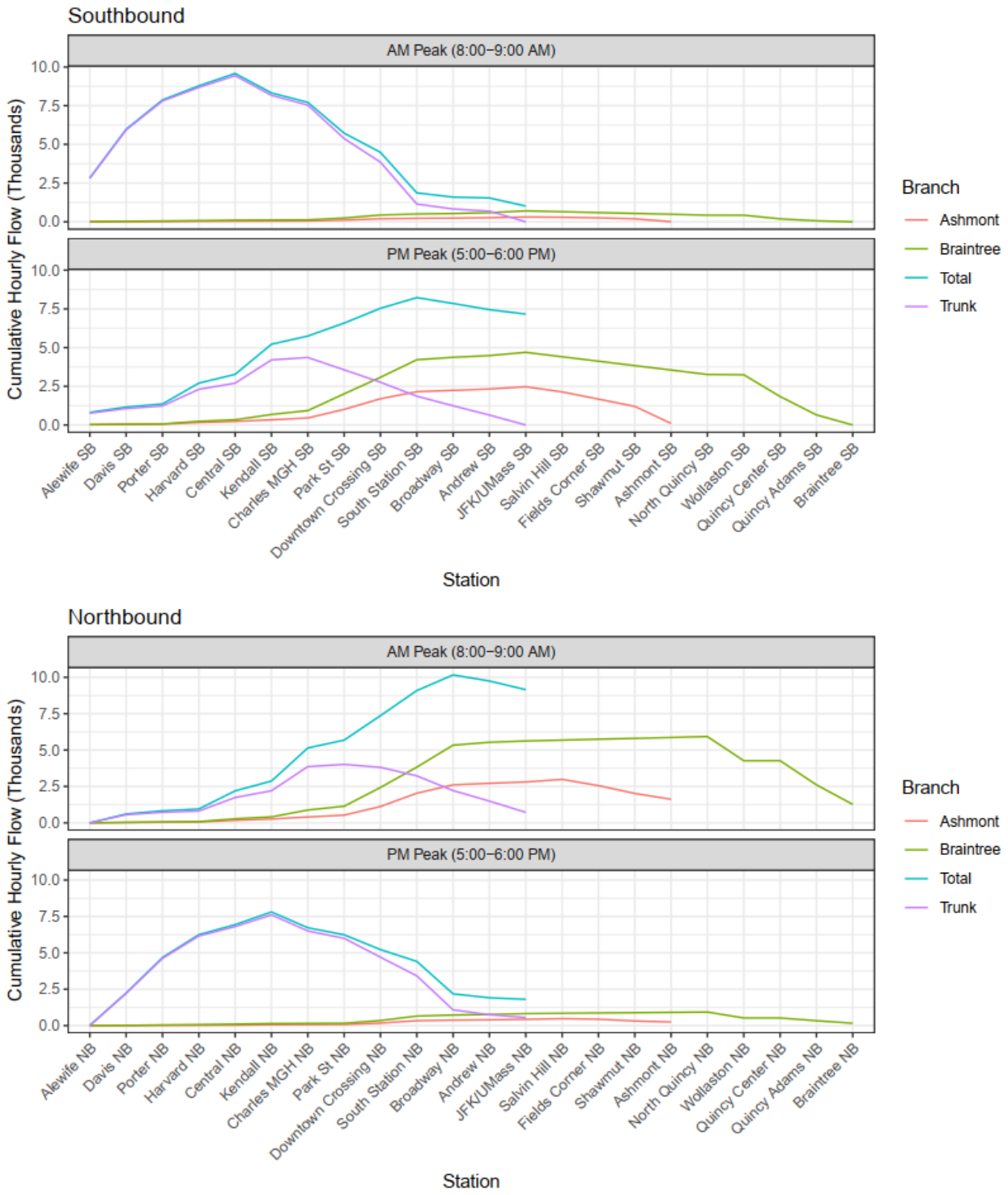

Figure 1-2: Peak Hour Passenger Flow

1.6.3 Line Bottlenecks

This section presents an overview of the typical bottlenecks in high-frequency rail transit, with specific reference to the Red Line, including station dwell time, terminal stations, junctions, and headway regularity. Finally, the relationship between these bottlenecks will be discussed and their connection to the line capacity.

Station Dwell Time

It is well recognized in practice that critical station dwell time is typically the constraining factor on total line capacity in high-frequency rail transit *(TCQSM*, 2013). The main components defining minimum station headway are station stop time and station close-in time, where the former is defined by passenger demand and operating practices, while the latter is primarily a function of signaling system characteristics. Improvements to station dwell time typically target passenger behavior (crowding, demand management, etiquette) and operating practices (consistency of operations, station attendants), which are inexpensive compared to infrastructure changes required to affect station close-in time.

In the case of the MBTA Red Line, several factors accentuate the bottleneck effect of station dwell time; namely passenger demand patterns, station spacing, and branched design.

Due to the strong central business district (CBD) oriented commuting patterns, and the radial heavy rail system layout with all transfer points located in the CBD, passenger demand on the Red Line is heavily concentrated, at three downtown stations; Park Street, Downtown Crossing, and South Station. Due to the concentration of demand, these downtown stations have significantly longer dwell times than outlying stations. Because each of these downtown stations is also a transfer point between rapid transit lines within the MBTA network, peak hour passenger flow is bi-directional, reducing efficiency of the boarding and alighting process compared to primarily uni-directional flow.

The signaling system and line layout on the Red Line is another limiting factor on station bottleneck capacity, due to the conservative system design and downtown station proximity. Downtown Crossing and Park Street are separated by just a single track block, meaning that trains occupying one station cause the upstream station block and station approach block to receive slower speed codes. This has a negative impact on station close-in time and bottleneck capacity, in addition to resulting in slower speeds and higher cycle times during peak periods. Such dense stop spacing would be atypical in more recently constructed heavy rail systems due to the line-capacity implications. Harvard is another key station where the line geometry and conservative speed codes negatively impacting pull-in and pull-out time and thus overall capacity. In both direction, trains entering Harvard at low headways may be required to come to a complete stop before proceeding into the station. The aforementioned issues of limiting signaling system configuration will see some improvement following the introduction of two extra speed commands enabled by planned upgrades to the signaling system and complete replacement of the rolling stock

(estimated completion 2023-2025). The extra speed codes are expected to allow trains to operate at 15 mph rather than the current 10 mph in some constraining situations.

Finally, the branched nature of the line contributes to increased dwell times in two ways. Firstly, an imbalanced demand between branches causes more extreme dwell times for Braintree trains and a corresponding reduction in capacity compared to a non-branched line with even demand between trains. Secondly, since not all southbound passengers are able compatible with all trains, passengers are left-behind on the platform after every departure ("incompatible passengers"). The increased platform load from incompatible passengers contributes to increased friction between boarding and alighting movements and thus longer dwell times for all trains at crowded downtown stations during the PM peak period.

Terminal Stations

According to the *TCQSM* (2013), it is unusual for terminal turnarounds to be the limiting factor in high-frequency transit operations. They suggest that a well-designed stub-end terminal should be capable of handling headways down to 2-minutes, and suggests an equation for calculating the maximum allowable layover time given fundamental attributes of the terminal geometry and a given headway. For further study on terminal capacity, Lee (2002) provides a framework for the simulation of high frequency rail transit terminals, and suggests the three key factors in determining terminal capacity and performance are: the train arrival process (and variability), the scheduled train recovery time, and the length of peak operations. These factors are important to consider when evaluating the viability of a given terminal operation.

On the MBTA Red Line there are three terminal stations; Alewife Station is the northern terminus, and Braintree and Ashmont Stations are the southern termini for the two branches. Terminal track schematics are shown in Figure 1-3, and peak period scheduled layover times for each terminal in **Table 1-3**. The track schematics depict crossovers, tail tracks, and the location of platforms (island or side).

The Alewife terminal is of primary concern under a future service increase. Alewife currently operates as a stub-end terminal with all crossover movements occurring in-front of the station. The crossover behind the station is used only for layups and put-ins due to the slow speed limit (approximately 5 mph). Under a current 4.5-minute peak headway, Alewife experiences congestion entering the terminal area, and irregular departure headways due to uneven speed limits on the two forward crossover paths. Additionally, Alewife has the shortest scheduled layover of the three terminals at approximately 6-minutes, impacting the ability for trains to make a complete recovery following northbound delays. The feasibility of a sustained 3-minute headway at Alewife will be explored in future chapters.

The branch terminals at Braintree and Ashmont, operating as stub-end and relay terminals, respectively, will see maximum sustained headways of 6-minutes in future scenarios, assuming an even distribution between branches. Even in the case of back-to-back pairs of branch trains at a 3-

minute headway, the capacity will likely not be critical, since the short headways would not be sustained for an extended period, as they are at Alewife.

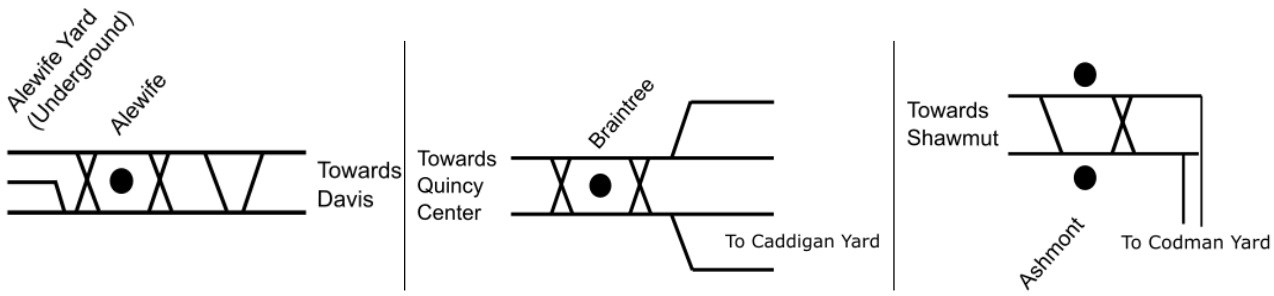

Figure 1-3: MBTA Red Line Terminal Track Schematics

(From left to right: Alewife, Braintree, Ashmont). Adopted from *MBTA Ridership and Service Statistics* (2014).

Table 1-3: Spring 2018 Scheduled Layover Times

Terminal	Time Period	Median	Mean	Range
Alewife	AM Peak	6	6.5	3-11
	PM Peak	7	6.7	5-7
Ashmont	AM Peak	11	11.7	9-15
	PM Peak	10.5	11.3	7-16
Braintree	AM Peak	13	13.5	11-16
	PM Peak	13	13.1	7-21

Junction

According to the *TCQSM* (2013), flat junctions can be operated down to extremely low headways of about 2-minutes, although grade separation is typically recommended for headways below 3-minutes. The *TCQSM* provides an equation for determining the achievable minimum headway at flat junctions given system geometry and performance attributes. In grade separated junctions, merging and diverging movements can be made without conflict, and should not represent a capacity constraint unless the junction track geometry requires very slow speeds and is in immediate proximity of a headway limiting station (*TCQSM*, 2013).

The Red Line contains one mainline junction, where the Ashmont and Braintree lines merge and diverge, called the Columbia Junction. Because the junction is configured as a grade-separated flying junction, the capacity is not a limiting factor, nor does it cause significant interference, such as at flat junctions. Today, uncoordinated branch trains arriving nearly

simultaneously to JFK can achieve approximately a 2-minute headway through Andrew Station, showing that junction capacity is not a limiting factor. However, the branched layout of the line does introduce scheduling complexity and headway irregularities into the trunk section, which can reduce the trunk capacity. In this way the branched layout of the line can be seen as a line bottleneck. The variability of headways caused by the branch merge will be explored in Chapter 2.

Headway Regularity

Headway regularity relates to the ability for an agency to reliably operate the scheduled level of service. Systems with high headway irregularity may have reduced overall capacity, because the maximum throughput is dependent on the consistent utilization of the bottleneck point. Additionally, systems running close to, or at-capacity may not be able to recover from gaps in service for an extended period.

In case of the Red Line, headway irregularities are introduced to the line at discrete points, in addition to the typical variability of high-frequency transit originating from stochasticity in demand and dwell times. These points include the branch merge, and the routinely uneven departure headways from Alewife. Large gaps introduced at these discrete sources may limit overall capacity by reducing the bottleneck utilization.

Red Line Bottleneck Summary

In summary, the main challenges of operating a future 3-minute headway on the MBTA Red Line can be discussed under the themes of headway irregularity and station dwell times, as shown in **Figure 1-4**, along with contributing line characteristics. In particular, it is critical to recognize the positive feedback loop between headway irregularity and limiting dwell times.

Headway irregularity is caused primarily by the branched line layout, irregular dispatches from the Alewife Terminal, and inconsistent and extreme dwell times, summarized as follows. The branched line layout contributes to significant headway irregularity in both directions. In the northbound direction, the lack of real-time coordination between dispatches and stochastic branch travel times results in northbound trunk headway irregularity. In the southbound direction, the branched layout contributes to uneven demand and dwell times between successive trains, and large gaps which occur when reordering trains at Alewife. The Alewife Terminal contributes to headway irregularity by failing to fully recover from northbound delays, owing to the schedule based dispatching regime, lack of flexibility in reducing layover times, and slow and irregular crossover speeds. Finally, station dwell times contributes to headway irregularity as one of the major sources of variability in line running times between successive trains, since initial headway differences tend to be reinforced by dwell time differences.

Limiting station throughput is caused primarily due to characteristics of passenger demand and line layout, with contribution from the aforementioned sources of headway irregularity.

Passenger demand characteristics which contribute to limiting station dwell times include an imbalance in demand between branches, the branched layout which causes left-behind related crowding in the southbound direction, and the concentration of demand at the downtown stations. Line characteristics which contribute to limiting dwell times include restrictive station geometry and signaling system characteristics. Finally, headway irregularity contributes to limiting station dwell times, since long headways are associated with higher demand and thus longer dwell times.

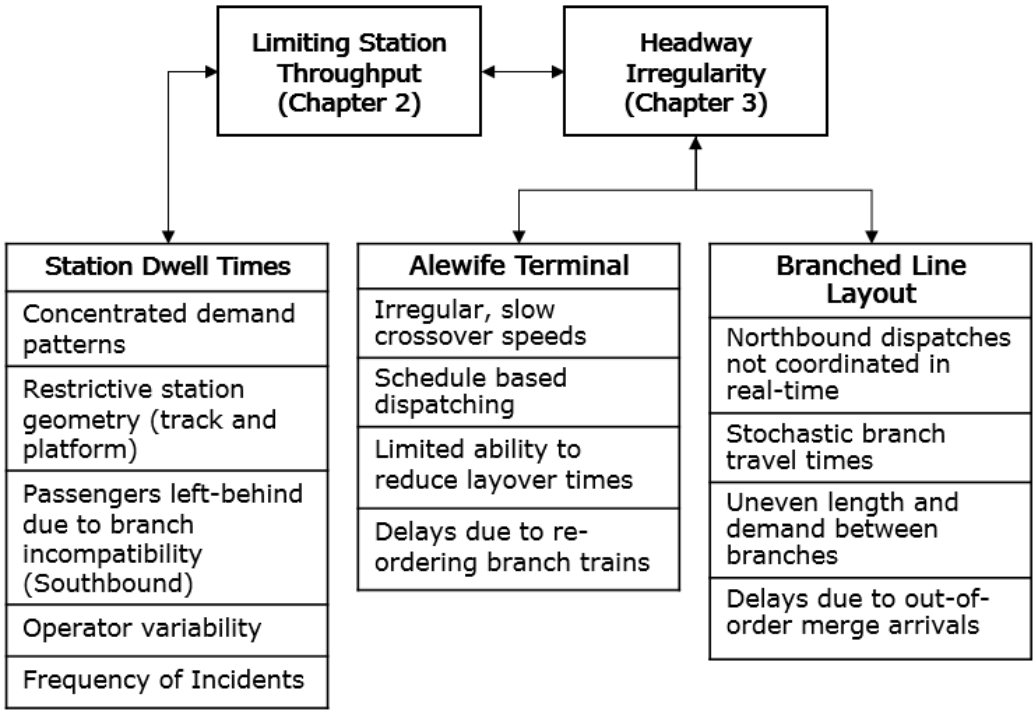

Figure 1-4: Overview of Line Bottlenecks and Interactions

1.6.4 Future Service Plan

Many important details of the future service plan are unclear at the time of writing, including for example, how additional trains will be distributed between the two branches, span of peak frequency, or the timeline for implementing the increased frequency (incrementally vs. all at once). Furthermore, information on the programmed improvements to the signaling system, new rolling stock attributes, and other capital investments were not available in detail. Another crucial element of modeling future service is projected ridership growth scenarios, which have consequences for dwell times and thus cycle times. With many unknowns, this research does not set out to identify full solutions to the future operation conditions, but instead to suggest incremental operational changes, which ultimately contribute to successful implementation of the proposed frequency. Parallel research at the MIT/NEU Transit Lab seeks to extend a previously developed simulation model of the MBTA Red Line (Wang, 2006) for application to future service planning, including different operational scenarios described in this document.

1.7 Thesis Organization

This section will outline the contents for the remaining chapters of this thesis.

Chapter 2 presents an exploration of the current state of headway regularity on the Red Line, and aims to highlight the negative impacts that a lack of regularity can have on service quality and resource efficiency. The review also identifies several causes of headway irregularity and their relative frequencies on the Red Line. The relationship between headway regularity and service frequency is emphasized, showing how the frequency improvements promised by the MBTA will likely fall short if not accompanied by regularity improvements.

0 presents an exploration of the current state of station dwell time on the Red Line. It aims to highlight stations that will require particular attention to ensure the success of future capacity objectives, explore the use of automated data for dwell time measurement and prediction, and outline strategies which can be taken to reduce the impact of dwell time on system reliability.

Chapter 4 introduces three service interventions ("Pilots"), aimed at improving headway regularity, as it relates to current and future service delivery objectives. The pilots demonstrate the incremental approach to service improvements which is necessary to achieve future service targets, while simultaneously providing benefit under current operating conditions.

Chapter 5 presents the results of the Alewife Drop-back Pilot, a crew management change intended to increase schedule recovery time, reduce queuing delays, and demonstrate the effectiveness of the technique for future operating conditions.

Chapter 6 presents the results of the Davis Southbound Holding Pilot, a headway-based holding strategy which was implemented in order to reduce headway differences between successive dispatches caused by crossover speed limitations, and to demonstrate the impacts of increased dispatch regularity on line performance.

Chapter 7 presents the result of the Andrew Northbound Holding Pilot, a headway-based holding strategy which was implemented in order to improve headway regularity following the branch merge, which has implications for future operating conditions which likely will not be able to recover from high service variability.

Chapter 8 offers conclusions and recommendations from the exploration of current headway and dwell time, and from the pilot studies, with implications for future service provision.

Chapter 2. Headway Regularity

2.1 Introduction and Motivation

This chapter presents a discussion of headway regularity, both in its impacts on service quality and line capacity. Headway regularity is a vital concept in transit capacity and quality of service analysis. At any level of service, headway regularity is closely linked to frequency as experienced by the customer, since it influences the average wait time and train crowding. At near-capacity operations, headway regularity at a line bottleneck is a pre-requisite for delivering the scheduled service frequency, as lost capacity due to irregular headways may not be possible to recover from without cancelling trips.

Providing high-frequency service with poor regularity also represents an inefficient use of valuable peak period resources, since improvements to headway regularity can typically be implemented with minimal costs compared to capacity improvements from capital expenditure. With large capital improvements planned on the MBTA's Red Line (signaling system improvements, full fleet rolling stock replacement, Alewife crossover reconstruction), a strategy to improve headway regularity represent an important complement to ensure the investments deliver their stated capacity benefits and for passengers to experience interim benefits.

Section 2.2 will discuss factors influencing headway regularity, including initial headway, headway stability, and contextual variables. Sections 2.3 and 2.4 will review initial headway regularity and headway stability on the MBTA Red Line, including reference to relevant contextual variables. Section 2.5 will discusses the implications of headway irregularity for passengers. Finally, Section 2.6 will discuss strategies to improve regularity.

2.2 Factors Affecting Headway Regularity

Outcomes of headway regularity of an existing rail transit service can be explained by considering initial headways, stability of line headways, and different contextual variables which influence passenger demand or train performance. This section will give a general overview of the three aforementioned factors.

2.2.1 Regularity of Initial Headways

The regularity of initial headways has a large impact on line performance. Here, the term "initial headways" is used rather than "dispatch headways" to include the effect of branch merges, and terminal crossovers which affect how the dispatch headway is experienced downstream. Without interruption from service controllers or from incidents, the initial headway of a trip tends to define a trip's outcomes; large dispatch headways become magnified due to increasing

passenger demands and dwell time, while short dispatch headways tend to further bunch due to relatively low passenger demand compared to the leading train. Dispatch control also deserves special consideration because unlike in-line headway control, headway adjustments made at the terminal are not generally perceived as delays by onboard passengers. For these reasons, improving dispatch headway regularity can cause significant improvements downstream, without significant negative customer impacts.

There are two primary regimes for dispatching in fixed route transit; headway-based and timetable based. Although passengers perceive high-frequency routes to run on headways, a survey of very high-frequency metros worldwide reveal that most in fact operate on highly precise timetables, scheduled to at least the nearest 15 seconds, due to the high amount of resource coordination required in such operations (Canavan et al., 2019). However, study of the operation control environment shows that service controllers may still focus primarily on controlling headways due to the difficulties of incorporating all schedule and crew considerations under time pressure (Carrel, 2009). These contrasting methods of service delivery and passenger perception should be considered when evaluating dispatch headways.

2.2.2 Headway Stability

The stability of headways is another important factor in determining line headway regularity. Headway stability is defined in this context as the ability for a train to maintain its dispatch headway within a given tolerance, and represents the relationship between dispatch headway and subsequent in-line headways. Headway stability is primarily influenced by the dispatch headway, and the resulting feedback loop between headway, passenger demand, and dwell time. These factors may in turn be influenced by contextual factors described under the following heading. Highlighting the disparate outcomes of different dispatch headways can help make the case for improvements to dispatch regularity and for more precise dispatch under different contexts.

2.2.3 Contextual Variables

In addition to the initial headway, contextual variables including train itinerary, time of day, rolling stock, and incidents can influence the probability of different outcomes at a given dispatch headway. The relation between dispatch headway and stable headway region will be unique for each context. These different factors will be outlined below.

Itinerary

The itinerary of a train (i.e., express/local, branch destination) is closely associated with its demand patterns during a particular time of day. Demand is of critical importance when considering headway evolution. For example, a train with lower demand (e.g., destined for a lower demand branch) may be able to maintain a stable headway even in context of a long dispatch,

whereas a train with higher demand (e.g., higher demand branch) may fall behind the leading train even when dispatched at the scheduled headway.

Time of Day

The time of day has a significant impact on passenger demand, which can in turn impact headway stability. During the peak-of-peak, headways tend to be the most sensitive to initial conditions, since high demand can quickly deteriorate performance through crowding and increased dwell time. Sensitivity may decrease during shoulder periods, when excess demand does not cause trains to reach a high level of crowding which influences dwell times. During off-peak periods, sensitivity to initial headway is generally expected to be much lower, with only large changes in initial conditions or demand fluctuations causing headways to become unstable.

Rolling Stock

Within a non-homogenous fleet, differences in rolling stock characteristics impacting headway evolution can produce differences in dwell time and running time performance. Factors affecting dwell times may include number of doors, number of cars, seating configurations. Factors affecting running time may include acceleration and deceleration performance, and performance under heavy passenger loads. Fleets of different ages can also vary in their susceptibility to mechanical issues. Each of these factors can influence headway adherence, especially in the context of a mixed fleet. In the case of the Red Line, a mixed fleet containing three separate generations of rolling stock are operated (Type 1-3). Most critically, Type 1 and 2 cars have three doors while Type 3 cars have four doors, resulting in differences in station stop passenger flow rates.

Incidents

Train or passenger incidents can cause significant discontinuities in headway stability, and act contrary to the other influences discussed herein. Without intervention from service controllers, incidents cause an increase in headway proportional to the incident time for the delayed train, and a decrease in headway proportional to the incident time for the following train (constrained by safe-separation distance). Incidents can impact the headways of not only the affected train, but also following trains. If there is little slack to absorb delays, a single incident can continue to impact trains, causing them to follow at minimum headways until slack is introduced into the system, whether by scheduled or ad-hoc pull-outs. It is important to consider the effect of incidents on headway stability, because they may act contrary to the anticipated direction of headway change caused by passenger demand, obscuring that connection.

Minimum Train Spacing

Minimum train spacing and other headway bottlenecks have an important impact on headway stability, because they prevent the expected result of bunching for closely spaced trains

with little demand. Minimum train spacing constraints may cause very short dispatch headways to appear stable.

Operator Differences

Although some operating characteristics such as maximum speed may be governed by the signaling system, there is still significant leeway for operator actions to shape headway stability. Some operator differences may include time switching ends at the terminal station, assertiveness in closing doors at busy stations, acceleration out of stations, and importantly, re-occupation time (i.e., the time required to proceed into the next signal block or station when following a train at the minimum allowed distance) in near-capacity situations. The impact of operator differences will vary significantly by agency context, depending on operator training, passenger behavior, and level of automation. No quantitative evidence is available to support the theory of operator differences on the Red Line, although MBTA management believes that differences in operator driving practices can impact headways significantly on the Red Line.

2.3 Initial Headways Exploration

This section will examine the Alewife Terminal and Columbia Junction and their effect on initial trunk headways. The focus of this section is placed primarily on the trunk section, which operates at a combined headway twice that of the branches, and is thus more critical for determining limiting line capacity.

2.3.1 Alewife

Due to its position as the trunk terminus, operations at Alewife play a significant role in delivering reliable southbound operations. Among the key issues facing Alewife operations are the variable headways from the northbound branch merge, re-sequencing of out-of-order branch trains, and inconsistent crossover speeds for trains entering and leaving the terminus. After giving a general overview of Alewife operations, these specific issues will be discussed.

Overall Headway Distribution

Table 2-1 shows the distribution of the scheduled headways arriving to and departing from Alewife during peak periods. Note that Red Line schedule is written only to the nearest minute, so a combination of clock headways is required to achieve the desired average headway. Although there is some variation in scheduled arrivals, trains in the AM Peak are scheduled to depart with a uniform headway. The situation is similar in the PM, with some minor variations in scheduled departure headways.

Table 2-1: Alewife Scheduled Headway Distribution (2018)

Headway	AM Peak (7:00-9:00 AM)		PM Peak (4:00-6:00 PM)	
(min)	Arrivals	Dispatches	Arrivals	Dispatches
2	-	-	4%	-
3	10%	-	4%	7%
4	69%	100%	56%	73%
5	21%	-	37%	20%

Observed headways arriving to and departing from Alewife Terminal in the peak periods are shown in **Table 2-2** for all weekdays in October 2018. Here, headways are measured as they arrive/depart from the platform, drawing the boundaries around the platform as closely as possible with the available train tracking data.

The pattern of arrival headways shows the significant variability that Alewife operations must contend with due to variability in northbound operations. Only 65% of trains in AM and 54% of trains in PM arrive within 1-minute of the mean headway. Contrary to what would be expected from Alewife as a schedule recovery point, no significant reductions in headway variability are observed between arrivals and departures; for example, the 15th and 85th percentiles show minimal improvements of no more than 0.2-minutes. Slightly larger improvements are seen in the 95th percentile.

Table 2-2: Alewife Terminal Headways

Time Period	Case	Mean	SD	15th Pctl	Median	85th Pctl	95th Pctl	Within 1-min of Mean Headway
AM	Arrivals	4.2	1.8	2.9	3.9	5.2	7.4	65%
Peak	Departures	4.2	1.4	3.1	4.0	5.4	6.9	66%
PM	Arrivals	4.4	1.9	2.9	3.8	5.9	8.1	54%
Peak	Departures	4.3	1.5	3.0	3.9	5.8	7.3	58%

The lack of significant improvements to headway variability and lack of reduction in large headways at the terminal suggests inefficiencies in the terminal procedures at Alewife, including lack of significant scheduled recovery time, conflicting terminal movements, and a schedule-based rather than headway-based dispatching policy. Schedule-based dispatching is typically not very robust and relies on compression of the scheduled layover time to achieve on-time departures for late-arrivals. In cases where layover times are already close to the minimum, such as at Alewife, reduction in layover times may not be possible and arrival delays are maintained without significant reductions, as seen in this section.

Although it is difficult to assign a single attribution to each long dispatch headway, it is noted that among peak period dispatches with headways 6-minutes or greater, approximately half

the trains had an Alewife layover time less than the peak period median, and half larger than the median. Trains with short layover times and long dispatch headways could indicate that operators were simply not able to recover enough time at the terminal, while trains with long layover times and long dispatch headways could indicate the presence of conflicting terminal movements.

Effect of Alewife Crossover

Investigation of headways downstream of Alewife reveals that the design and condition of the crossover used by trains departing Alewife has a significant impact on downstream headway distribution resulting from a difference in allowable speed for trains departing Alewife from Platform 1 ("P1 trains") and Platform 2 ("P2 trains") (**Figure 2-1**). Trains departing from P1 are limited to 10 mph, as compared to a speed limit of 25 mph for P2 departures. By the time trains reach Davis, manifests in a running time difference of approximately 60s between dispatch platforms, effectively increasing the headway of P1 trains by 60s and decreasing the headway of P2 trains by 60s. For example, an even 4:00 dispatch headway results in headways of 5:00 (+60s) and 3:00 (-60s) for P1 and P2 trains, respectively, at Davis Station Southbound.

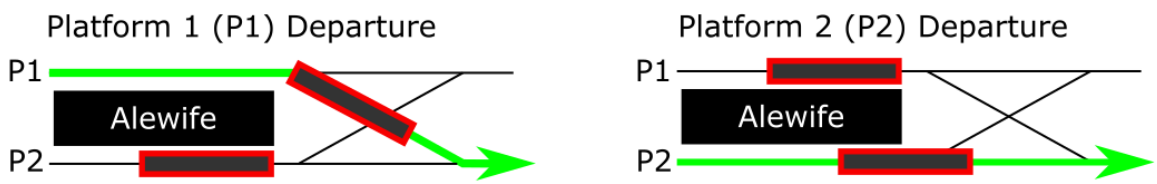

Figure 2-1: Alewife Departure Movements

The distribution of peak period headways once trains arrive at Davis, compared to the headways measured departing the Alewife terminal are shown in **Figure 2-2**. The outcome of the different speed codes is most noticeable in the third panel showing the combined headway distribution; measured at Davis, the combined headway displays a bi-modal distribution with modes at approximately 2.5 and 4.5-minutes. In the first two panels, smaller but still noticeable differences in headways by platform measured at Alewife show that the closest track circuits to the platform already include some pull-out time, which is different between the two platforms.

Based on commonly understood effects of bus/train bunching, the bi-modal headway distribution can be expected to lead to increased average wait times, uneven passenger loading between successive trains, and stop-and-start speed codes for the following (bunched) train. Although it is understood by operating staff that the two platforms receive different speed codes, there is no attempt by dispatchers to adjust/correct for the impact on headways. A headway-based holding intervention described later in Chapter 6 attempts to address these differences.

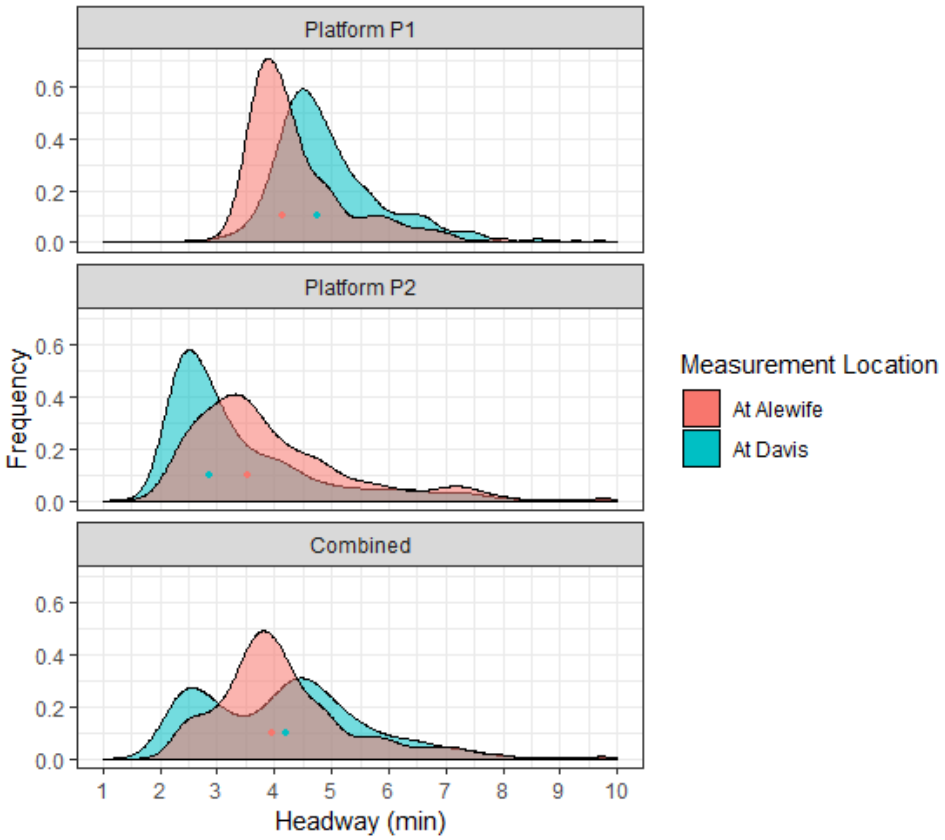

Figure 2-2: Effect of Alewife Crossover Speed Difference on Headways at 1st Downstream Station

Terminal Queuing Delays

The lack of flexibility in terminal operations contributes to another major problem; queuing delays to enter the terminal. The limits of the analysis are chosen as departing Porter Northbound (two stops prior to terminal) to arrival at Alewife terminal, based on analysis that shows delays are primarily concentrated between these points. The queuing delays are primarily concentrated between Porter Station (two stops away) and Alewife, and so the limits of the analysis for the following delay metrics is chosen as departing Porter to arriving at Alewife. Notably the analysis includes dwell time at Davis Northbound, which is assumed to be of minimal impact compared to the queuing delays. The aforementioned boundaries will be referred to as the Terminal Area for the remainder of this analysis.

To communicate the delays, a metric of excess running time (ERT) is used, which represents the running time minus the minimum observed running time (approximately 200s in this case). **Table 2-3** shows summary statistics of for the terminal area excess running time for the month of October 2018. **Figure 2-3** shows individual observations of terminal area excess running time for a sample of days, which is valuable for understanding how the delays propagate and dissipate.

The median ERT is found to be relatively minor at about 1-minute in both AM and PM peaks. Larger percentile ERT are more severe; in the AM peak exhibits 85th percentile delays of 4.1-minutes and 95th percentile delays of 6.1-minutes. Extreme delays in the PM peak are slightly smaller, possibly because service controllers are more conscious about limiting delays to in-vehicle passengers in the PM, when northbound passenger volumes are higher. The disaggregate sample shows a common pattern of mounting delays, which eventually dissipate, either through dispatcher action or a long northbound gap. For example, in the case of the AM peak on October 18th, a buildup of ERT of 4-5 minutes occurs, lasting for 30-minutes. In addition, some large delays occur without any buildup are visible (ex. October 16th, PM peak). This is attributed to improper prioritization of terminal movements. These regular delays represent an inefficient use of operator/rolling stock resources, and substantial inconvenience to passengers.

Table 2-3: Alewife Terminal Area Excess Running Time (ERT)

Time Period	Mean	SD	15th Pctl	Median	85th Pctl	95th Pctl
AM Peak	2.0	2.0	0.4	1.2	4.1	6.1
PM Peak	1.4	1.3	0.5	0.9	2.5	3.9

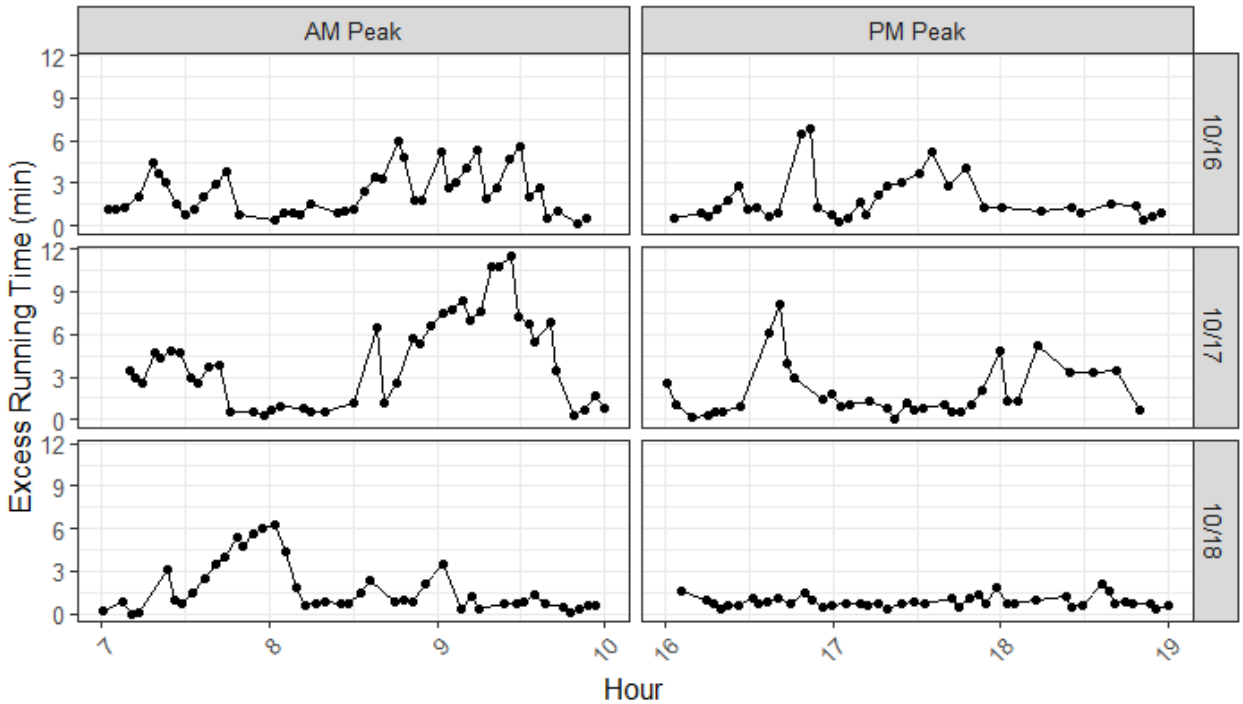

Figure 2-3: Alewife Terminal Excess Running Time (Sample of Days)

2.3.2 Northbound Branch Merge

Next, the northbound branch merge south of Downtown Boston is examined due to the significant impact it has on trunk headway variability. First, an overview of the observed merge headways will be presented, showing significant irregularity and a pattern of day-to-day variation. Second, branch running times are examined, showing the potential for dispatch coordination.

Due to passenger demand patterns on the Red Line, only the PM Peak merge operations are examined; in the PM, 88% of northbound passengers originate on the trunk compared to just 44% in the AM. These demand patterns imply that management of northbound trunk headways is significantly more important with respect to crowding in the PM, and can be done with less delay to in-vehicle branch passengers. Although not shown here, it should be noted that AM merge headways follow largely the same patterns and procedures as do the PM.

Merge Arrival Headways

A summary of headway statistics both before and after the merge is shown in **Table 2-4**. Overall, there is evidence of significant irregularities in merge headways. Recall that **Table 2-1** showed roughly uniform northbound scheduled headways of 4-5 minutes. Headways arriving at JFK have a spread of 5.0 minutes for the middle 70% of observations, and a CV of 0.58. Headways at Andrew are observed to improve slightly compared to at JFK, however the improvement is attributed only to the effect of the minimum safe separation on spacing out near-simultaneous arrivals. Andrew Northbound has a minimum headway of approximately 100s, as observed through empirical headway distributions. At JFK, the branches berth on separate platforms, so there is no minimum separation time there for two trains of different branches. Trains merge onto a common track at a flying junction between JFK and Andrew stations.

From discussions with Red Line management, it was discerned that branch merge headways are primarily managed through timetabling, rather than through real-time coordination. When real-time coordination does occur, it is in the form of in-line holding rather than coordinated dispatching, and mostly focuses on ensuring the correct sequence of branch trains rather than achieving even headways.

Table 2-4: Headways Before and After Northbound Merge (4:00-6:00 PM)

Station	Mean	SD	CV	15th Pctl	Median	85th Pctl	95th Pctl
JFK	4.2	2.4	0.58	1.7	3.9	6.7	8.4
Andrew	4.2	2.1	0.50	2.0	3.7	6.5	8.1

Figure 2-4 shows the headway distribution by day and by branch arriving at Northbound JFK for weekdays in October 2018. The figure demonstrates significant *daily* variations in PM Peak headways arriving to JFK. The difference in arrival headways between Ashmont and Braintree trains varies by day, with the worst days showing a gap of about 3-minutes between

branches; for example, October 1st shows a median Ashmont headway of 6-minutes and a median Braintree headway of approximately 2.5-minutes. October 10th shows the opposite median headways – 3-minutes for Ashmont and 6-minutes for Braintree. Disaggregate observations for the two aforementioned days, along with one day of nearly equal median headways (October 12th) are shown in **Figure 2-5**. The disaggregate observations allow us to see how bunched arrivals are extremely prevalent and persistent. *These finding confirms that northbound branch headway coordination is not a major focus of service control today.*

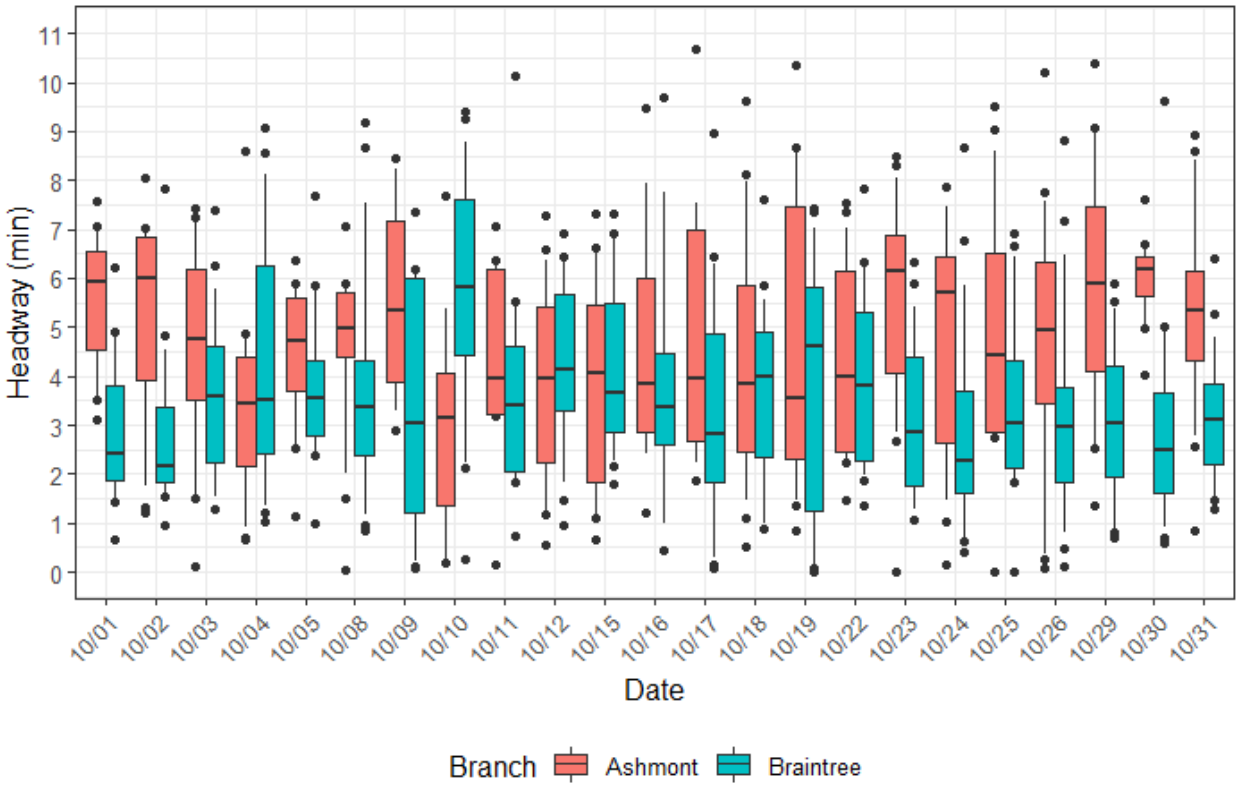

Figure 2-4: Daily Variation in Northbound Branch Merge Headways (4:00-6:00 PM)

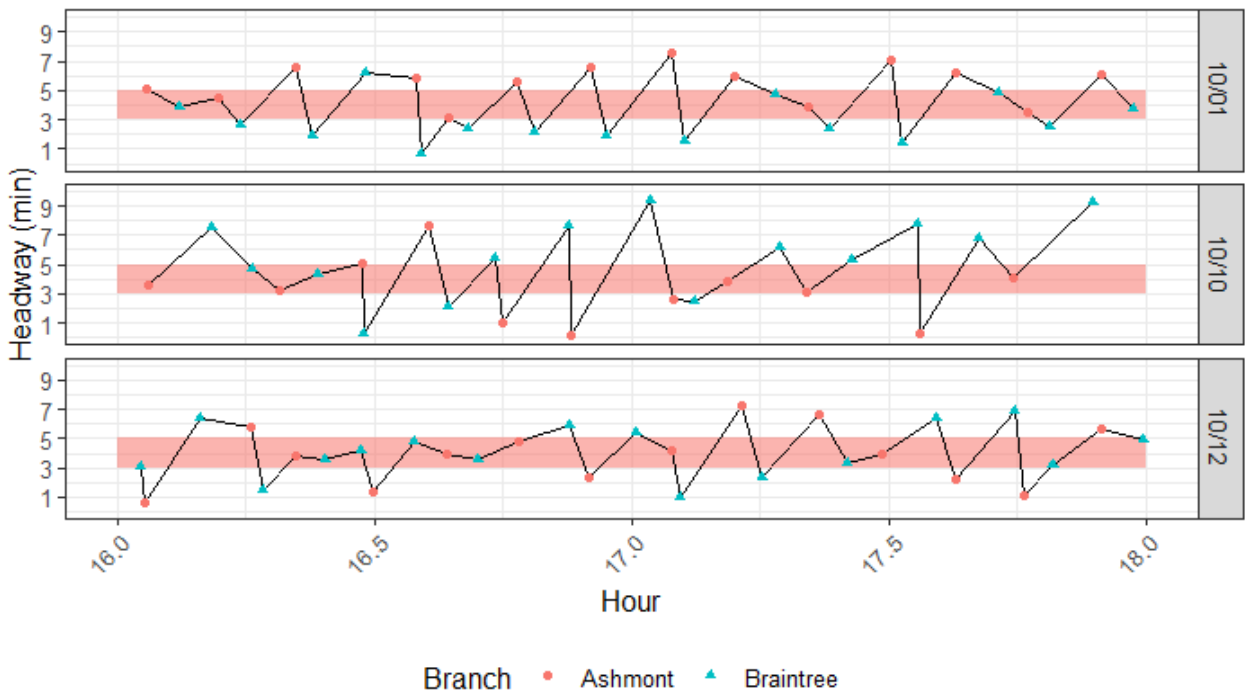

Figure 2-5: Northbound Branch Merge Headways - Sample of Days (October 2018)

Branch Running Times

Despite a scheduling process aimed at providing uniform trunk headways by offsetting branch departure times by the difference in running time along each branch, significant variations occur due to the effect of stochastic running times and other delays.

Table 2-5 shows the running time from departing the branch terminal stations to arriving at the branch merge in AM and PM peak periods. The larger magnitude and variability of running times for the Braintree branch is a result of the higher demand and length of that route. The narrow range in Ashmont running times (approximately 1-minute spread between 15th and 85th percentiles) suggests the opportunity to dispatch Ashmont departures in real-time relative to on-route Braintree trains in order to achieve the desired merge headway at JFK. The time of an Ashmont departure could be set dynamically when Braintree trains are about 5-minutes away from the merge, giving a reasonable certainty for more uniform merge arrival headways. The difference in running times between the AM and PM peaks seen below highlights the need to dispatch using different offsets by time of day, which can be determined through historical running time analyses.

Table 2-5: Running Time from Branch Terminals to Northbound Merge

Time Period	Branch	Median	Mean	SD	15th Pctl	85th Pctl
AM Peak	Ashmont	9.0	9.4	1.6	8.6	9.6
	Braintree	20.4	20.8	1.9	19.6	21.9
PM Peak	Ashmont	8.6	9.4	2.4	8.3	9.4
	Braintree	19.3	19.7	2.1	18.5	20.5

2.4 Headway Stability

This section presents the result of an analysis of headway stability on the MBTA Red Line and explores different contextual variables affecting stability. It begins with a discussion of the methodology applied to measure headway stability.

2.4.1 Methodology

As a methodology to assess the headway stability of a given trip, it is proposed that linear regression can be applied to rapidly extract information about the intercept, slope, and goodness of fit from a linear regression of the headways observed at each stop along a trip. Modeling the headway pattern using coefficients from a linear regression is more robust than using the actual dispatch and final headways, because it dampens the headway variability observed at those specific stops. Contextual variables including dispatch headway, branch, rolling stock, time of day, and presence of incidents will then be added, and the headway stability outcomes can be examined within each context.

For each trip, a linear regression model is estimated with the independent variable as the station index along the line (0 representing the first station) and headways departing each station as the dependent variable. The procedure is shown in **Figure 2-6** with three sample regressions. Taking the first station as x=0 allows for the y-intercept to be interpreted in a straightforward manner as the dispatch headway and the slope ($\beta_{Headway}$) to be interpreted as the average headway change per stop. From each model, the estimated coefficients are extracted, along with the R^2, a measure of the goodness of fit. In this case the first stations are chosen as Andrew in the northbound direction and Davis in the southbound direction, following the findings of large discontinuities in headways immediately prior to these points.

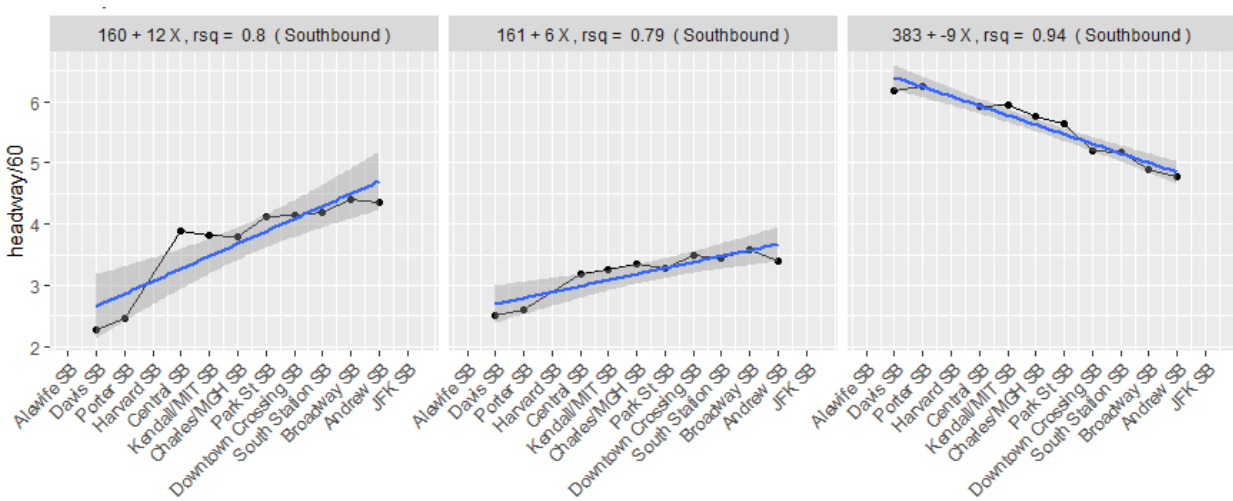

Figure 2-6: Sample Linear Regression of Trip Headways

A rule-based approach is then applied to classify the stability of each trip using the regression results, as shown in **Table 2-6**. In this case, Δ, interpreted as the per-stop average headway change is chosen as 6s, which represents a 1-minute stability threshold over the 10 trunk stops (excluding Alewife/JFK and the initial station).

Table 2-6: Classification of Headway Stability by Regression Coefficients

$\beta_{Headway}$	Classification
$\beta_{Headway} < -\Delta$	Unstable, decreasing
$-\Delta < \beta_{Headway} < \Delta$	Stable
$\beta_{Headway} > \Delta$	Unstable, increasing

2.4.2 Results of Stability Analysis

Initial Headway

The results of a headway stability classification for peak period trips is shown in **Figure 2-7**, with trips grouped by their initial headway at Davis (Southbound) and Andrew (Northbound). Only initial headways with at least 30 observations for the time period and direction are included. In general, the stability trends are as expected:

(1) Initial headways close to the scheduled headway of 4-minutes exhibit the most stable outcomes, and least likelihood of increasing.

(2) Beginning with the scheduled headway, each additional minute of initial headway contributes to a decreasing likelihood of stable outcomes. Notably, both unstable, increasing and unstable, decreasing results become more prevalent above 4-minutes initial headway. This highlights the opposing forces acting on these trains; increased passenger demand threatening to

increase dwell times and headways, and decreased train congestion allowing the train to travel more quickly.

(3) Initial headway below the scheduled headway (2-3 minutes) have a high proportion of unstable, increasing headways, showing the effect of line bottlenecks.

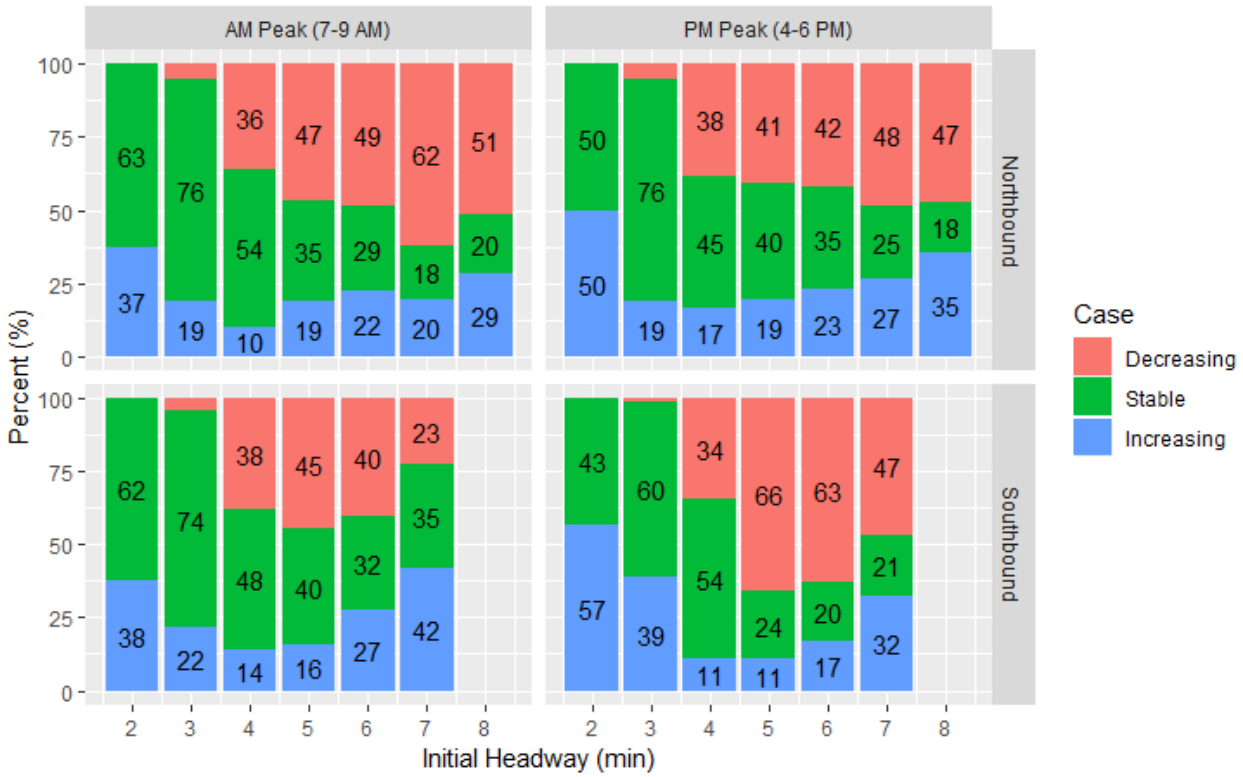

Figure 2-7: Peak Period Headway Stability Grouped by Initial Headway

Effect of Rolling Stock

Figure 2-8 shows the effect of rolling stock on headway stability in the AM peak, Southbound direction. The differences between 3 and 4-door trains are most pronounced for headways greater than or equal to 5-minutes. At these headways above the scheduled headway, the 3-door trains appear less capable of handling the higher demand associated with a long headway, and exhibit a higher proportion of unstable, increasing outcomes. Meanwhile, 4-door trains with 5-6 minute headways have the highest proportion of unstable, decreasing headways – despite the long initial headway, 4-door trains perform well under the higher demand.

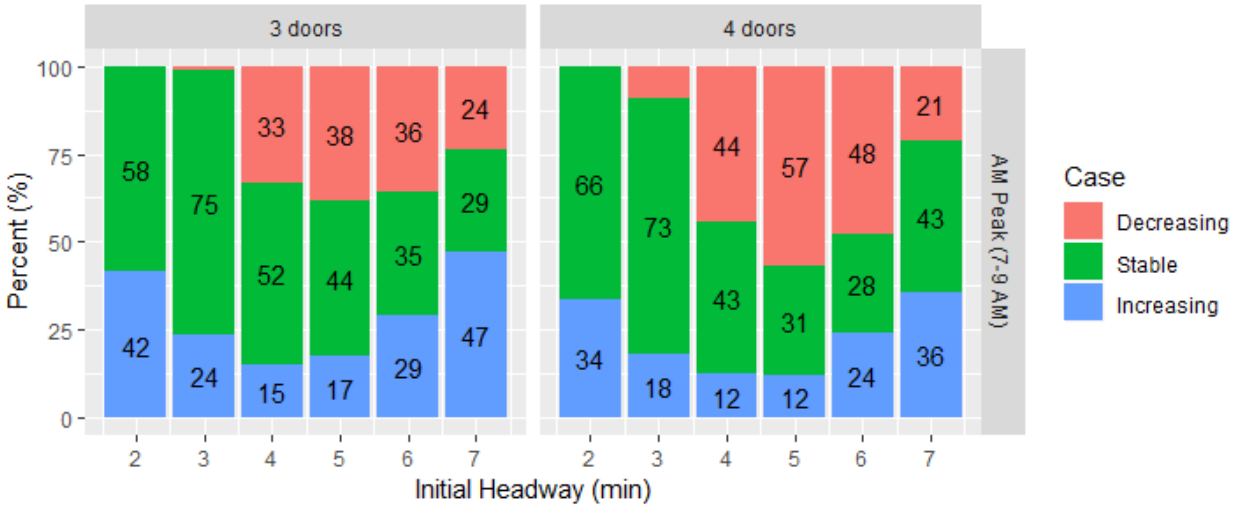

Figure 2-8: Effect of Rolling Stock (Southbound, AM Peak)

Section 2.3 demonstrated the large impact that the Alewife crossover and branch merge have on initial trunk headways in the north and southbound directions, respectively. This section provided quantitative evidence of how initial headway irregularities contribute to further negative outcomes in the form of headway instability.

2.5 Passenger Implications of Headway Regularity

In addition to the aforementioned passenger agnostic metrics, headway regularity should be considered from a passenger perspective. This section will explore the effect of headway irregularity on the MBTA Red Line as experienced by passengers in terms of wait times, crowding, and reliability of journey times.

2.5.1 Wait Time

The effect of headway irregularity on wait times is well understood in the literature; because more passengers arrive during long headways, an increase in headway variability results in an increase in average wait time, even when the mean headway remains constant. In addition, time spent waiting for public transit is frequently cited as more onerous than in-vehicle travel time (Raveau et al., 2014; Wardman, 1998), lending importance to wait time metrics. This section will present the metrics of average and percentile wait time, which can be used to describe passenger experience of headway regularity by station or for the entire line.

Average Wait Time

Average wait time can be an effective metric to communicate headway regularity, because it is readily interpretable by all stakeholders. Assuming passengers arrive uniformly with a rate of λ, the average wait time, $\overline{w}$, can be calculated for a given period and location as follows.

$$\overline{w} = \frac{\sum_i (\lambda h_i \cdot h_i/2)}{\sum_i \lambda h_i} = \frac{\sum_i {h_i}^2/2}{\sum_i h_i},$$

Where h_i = headway of train i, and

λ = passenger arrival rate

Percentile Wait Time

In addition to average wait time, extreme wait time (for example, 95th percentile) is an important metric to communicate the extreme events that are critical to customer perceptions of transit reliability. Percentile statistics are also more robust to extreme values compared to the average, and important for communicating skewed distributions. For any percentile, the percentile wait time, W_p, can be found as follows:

Consider a period with uniform passenger inter-arrival time Δ, and headways h_i. The wait times experienced by all passenger boarding train i, are given by the ordered series, W_i:

$$W_i = \left\{\Delta, 2\Delta, \dots, \left(\frac{h_i}{\Delta} - 1\right)\Delta, \frac{h_i}{\Delta}\Delta\right\},$$

Where Δ = passenger inter-arrival time

Then W is defined as the ordered multiset containing all elements of W_i for all trains in the period of interest. Then, any percentile (p) wait time, W_p, can be found by selecting the value in position P from set W, where:

$$P = p\left(\frac{N+1}{100}\right),$$

Where N = number of elements in set W, and

p = percentile of interest

Effective Headway

In addition to passenger wait times, headway metrics are useful because they can help bridge the gap between passenger and agency perspectives (Tribone, 2013). Tribone suggests the metric of effective headways (H_e), which is the average headway weighted by the number of passengers experiencing each headway. Again assuming a uniform passenger arrival rate, the formulation is simply twice the average wait time:

$$H_e = \frac{\sum_{i=1}^{N} {h_i}^2}{\sum_{i=1}^{N} h_i}$$

The concept of effective headway was extended by Maltzan (2015) to communicate the effective impact on the entire line, weighting each stop by the historical passenger arrival rate to develop a line-effective headway, H_E.

$$H_{E,i,p} = \frac{\sum_i \sum_o \lambda_p^o (h_i^o)^2}{\sum_i \sum_o \lambda_p^o h_i^o}$$

Where: p = time period of interest, and

o = origin station

Results

Results of a peak period wait time analysis for the MBTA Red Line are shown in **Figure 2-9**, including average and upper percentile wait times. The half average and average headways are shown as guidelines, representing, respectively, the average and maximum wait times in a hypothetical case with zero headway variability at today's headway.

Overall, it is evident that passengers must budget significant extra travel time due to irregular wait times. In all cases shown, the 85th percentile wait time is approximately equal to the average headway, meaning that 15% of passengers wait longer than the mean headway during the peak periods. In the southbound direction, average and upper percentile wait times are observed to continually increase throughout the line, caused by the instability of headways, as seen in the previous section. The 95th percentile exhibits the most rapid increase, attributed to the effect of passenger related incidents and irregular dwell times at the downtown stations. In contrast to the southbound direction, the northbound direction wait time percentiles remain mostly stable throughout the line, which is attributed to the bottlenecking effect of the terminal.

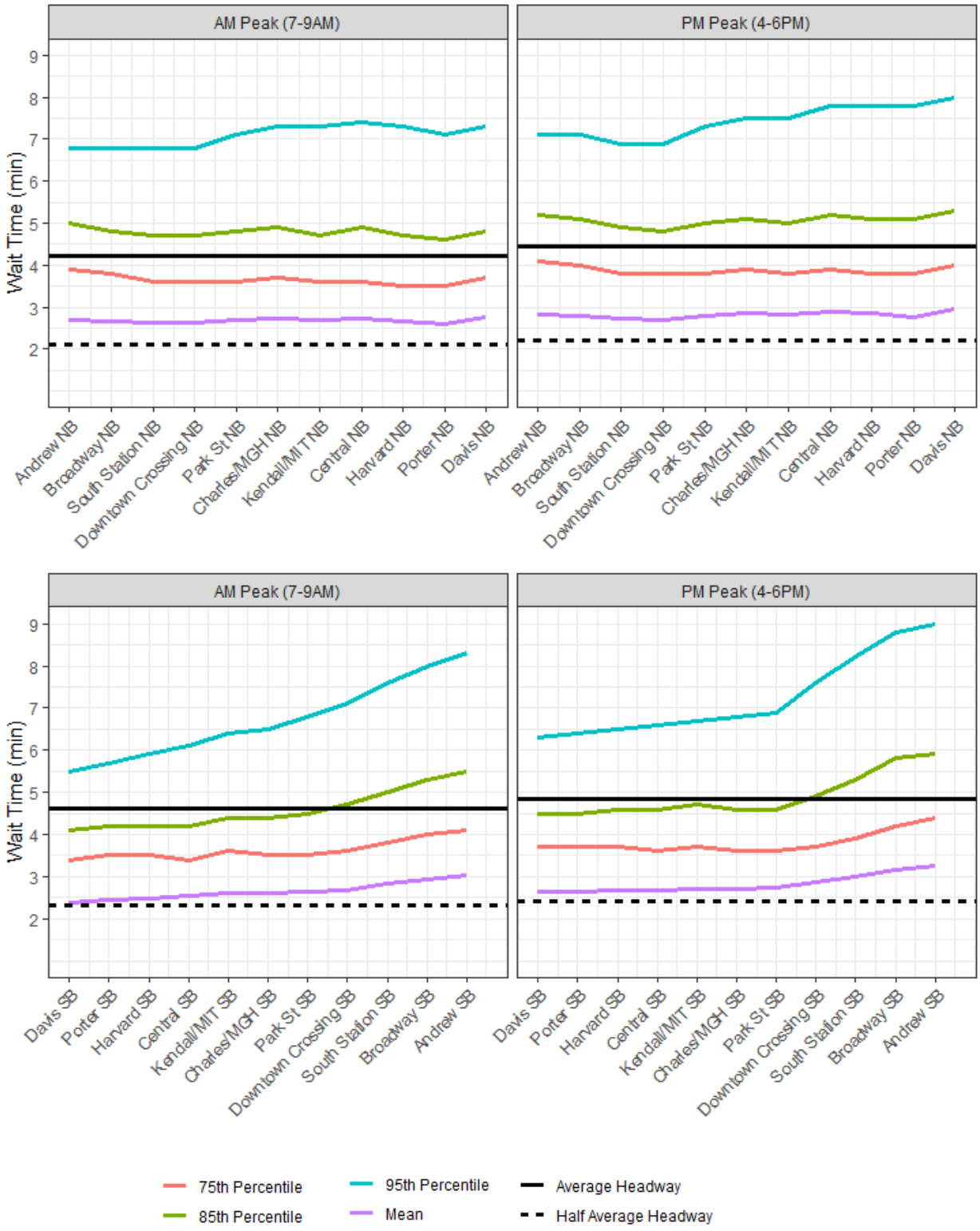

Figure 2-9: Trunk Line Waiting Times (Top – Southbound, Bottom – Northbound)

The effective line headways are shown in **Table 2-7** for the trunk portion of the MBTA Red Line. The mean line headway is also included as a reference point, which presents the headway without any passenger weighting. A large difference is observed between the mean line headway and the effective line headway, indicating significant variability in service delivery, and an inefficient application of peak period resources. The difference between the effective line headway and mean line headway is slightly better in the southbound direction, attributed to being closer to the point of dispatch, compared to the northbound direction which inherits the variability from the branch merge. This metric is effective as an aggregate measure in communicating the degradation of customer experience from irregular headways.

Table 2-7: Effective Line Headway

Direction	Time Period	Mean Line Headway	Effective Line Headway	Difference
Northbound	AM Peak	4.3	5.3	1.1
	PM Peak	4.5	5.6	1.1
Southbound	AM Peak	4.3	5.0	0.7
	PM Peak	4.6	5.4	0.8

2.5.2 Passenger Crowding and Effective Utilization

Headway irregularity, especially long gaps and bunching results in unevenly loaded trains. The effects of headway irregularity on crowding can be communicated using metrics such as the variability in passenger loading at the peak load point and frequency of left-behinds. More generally, a metric of effective capacity utilization will be proposed to communicate the effects of headway irregularity on train loading.

Passenger Train Load

The expected passenger load for each train can be estimated using historical passenger origin-destination arrival rates and assigning passengers to trains according to the number of passengers expected to arrive during the trains preceding headway at each station, assuming no left-behinds. A distribution of expected train loads can then be determined for each station, as would be viewed by a stationary observer. Summary statistics of passenger loading can then be used to communicate crowding, or the effects of headway irregularity.

More formally, the passenger load is estimated as the summation of origin-destination passenger arrivals rates for all stops with origin less than or equal to current stop and destination greater than current stop, for all served itineraries.

$$L_{iJ} = \sum_{b}^{B} \sum_{o=1}^{O} \sum_{d=o+1}^{D} h_{i,b,o} \lambda_{O,D,p}$$

Where L_{iJ} = train load of train i departing stop J,

i = train,

o = station,

b = train itinerary (e.g. destination, branch),

K = last stop in itinerary b,

B = all itineraries served by train i,

$h_{i,b,o}$ = departure headway of train i at stop J for itinerary b (minutes), and

$\lambda_{O,D,p}$ = historical arrival rate between stop j and stop k during time period p (passengers/minute)

Effective Utilization

A new metric of effective utilization, $U_{E,i}$, is proposed, which measures the passenger demand experienced by a train i over the line as a fraction of the average demand experienced by a train in the same period.

$$U_{E,i} = \frac{\sum_b \sum_o \lambda_{b,p}^o h_{b,i}^o}{\sum_b \sum_o \lambda_{b,p}^o \bar{h}_{b,i}^o}$$

Where $\lambda_{b,p}^o$= historical passenger arrival rate at station o during time period p with a destination on itinerary b, and

$h_{b,i}^o$ = headway of train i at station o for itinerary b.

This metric further attempts to bridge the gap between passenger and agency perspectives, by communicating the undesirable effects of bunched trains and long gaps on passenger loading on an individual train basis. The metric can be understood as follows; when $U_{E,i}$ is equal to 1.0, the demand experienced by train i is equal to the period average. Thus a system which is operating at capacity and making optimal use of available train assets would have an effective utilization of 1 for all trains during the peak period. Effective utilization greater than 1 indicates over-utilization of the asset, and would result in passengers left behind on the platform in at-capacity systems. Conversely effective utilization less than 1 indicates under-utilization, which must be absorbed by future trains.

Compared to effective headway, the fractional nature of this metric helps to communicate the undesirable effects of non-uniform headways, both for short and large headways. For example, at an average headway of 4-minutes, a train with an *effective headway* of 3-minutes appears to be positive from an agency perspective, since lower headways mean lower wait times. In reality, this

train is bunched and experiences lower demand. In this case, the train would have an *effective utilization* of 0.75, indicating that the asset is underperforming.

Results

This section presents calculations of effective utilization for the MBTA Red Line under various scenarios. The effective capacity calculations demonstrate the capacity implications of the previous findings of initial headway irregularity and instability.

The distribution of effective utilization for all peak period trips, including branch and trunk demand is presented in **Table 2-8**. As expected due to the variability in line headways observed previously, a wide range of effective utilization values are observed. The minimum observed effective utilization of 0.53 approximates the natural limit imposed by safe separation requirements; implying that the minimum weighted headway a train can maintain over the line under ideal conditions is approximately half the average peak period headway. Conversely, there is no maximum effective utilization – under large delays the value would be unbounded since the method does not account for left-behinds.

Overall, just 53% of all peak period trips are found to be within 25% of the period average utilization (0.75 to 1.25). The fractions above and below this 25% threshold are approximately equal; 27% of trips have an effective utilization below 0.75 and 20% of trips have an effective utilization above 1.25. This finding indicates in the system overall, there is significant potential to improve system capacity by simultaneously reducing the number of bunched trips and long gaps.

Table 2-8: Effective Utilization (All Peak Period Trips)

Percentile	Min	5%	15%	25%	50%	75%	85%	95%
Effective Utilization	0.53	0.63	0.69	0.74	0.90	1.17	1.34	1.68

Effective capacity will now be explored in more depth, relating to the previously discussed factors affecting headway stability, including rolling stock, Alewife dispatch platform, and branch.

Effect of Alewife Platform

In the disaggregate sample, the routine effect of the dispatch platform on passenger loading is visible. The effect is most pronounced in the first hour of service on 10/16, where service appears to be stable with no long gaps; dispatch platforms alternate, with P2 trains consistently seeing utilization around 60-75% and P1 trains seeing utilization between 100-125%.

Table 2-9 shows the distribution of effective utilization for southbound AM Peak trips by Alewife Dispatch Platform. The results are not differentiated by branch since differences in branch demand are very minor in the AM peak, southbound direction. A disaggregate sample is also provided in **Figure 2-10**.

The differences observed between P1 and P2 dispatches is substantial; the median effective utilization is 108% for P1 dispatches compared to only 75% for P2 dispatches. The differences at the upper percentiles, for example the 75th is also striking; a 75th percentile P1 train will experience boardings around 125% greater than the average, while a 75th percentile P2 train has an effective utilization of 0.94, still below the average utilization (1.0). Conversely, the 95th percentile utilization is quite similar between platforms, suggesting the influence of incidents rather than routine demand on these values. In the disaggregate sample, the routine effect of the dispatch platform on passenger loading is visible. The effect is most pronounced in the first hour of service on 10/16, where service appears to be stable with no long gaps; dispatch platforms alternate, with P2 trains consistently seeing utilization around 60-75% and P1 trains seeing utilization between 100-125%.

Table 2-9: Effective Utilization Percentiles by Alewife Dispatch Platform (AM Peak, Southbound)

Alewife Dispatch Platform	5%	15%	25%	50%	75%	85%	95%
Platform P1	0.79	0.89	0.94	1.08	1.24	1.36	1.67
Platform P2	0.60	0.64	0.66	0.75	0.94	1.12	1.69

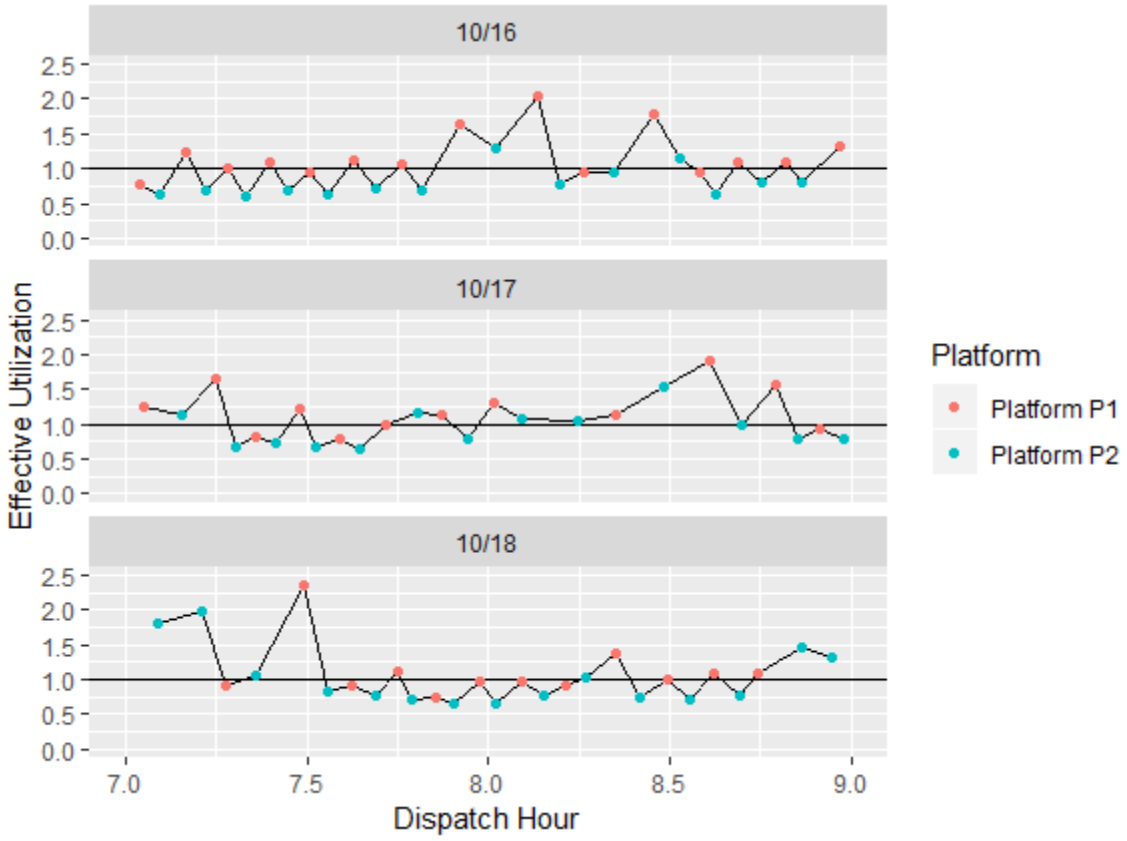

Figure 2-10: Effective Utilization by Alewife Dispatch Platform (AM Peak, Southbound) – Sample of Days

Effect of Northbound Merge Headway

In Section 2.3.2, the northbound merge headways were examined, finding significant variability due to minimal real-time coordination between branches. Here, the effects of the uncoordinated merge are explored in terms of effective capacity.

Figure 2-11 shows the effective utilization for northbound PM Peak trips. Because branch demand is very low relative to trunk demand, a trip's utilization is primarily defined by the initial headway after the merge. Trips with headways under 3-minutes experienced minimal utilization, exceeding an effective utilization of 0.85 just 8% of the time. The effective utilization for trips with initial headways beyond 3-minutes grew linearly in relation to the initial headway. Trips with initial headways beyond 5-minutes made up just 20% of all trips, but contributed 32% of the effective utilization over the period. These findings provide strong evidence that the initial headway of a trip is a major factor in determining its outcomes, and suggests significant improvements to northbound capacity and crowding are possible with improved merge control.

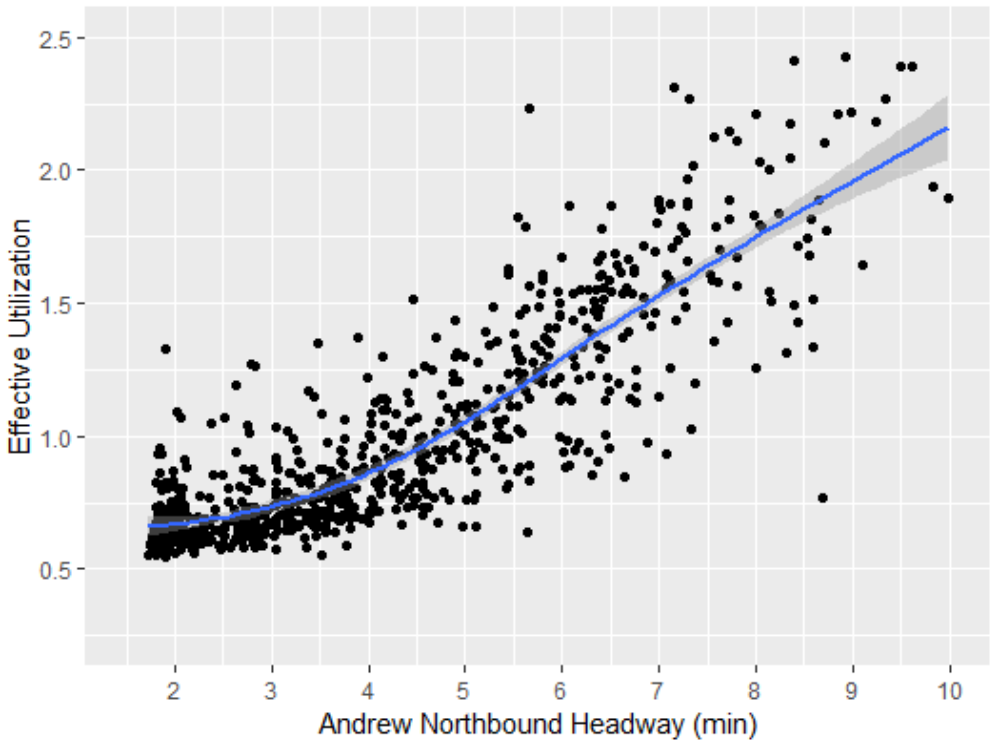

Figure 2-11: Effective Utilization of Northbound PM Trunk Trips

2.5.3 Journey Time Reliability

Journey time reliability is an important measure of passenger experience in transit. Depending on the available data sources, passenger journey time can be measured using either Automatic Vehicle Location (AVL) or card-based approaches as described by Wood (2015). Card-based journey time metrics such as Excess Journey Time (London Transport, 1999; Frumin, 2010) measure individual journey time based on tap-in and tap-out times, and include elements important to individual experience such as left-behinds, route choice, and access and egress times. In contrast, AVL based approaches use only vehicle movements to calculate distributions of wait times and travel times, and excluding some of the aforementioned factors critical to individual experience. However, AVL based measures are valuable from the operator's perspective because they are simple to apply, and can be used in open systems without tap-out.

The effects of headway irregularity on passenger journey time reliability can be captured using the concept of reliability buffer time (RBT), developed by Chan (2007). The RBT represents the buffer time a typical passenger must add to their trip to ensure an nth percentile probability of on-time arrival. The RBT can also be expressed as a fraction of median travel time, Reliability Buffer Factor (RBF) which communicates what portion of travel time a passenger with a certain OD pair must factor in due to unreliability in journey time.

Reliability Buffer Time (RBT)

Chan (2007) defined the Reliability Buffer Time as the difference between the median and n^{th} percentile journey time (JT), for a particular OD pair and time period (t), as follows:

$$RBT = N^{th} \text{ Percentile } JT_{t,OD} - 50^{th} \text{ Percentile } JT_{t,OD}$$

In addition, Chan proposed a Reliability Buffer Factor (RBF), as the ratio of the RBT to the median journey time. The RBF can serve as a relative factor, indicating the proportion of time a passenger must allocate for a reliable arrival time, relative to the expected journey time:

$$RBF = (N^{th} \text{ Percentile } JT_{t,OD} - 50^{th} \text{ Percentile } JT_{t,OD}) / 50^{th} \text{ Percentile } JT_{t,OD}$$

Results

Reliability Buffer Time and Reliability Buffer Factor were calculated for trunk OD pairs in AM and PM peaks. **Figure 2-12** shows the RBF for the AM peak, southbound direction – other results for the RBT and RBF factors are included in **Appendix 1**.

In general, a trend of high RBF for very short trips, and decreasing RBF as trip length increases is observed. This finding indicates that wait time represents a larger portion of the difference between 95th and median JT compared to travel times. Passengers traveling from north Cambridge into downtown generally require 20-40% (4-6 minutes) RBT to ensure an on-time arrival. In contrast, passengers traveling Downtown Crossing to South Station, a popular 1-station trip for passengers transferring from the Orange Line, must budget an additional 200% (around 5-minutes) travel time compared to the median. This is almost entirely due to irregular wait times; it was shown previously that a southbound passengers at Downtown Crossing the AM peak has a 95th percentile wait time of approximately 7-minutes (**Figure 2-9**).

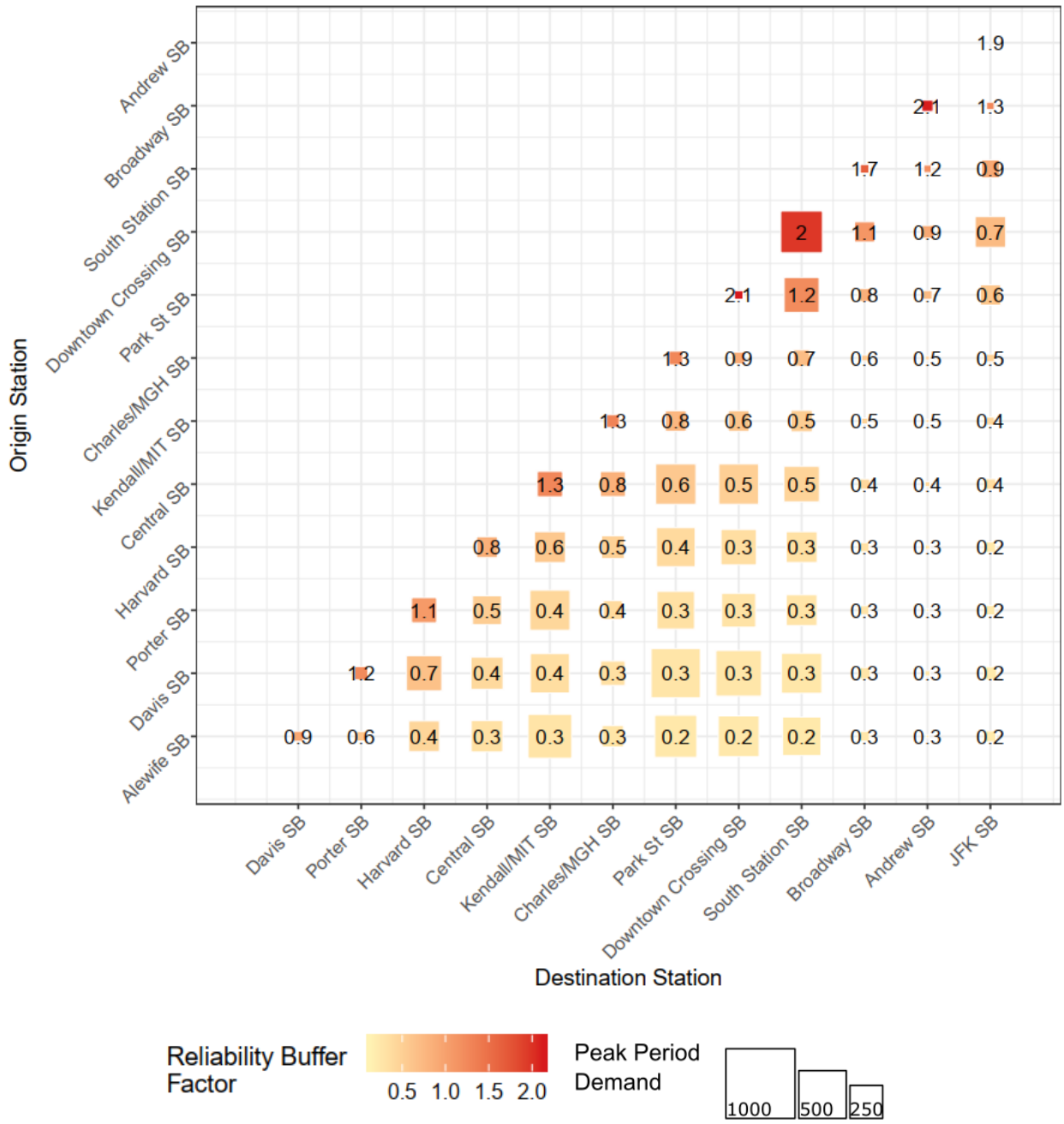

Figure 2-12: Reliability Buffer Factor (AM Peak, Southbound)

2.6 Discussion on Improving Headway Regularity

Overall, the findings in this chapter suggest a need to increase the prioritization of headway regularity in order to provide a cost-effective and reliable service to customers. During a time of intensive capital investment into system infrastructure, it is important for the agency to maintain awareness of fundamental aspects of service provision including headway regularity, which can deliver system improvements in the short-term, well in advance of large capital investments.

In this chapter, the key characteristics of the MBTA Red Line which contribute to difficulty in maintaining regular headways were examined. These include irregular initial trunk headways due to northbound branch merge and Alewife crossover SGR, alternating branches (with imbalanced demand), and a mixed fleet with different rolling stock characteristics. While some of these issues can be addressed through technical solutions such as the targeted service interventions discussed in the coming chapters, taken together they point to larger cultural issues which require a different set of solutions.

The issues uncovered in this chapter point towards minimal agency awareness and/or concern of how irregular headways degrade service for customers, and few staff incentives to improve headway regularity. These cultural issues suggest the need for a larger agency-wide reorientation to prioritize on-time performance and operating precision. Specific solutions could include improving operations reporting, and providing staff incentives or operational targets.

Reporting and transparency are important agency aspects influencing on-time performance. The MBTA currently operates a public facing dashboard which reports on reliability of each route. In this context, the MBTA defines reliability as the percentage of passengers waiting less than the scheduled headway. This definition masks many facets of customer reliability (even without accounting for passenger left-behinds), masking the irregularity of wait times below the scheduled headway, and forgoing the more thorough journey time metrics used by other agencies (TfL). Detailed headway dashboards such as those designed by Tribone (2013) could be applied to improve transparency of headway regularity both within and without of the agency, and bring attention to specific incidents rather than masking them behind a single daily metric. Detailed and transparent reporting is an important step in incentivizing service improvement within the agency, as is establishing efficient communication channels between service providers and those with the tools to analyze it.

In most available literature on performance incentives in public transportation, the focus is placed on design of private sector contracts (Laidig, 2010; Pyddoke & Lindgren, 2018) rather than on performance at the staff level. In one of the only available studies on staff level incentives, Hartman et al. (1994) details two case studies from Capital Metro Transit Authority (Austin, Texas) and the Toronto Transit Commission (TTC). Both agencies provided financial incentives to all staff for suggestions leading to budget savings, improved customer service, on-time performance, or safety. The objective of the TTC Suggestion program was to “link employee performance and participation to the overall mission and objectives of the agency” and was viewed

as “highly successful” by management two years after implementation. Such incentive programs could provide great benefit to the MBTA by involving operating staff in the process of transformation. Operating staff may in many cases be the most aware of the root causes and solutions to issues affecting headway regularity, and could provide vital knowledge to service planners if given the opportunity and correct incentive.

Technical solutions to improve headway regularity are numerous, including preventative and corrective measures. Preventative measures aim to prevent issues of headway irregularity before they occur; these solutions could include reducing rolling stock differences, real-time headway-based dispatching, real-time coordinated branch dispatching, or more precise scheduling to account for anticipated differences in demand/performance. Chapter 5 describes a crewing intervention at Alewife which, among other objectives detailed later, aims to reduce initial headway irregularity. Responsive measures aim to stem irregularity after it occurs; these solutions could include in-line headway-based holding or others such as expressing or short-turning, for which Carrel (2009) provides a valuable inventory. Chapters 6 and 7 provide details of the implementations and results of two headway-based holding service interventions, intended to reduce initial variation in southbound and northbound trunk headways, respectively. The pilots are also intended to increase awareness of headway regularity and some potential solutions to address it within the agency. The straightforward strategies chosen represent a starting point to be built upon by agency staff following the end of the pilots.

Chapter 3. Station Dwell Time

It is evident from the state of service today that dwell times will be one of the major barriers to reliably reducing the minimum headway on the Red Line. This chapter will attempt to quantify the state of station dwell time in the system today and its consequences for future performance. Section 3.1 will explain the motivation; the relation between dwell time and minimum headway. Section 3.2 will outline different approaches to understanding dwell time including with and without passenger regard. Section 3.3 will discuss the motivation, applications, and issues of using automated data for dwell time measurement and modeling. Sections 3.4 and 3.5 will propose two methodologies for exploring dwell times based on automated data, and present results of such exploration for the Red Line. Lastly Section 3.6 will explore methods for dwell time reduction with specific attention to the case of the Red Line.

3.1 Motivation: Determining Station Bottleneck Capacity

The *TCQSM* (2013) provides a useful reference for practitioners looking to estimate the minimum headway achievable through constraining stations; the minimum possible headway is approximated by the sum of dwell time, station re-occupancy time, and an operating margin. Dwell times can be taken from existing operations data, or from similar systems in the case of insufficient historical data, and is effected mainly by passenger demand and agency operating practices. The minimum re-occupancy time can be calculated from system attributes including the signaling system and rolling stock, or from historical data in a system operating close to capacity. Finally, the operating margin can be a standard value such as 15s depending on the reliability targets of the agency, or it can be excluded if sufficiently rigorous values are used for the other components.

While this approach can be supplemented with simulation models to answer more complex capacity and reliability questions, the basic approach is still an invaluable for initial investigations into line capacity. Of the components of minimum headway, dwell time is of the primary interest, due to its large magnitude, high variability, and low cost strategies for improvement. Thus, this chapter will focus on understanding station dwell time.

3.2 Approaches to Estimating Dwell Time

This section will outline the two main approaches to understanding dwell time of a system with historical operating data: (1) a statistical approach (without passenger regard), and (2) a regression approach (with passenger regard).

(1) Statistical approach (without passenger regard)

A statistical approach without passenger regard is a valuable tool for rapid identification of bottleneck stations under current operating conditions. A set of peak period dwell times can be

quickly assembled for each station from historical train tracking data, and summary statistics such as mean plus two standard deviations (*TCQSM*, 2013) can be extracted easily. This approach can also be applied to measure dwell time changes following a service intervention (such as headway control or change in train operator procedures).

While a statistical approach is sufficient for rapid identification, it is not easily applicable to other operating conditions due to its sole reliance on historical observations. For example, applying a statistical approach to estimate the constraining dwell time at a different headway is problematic due to the relationship between headways and dwell times; in the short term, an increase in service will result in less passenger demand per train and lower dwell times. This could be addressed somewhat by subsetting dwell times prior to statistical analysis; for example, sampling from the observed dwell times based on headway to obtain a distribution representative of future conditions.

(2) Dwell Time Regression (with passenger regard)

Conversely, dwell time regression models with passenger regard can be invaluable for predicting dwell time in a multitude of service planning applications, because they capture the fundamental relationship between passenger activity and dwell time. Regression models are also used extensively in rail transit simulation modeling to accurately capture the interaction between service and dwell times. A fundamental approach with passenger regard is necessary to capture dwell time changes with respect to large changes in service, such as the frequency increases planned for the MBTA Red Line, which cause significant changes in passenger activity. While large volumes of data can help to estimate a more robust model, regression can also be applied to more limited manual studies and then used for predicting dwell in other contexts.

3.3 Dwell Time Modeling with Automated Data

This section outlines the main motivations, applications, and issues of modeling dwell time with automated data sources.

3.3.1 Motivation

Using ADCS for dwell time modeling has a number of advantages compared to manually collected studies. It allows us to rapidly examine a large number of stations and time periods without regard for data collection costs, whereas past manual studies have been limited to the specific – most likely busiest – stations. Due to the low cost of gathering the data, study of dwell time using automated data also allows for continuous monitoring and evaluation, in itself of significant value, especially in face of dwell time reduction targets.

3.3.2 Applications for Automated Dwell Time Modeling

There are two important applications for dwell time modeling with automated data which will be essential for achieving future capacity targets: (1) simulation modeling to determine effect of dwell time and target reductions, and (2) monitoring the status of interventions to reduce dwell time.

(1) The approaches described in this chapter can be used to develop a dwell time model for use in simulation. Simulation will be necessary to determine line constraints, and can be used to test operational strategies and improvements which are impractical to test in the field. In particular, simulation modeling relates to the dwell time constraint in that it can be used to test different dwell time scenarios. Scenarios could include reductions only in the form of strictly enforced maximum dwell times, or in the form of increased uniformity of dwell times such as those resulting from automation. Simulation can also identify the effects of changes to the signaling system on minimum re-occupancy time.

(2) Monitoring and reporting are necessary steps to make dwell time improvements quantifiable and accountable. Gathering evidence of dwell time improvements following interventions can help to gain broader institutional support for future dwell time reduction initiatives. The methods discussed here, either statistical or regression approaches with automated data could be applied continuously for monitoring and reporting purposes at significantly lower cost than manual surveys.

3.3.3 Issues

Issues with modeling dwell time with automated data can be separated broadly into two categories: (1) issues with the available data sources, and (2) factors not captured in available ADCS.

(1) Issues with the available data sources are primarily in measurement of passenger demand and measurement of train movements. Passenger demand obtained from fare card sources may contain errors in origin/destination inference, from passengers not interacting with the fare system (non-interaction/fare evasion), and from passenger to train assignment models. Some of these issues are reduced or eliminated when analyzing closed systems with tap-in and tap-out, rather than tap-in only systems. In tap-in only systems, estimating left-behinds with only fare-card entry/boarding information is less accurate because information on egress station and time must be inferred, and there is no way to validate inference with the fare card data.

Dwell times obtained from train tracking system may contain inaccuracies, primarily in that dwell time is measured from the occupation of track blocks which may not exactly represent the boundaries of the station. Furthermore, it is impossible to identify components of the block occupancy time, for example, station run-in time, passenger movement time, and door open time. While some non-passenger components of station stop time can be assumed to be random variables, run-in and run-out time depend on the exact locations of the block track circuits and the

speed code received due to position of downstream trains, which is difficult to model and would require the line speed code logic. As discussed previously, pre- and post-shunting are another factor influencing the quality of automated dwell time measurements, although the frequency of such issues is impossible to estimate.

(2) Factors not captured with the available ADCS may include discrete events affecting only certain trains, or unobservable factors which affect all trains. Discrete events affecting dwell time can be broadly grouped into train and passenger incidents. Passenger incidents could include sick or unruly passengers, or passenger requiring assistance from the operator to board. Train incidents could include mechanical issues or service control such as headway-based holding. Unobservable factors which affect all observations rather than discrete events could include the uneven distribution of passengers along the platform in selected stations, within the train, or at specific doors. These factors may follow predictable patterns, such as passenger-platform distribution which is inherent to station attributes, or headway control which is related to the position of other trains.

Although many of these issues would be avoided through the use of labor intensive, manual surveys, the low cost and breadth of analysis possible with ADCS presents a significant opportunity to understand system dwell time on a scale not previously possible.

3.4 Statistical Approach

3.4.1 Methodology

The methodology presented here is borrowed from the *TCQSM* (2013), with the key difference being that data is obtained from automated track circuit based measurements, rather than from manual surveys. The *mean plus two standard deviation* method is used to determine the limiting dwell time, which represents the 97.5th percentile dwell time assuming a normal distribution. Only dwells from the PM peak hour (5:00-6:00 PM) are selected, since the PM peak hour shows significantly longer dwell times downtown compared to the AM. The safe separation time is modeled as the interval between a train passing over the departing circuit and the following train passing over the arriving circuit, termed here as the "re-occupancy time". For the re-occupancy time, records are taken from all times of day rather than just the peak hour, since trains following at minimum possible headways are of concern, regardless of dwell time. It is assumed that due to frequent bunching which occurs on the Red Line, the minimum observed re-occupancy times represent a good approximation of the theoretical minimum times. The minimum re-occupancy time is selected, except where the values are judged to be unreasonably low compared to the expected value of around 45-60s. Extremely low values are attributed to issues of pre- and post-shunting. For these stations, the lowest recorded values of re-occupancy were examined manually and the minimum "reasonable" value was selected. The values are highlighted in the results below.

It should be noted that although the equivalent formulation with automated track circuit data results in different proportions of station occupancy to re-occupancy times compared to a manual survey, the summation of the two components under either method is equivalent assuming equal close-in speed for both trains. This is illustrated in **Figure 3-1**. This last assumption may not always be true, but contributes minimally to the overall calculation.

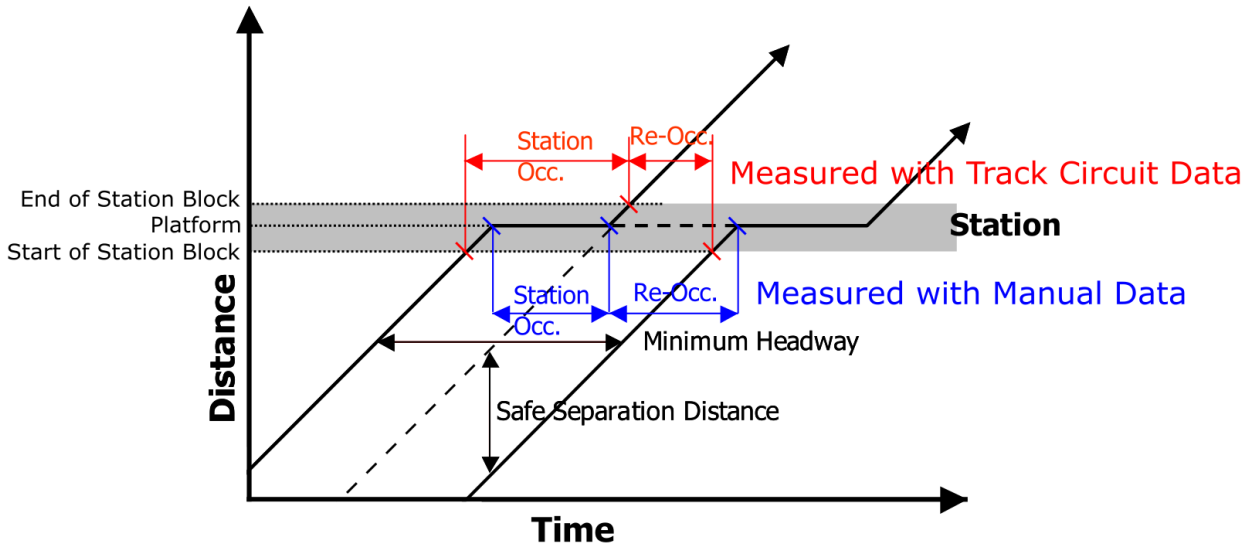

Figure 3-1: Measurement of Limiting Station Headway with Automated and Manual Approaches (Adapted from TCQSM (2013))

3.4.2 Results

Results from a statistical analysis of station occupancy times and re-occupancy times for 1-month (April 2017) is shown in **Table 3-1**. The results are ordered by descending limiting station occupancy and re-occupancy time. Notably, three stations, South Station NB, Downtown Crossing NB, and Park Street NB do not have complete entries due to lack of available data for the relevant track circuits. According to a signal engineer at the MBTA, these circuits are "malfunctioning" and frequently report "false occupancies" and are thus discarded during pre-processing as to not interfere with the train tracking algorithm. PM peak dwell times in the southbound direction are generally understood to be more constraining than in the northbound direction due to the effects of uneven branch demand and crowding caused by incompatible passengers; thus the lack of data for northbound stations is not a major concern for identifying the minimum line headway.

A significant number of stations are observed to exceed 3-minutes limiting station occupancy plus re-occupancy. It is noteworthy that the busiest downtown stations are not the most constraining, but rather Davis NB, which is located immediately *before* the key bottlenecks of the Alewife Terminal. The fourth station on the list is a similar case, being located immediately before Park SB. The restrictiveness of these stations is owing to signal logic which prevents trains from

fully entering these stations until a downstream block is cleared, where the relevant downstream block is itself also a headway bottleneck. Thus, lowering the minimum headway at these bottlenecks may be more a function of signaling than dwell times. The more widely recognized bottleneck stations of Park and Downtown Crossing SB follow, each with limiting station occupancy times of approximately 3-minutes, in addition to re-occupancy times of 60-80s. Following the downtown stations, Harvard NB and SB appear, likely due to track geometry which requires extremely limited speeds of 10-15 mph both entering and exiting the station, significantly extending the station occupancy time.

Table 3-1: Statistical Station Occupancy and Re-Occupancy Results (s)

Station	Occupancy			Re-Occupancy	Combined
	Mean	SD	Mean + 2*SD	Min (Adjusted[1])	Mean + 2*SD + Min Adj
Park Street SB	116	34	183	80	263
Downtown Crossing SB	105	32	168	61	249
Charles MGH SB	97	45	187	2 (54)	241
Harvard NB	100	32	165	0 (68)	233
Harvard SB	86	40	166	56	222
Central NB	79	42	164	49	213
Charles MGH NB	79	39	157	0 (53)	210
Kendall SB	80	33	146	64	210
South Station SB	83	21	125	81	206
Broadway NB	66	33	132	67	199
Central SB	69	31	132	52	184
Davis SB	60	26	112	62	174
Kendall NB	76	28	132	0 (33)	165
Porter SB	62	23	108	3 (52)	160
Andrew NB	64	23	110	48	158
Porter NB	52	23	99	43	142
Andrew SB	59	12	83	2 (55)	138
Broadway SB	71	11	92	45	137
Park Street NB	97	43	183	-	-
Downtown Crossing NB	-	-	-	0 (27)	-
South Station NB	-	-	-	6 (22)	-

NB = Northbound, SB = Southbound

[1] Values in this column were manually selected from the minimum recorded re-occupancy times based on the author's judgement, excluding unreasonably low observations.

A number of caveats should be emphasized in interpreting the results from the statistical analysis of limiting dwell times: (1) dwell times and re-occupancy times today may not be representative of future signaling system and rolling stock improvements, and (2) dwell times today may not be representative of future dwell times due to short term reductions in passenger demand per train caused by an increase in frequency.

(1) Planned line upgrades are expected to reduce train safe separation distance through the addition of extra speed commands, and improved acceleration/braking rates. Thus the minimum re-occupancy times given in **Table 3-1** may not be representative of future signaling system and rolling stock characteristics. Since the station occupancy time as measured using ADCS also includes station run-in and run-out time, this too will see some improvement. In addition to the effects of safe separation time, improved passenger circulation is an anticipated benefit of the future rolling stock, since all vehicles will have 4 wider doors, compared to only a subset today. While the effects of signaling improvements will require simulation to estimate, the improvements in passenger circulation can be benchmarked using the 4-door vehicles in operation today.

(2) Improvements to the line frequency has the potential to reduce dwell times, at least in the short term, since a similar passenger demand will be distributed across more trains. Assuming passenger arrival rates remain constant in the short term, the dwell time of trains at a future 3-minute headway can be benchmarked by selecting only observations from today which are close to a 3-minute headway.

Following these assumptions, peak hour dwell observations with 4-doors and arrival headways less than 4-minutes are selected, which may represent dwell times more in line with future conditions. **Table 3-2** shows the results for this "filtered" analysis. Results from the initial, unfiltered analysis are included for comparison. The sample size for each station is included to emphasize the lower number of samples fitting the aforementioned criteria.

The findings from the filtered analysis are mixed; some critical stations exhibit an *increase* in limiting dwell time under these assumptions (Charles SB, Davis NB. Harvard SB/NB), while others exhibit a decrease (Park Street NB/SB, Downtown Crossing SB).

The substantial increase in limiting dwell observed at Harvard, Charles, and Davis has two possible interpretations; first, that the effect of close train spacing has a severely adverse effect on run-in and run-out times at these stations, or second, that the combination of closely spaced trains queuing to enter the station and the effects of pre-shunting causes dwell times to appear longer than in reality. *Further study is recommended to determine the current limiting capacity at these stations due to signaling constraints, and to what extent this will be mitigated by planned future improvements*. Despite some improvements to the limiting dwell time exhibited after filtering the observations, Park Street NB/SB, and Downtown Crossing SB, all still exhibit limiting station occupancy times in the range of 2.5-3 minutes, not including re-occupancy times. While some improvements can be expected at these stations following upgrades to the signaling system,

significant dwell time reductions still appear to be required for these downtown stations to support uninterrupted headways of 3-minutes.

Table 3-2: Filtered Statistical Analysis of Station Bottleneck Occupancy for 4-Door Trains with Arrival Headways Less than 4-Minutes (s)

Station	Mean	SD	N	Mean + 2*SD (Filtered)	Mean + 2*SD (No Filtering)
Charles MGH SB	103	70	57	243	187
Davis NB	87	63	65	213	199
Harvard SB	82	62	55	205	166
Harvard NB	100	51	69	201	165
Park Street NB	94	43	55	179	183
Downtown Crossing SB	95	29	63	153	168
Park Street SB	110	19	58	148	183
Broadway NB	68	35	49	138	132
Kendall SB	77	29	61	136	146
Central NB	70	29	60	127	164
Charles MGH NB	72	20	53	113	157
Porter SB	59	27	46	112	108
Andrew NB	62	21	47	104	110
Davis SB	58	22	46	103	112
Central SB	62	20	61	102	132
South Station SB	72	10	71	92	125
Kendall NB	63	13	53	89	132
Broadway SB	65	9	82	83	92
Andrew SB	53	8	78	69	83
Porter NB	43	13	57	68	99
Downtown Crossing NB	-	-	-	-	-
South Station NB	-	-	-	-	-

NB = Northbound, SB = Southbound

3.5 Regression Approach

Following the finding of significantly limiting station bottlenecks in the previous section, it is clear that a regression approach to dwell time modeling will be required. This approach is necessary for two reasons. Firstly, a regression approach will allow us to make attributions of long dwell times to a specific cause, such as passenger demand or train congestion, which is necessary to develop targeted improvement strategies. Secondly, the effect of such extreme dwell times on capacity will need to be further investigated using a simulation approach; and a regression

approach can be applied to fully develop the necessary simulation model which incorporates passenger demand.

3.5.1 Methodology

Dwell observations for weekdays in April 2017 were compiled for non-terminal stations on the MBTA Red Line by combining track circuit data and passenger fare card activity. Outliers were omitted where measurement error is expected; observations under 20s were removed, representing 1.3% of observations, which are attributed to pre- or post-shunting error. Long dwell observations are retained, as measurement error cannot be confirmed, and it is not immediately clear whether these can be attribute to passenger demand.

A basic linear model formulation, shown below, with only the average number of passenger movements per door was used as an explanatory variable was selected in order to provide clear estimates of passenger flow rates, which are comparable to established ranges from other systems. Separate models are estimated for each dominant direction of passenger flow (predominantly boardings, predominantly alightings, and mixed flow) observed on each platform, where a threshold of 85% was used to classify flow direction as dominant. Models were not estimated for cases with less than 100 observations. Different measures of train crowding were trialed, but were excluded from the final model due to a limited dispersion of crowding observations, which produced many unreliable estimates of crowding impact.

$$DT = \beta_0 + \beta_{BAd} \cdot BA_d$$

Where,

DT = dwell time, and

BA_d = average number of boardings and alightings per door

Robust weighted ordinary least squares regression with bi-squared error function is applied to reduce the impact of outliers in estimating dwell time model coefficients. Robust regression is favored over ordinary least squares regression when a small number of outliers have a large impact on coefficient estimation, but observations cannot be categorically excluded (Yu & Yao, 2017). This approach proves to be particularly useful for estimating the impact of passenger volumes on dwell time in this case due to a significant number of unlabeled, train congestion related outliers.

Headway and block occupancy downstream of stations was also extracted from the train movements database, for the purpose of identifying and removing train congestion. Instances of train congestion were identified by determining which downstream block occupancy corresponded to large dwell time prediction errors.

3.5.2 Results

First, an overview of results for trunk stations is presented to demonstrate the breadth of analysis possible using ADCS. Finally, there will be a focused discussion on four key platforms; Davis Northbound, Charles Southbound, Kendall Southbound, and South Station Southbound to demonstrate the investigative power of this approach.

Overall, 120,119 observations were modeled across 37 platforms, with separate models by dominant passenger flow direction (predominantly boardings, predominantly alightings, and mixed flow). For all stations, the mean absolute prediction error is 9.7s; 50.7% of observations have an absolute estimation error of less than 5 seconds, and 79.1% of observations have an absolute estimation error of less than 10 seconds. Coefficient for passenger flow rate was found to be significant at the 95% confidence level for all platform and flow direction specific models. The coefficient estimates are given in **Table 3-3** for key stations.

Table 3-3: Dwell Model Coefficients

Station	Direction	Passenger Flow Direction	β_0 (Intercept)	β_{BAD} (Passenger Movements per Door)
Davis	Northbound	Mixed	49.4	0.41
Davis	Northbound	Offs	48.9	0.34
Harvard	Southbound	Mixed	56.6	0.44
Harvard	Southbound	Ons	57.1	0.48
Harvard	Northbound	Mixed	74.0	0.49
Harvard	Northbound	Offs	75.0	0.34
Charles MGH	Southbound	Mixed	51.0	0.87
Park Street	Southbound	Mixed	62.3	0.45
Park Street	Northbound	Mixed	63.7	0.37
Park Street	Northbound	Offs	66.0	0.20
Downtown Crossing	Southbound	Mixed	50.0	0.50

Coefficients for boarding and alighting time per passenger per door are reported in **Table 3-4** in comparison with the TCQSM. Note that values have been converted to a passenger time per door whereas the above coefficients represent the per passenger time for the entire train. While the values are generally quite close, the results from this study are larger for all flow types, suggesting more extreme observed values of loading diversity in the MBTA system.

Table 3-4: Comparison of Estimated Per Door Passenger Flow Rates With Established Ranges

Passenger Flow Type	MBTA Results*	TCQSM (*2013*)**
Predominantly Boardings	2.67	2.05
Predominantly Alightings	2.64	1.83
Mixed Flow	3.17	2.69

*Median observed coefficient from all platform specific models

**Median from reported range of flow rates observed across systems

Passenger Activity Time

The passenger activity time, defined in this context as the variable portion of dwell time prediction related to passenger volume, generally makes up a small portion of overall dwell time. The mean predicted passenger activity time across all observations is just 8.0s. These values are reasonable, since anecdotally, even trains dwelling for the minimum possible time can serve a large number of passengers. During the peak hours (8-9AM and 5-6PM), the average passenger activity time is 14.4s, with 23.6% exceeding 20s and 9.3% exceeding 30s. At Park Street, which experiences the longest dwell times on the line, the average passenger activity time during peak periods was 24.2s, with 1.4% exceeding 60s. This finding suggests that most extreme dwell times (2+ minutes) found in the previous statistical analysis are not simply a result of high passenger volumes, but other incidents.

Robust Regression

Figure 3-2 below shows a comparison of regression results using traditional Ordinary Least Squares (OLS) Regression and a Robust Regression with a bi-square loss function, applied to observations for the four selected stations, without any prior filtering of outliers. Simply, the robust regression technique works by iteratively fitting a weighted least squares regression, with the lower weights assigned to the points which exhibit the largest pull on the line of best fit after each iteration (Yu & Yao, 2017). For most stations, the impact on the slope is minor, with the exception of Davis Northbound (upper, right), which is heavily influenced by the presence of train congestion outliers. For the case of Davis, the estimated coefficient for passenger flow rate is closer to the anticipated values, as compared those found in the literature, after applying robust regression.

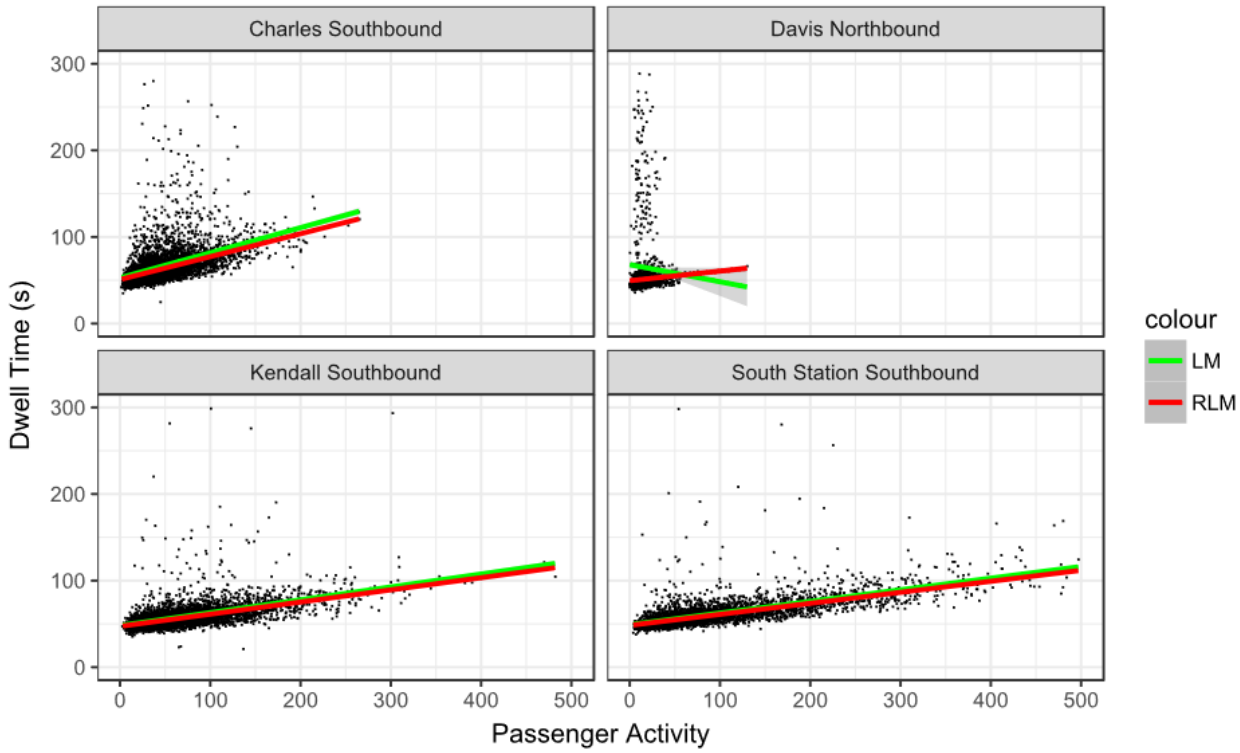

Figure 3-2: Regression of Mixed Passenger Activity By Two Methods

(LM = Linear Model, RLM = Robust Linear Model)

Congestion Identification

Unlabeled outliers such as congestion and train holding represent a major difference from manual dwell studies. Train congestion (in the form of slow speed codes and Red "do not proceed" signals) is identified by comparing prediction error from the initial regression to different downstream track occupancy conditions to identify those with a significant impact.

Previous rail dwell time studies relying on small sample sizes have avoided the unpredictable nature of dwell time in real-world system, including delays due to train congestion, passenger incidents, and operator behavior. In order to estimate a reliable dwell time model defined primarily by passenger activity, specific attention must be paid to the effect of outliers, especially when dealing with ADCS. Unlike other sources of error, such as passenger incidents, delays in dwell time due to train congestion can be identified using only analysis of track circuit data. Separating observations with train congestion for dwell time analysis is important to determine the effect of passenger demand, and to attribute the cause of long dwell times.

In order to identify train congestion, dwell time prediction error, calculated using only passenger volume and employing robust regression techniques to reduce the weight of outliers, is

compared to downstream block occupancy. For each station, a threshold for identifying train congestion was defined by manually observing the impact of downstream block occupancy on dwell time residual.

The results of the congestion identification process are shown in **Figure 3-3**. Observations highlighted in red shows the occasions when train congestion ahead is identified. As it appears from the figure, many observations at Davis Northbound and Charles Southbound that experience long dwell time have been subject to train congestions. In the case of Davis Northbound, 56.5% of observations with prediction error larger than 60s corresponded to identified instances of congestion.

While using this procedure helps to identify many outliers and enables more reliable estimates of dwell time with respect to passenger activity, many outliers are still observed which could not be attributed to train congestion. Further study is required for understanding the cause and identification methods for remaining outliers.

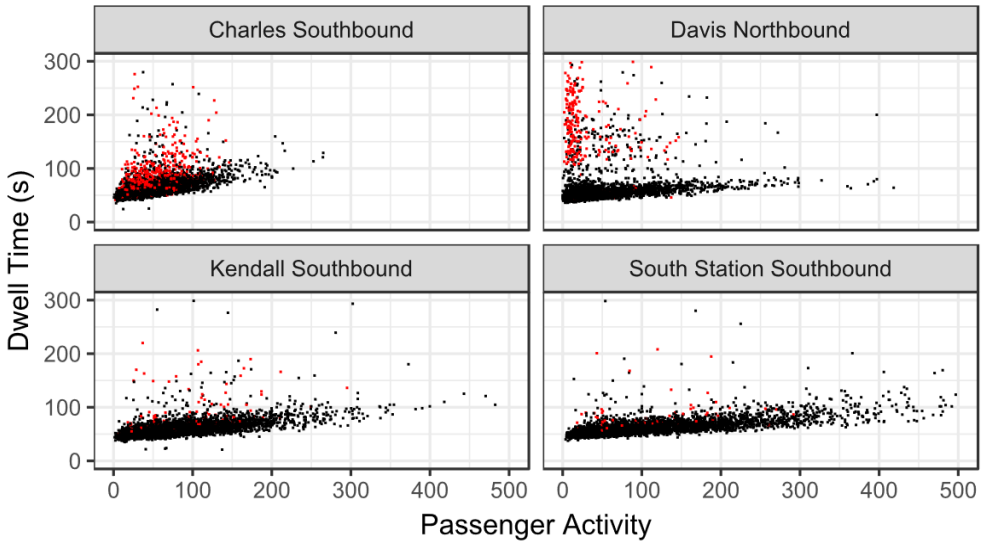

Figure 3-3: Results of Train Congestion Identification Shown Against All Dwell Observations

(Red = train congestion identified)

3.5.3 Discussion

Results from the regression analysis help to highlight individual differences of each station; for example: range of passenger activity, fraction of outliers, variation of dwell time with respect to passenger activity, and rate of change of dwell with respect to passenger activity.

Of particular relevance to the line capacity discussion is the cause of long dwell times. Here, regression provides two important findings, which the statistical analysis could not address: (1) stations differ largely in their proportion of outliers not explained by passenger volumes, and (2) different slopes are apparent between stations, indicating differences in how some stations perform under increasing passenger volumes.

(1) Regression with passenger regard allow us to identify high dwell observations which are not constrained by passenger volumes. The proportion of these outliers vary greatly between stations (**Table 3-5**), which was attributed partially to the frequency of train congestion ahead of the stations. This frequency of extreme dwells is closely related to minimum achievable headway at these points.

Table 3-5: Percentage of Observations Exceeding Given Prediction Error by Station

Prediction Error	Charles SB	Davis NB	Kendal SB	South Station SB
> 30s	5.5%	11.0%	2.6%	1.8%
> 60s	2.0%	9.4%	1.4%	0.8%
> 90s	1.1%	6.8%	0.8%	0.5%
> 120s	0.6%	4.6%	0.5%	0.3%

(2) Different slopes are immediately apparent between stations. The model developed here only takes into account global passenger demand. The differences between stations thus may be attributed to systematic differences in crowding between stations caused by station layout. While further study is required, the initial results suggest that stations with exits at only one end of the platform, such as Charles MGH, Andrew, and Broadway, exhibit a steeper increase in dwell with respect to passenger demand compared to other stations. This is interpreted as passengers not spreading out along the platform, causing a higher proportion of passengers at the busiest door. This trend appears for both predominant boarding and alighting models, suggesting passengers alighting at these stations also plan their position within the vehicle accordingly. Despite having exits at only one end, busy stations such as Park Street are less susceptible to this behavior, because heavy crowding causes passengers to spread out more evenly to avoid discomfort and because of the double platform at this station. This finding could help to highlight stations exhibiting this type of behavior as targets for intervention to improve passenger distribution along the platform.

3.6 Strategies for Reducing Dwell Times

An extensive body of literature exists on the factors affecting dwell times. However, literature on ways to reduce dwell time appears to be more limited, perhaps due to the extremely local and system specific factors influencing the success of a given strategy. Once the causes of

long dwell times have been determined, it is useful to develop a framework for identification of solutions.

In his research on crowd management and dwell time reduction in metro systems, Mensink (2017) provides a valuable framework for assessment of appropriate strategies, including a substantial list of applicable measures. Mensink places crowding reduction measures along a two-dimensional spectrum, from responsive to preventative, and from dynamic to static.

Another valuable way to classify strategies is in terms of which agent's behavior is being targeted. This can roughly be divided into strategies targeting passenger behavior, strategies targeting operator behavior, or strategies affecting the environment. Strategies targeting operator behavior can include retraining, changes to disciplinary rules, or changes to incentive structure. Strategies targeting passenger behavior can focus on discouraging unwanted behavior (crowding around doors, not moving down the platform, holding doors open), or influencing demand patterns to reduce peak crowding. Improvements targeting the dwell time environment could include expanding platforms, platform screen doors, or automated dwell time control such as in system with unattended train operations (UTO).

In terms of environmental solutions, special mention should be made of Park Street Station, which appears to be one of the most constraining stations in terms of dwell times, but has an underutilized dual platform layout which could represent a significant asset in reducing dwell times. The station currently operates without any separation of flows; in both the north and southbound directions, boarding and alighting passengers are free to use either the island or side-wall platforms. The double platform layout could be used to segregate boarding and alighting flows, as shown in **Figure 3-4**, enabling, enabling more efficient passenger movements – it is well recognized that one-directional flow is more efficient than conflicting bi-directional passengers flows (TCQSM, 2013). In the southbound direction, there are additional safety and flow benefits from separating alighting passengers from the significant platform crowds caused by branch passenger incompatibility in the PM peak. Due to the complex layout of Park Street Station, enforcement of this solution will require a significant customer communication campaign, however, a soft-implementation could begin by reversing the door-opening order, thereby nudging alighting passengers towards the desired (center) platform. *Design and implementation of this strategy should be made an immediate focus, given the opportune station configuration and line-limiting dwell times at this station*.

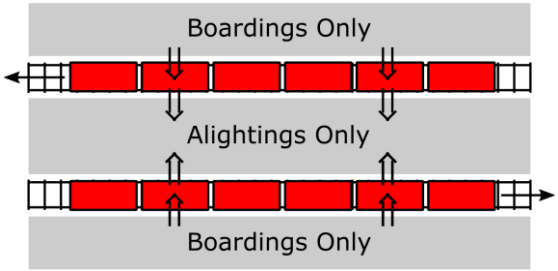

Figure 3-4: Proposed One-Way Passenger Flow at Park Street

Finally, at the MBTA, it appears that a great deal of the problems with dwell time are cultural in nature; lack of agency prioritization of on-time performance, operators preferring unhurried operations compared to the fear of disciplinary actions for closing the door on passengers, and passengers squeezing on to trains without regard for the system disruption. Appealing to passengers to reduce dwell time can appear to be insensitive when passengers view lack of on-time performance as an agency fault. It is clear that a great deal of effort will be required over an extended period in order to change the culture surrounding dwell times prior in order to enable a 3-minute headway.

Chapter 4. Pilot Case Studies Introduction

Following the finding of headway irregularity originating from the southbound trunk terminal and northbound branch merge presented in Chapter 2, three service interventions ("pilots"), listed below (1-3) were proposed. Taken together the pilots were intended to challenge current operating norms, and to test assumptions about future operating practices. The pilots were designed with the intent of rapid implementation, in order to demonstrate how small, incremental operating improvements can substantially rapidly improve reliability. The three pilots were trialed over the course of six weeks on the MBTA Red Line. This chapter will provide an overview of the pilot designs and motivations, pilot timeline, and an overview of the relationships between the pilots. The following three chapters will provide more detail about the implementation and results of each pilot.

(1) Alewife Terminal Crew Drop-backs ("Drop-back Pilot") - AM & PM Peak Periods
(2) Davis Station Southbound Train Holding to Target Headway ("Davis Holding Pilot") - AM & PM Peak Periods
(3) Andrew Station Northbound Train Holding with Variable Target Headway ("Andrew Holding Pilot") - PM Peak Period Only

4.1 Alewife Terminal Drop-back Pilot - Introduction and Motivation

Drop-back crewing (also known as "step-backs" or "fall-backs") are a scheduling technique where operators step off their train at a terminal station, allowing another operator to take over control of the train they arrived on. The operator proceeds to take a break and departs on a future train. By decoupling the operator from their train, drop-backs enable reductions in train turnaround time without compromising operator layover time, allowing terminal capacity to be increased. In addition to reducing layover time and improving terminal capacity, drop-backs can also help to improve consistency of departure headways, increase schedule recovery, and improve fleet utilization.

Discussions of drop-back crewing in the literature are limited, perhaps due to the heavily practical nature of such crewing techniques. The *TCQSM* (2013) suggest that crew drop-backs can be used to reduce layover time when the time for changing ends is prohibitive. In his review of high-frequency rail terminal capacity and performance, Lee (2002) also mentions drop-backs as a method of reducing the layover times employed both at the CTA and New York City Transit (NYCT) but does not provide any empirical support for the strategy. Within the MBTA, drop-backs have been implemented since 2016 on the Orange Line to aid in maximizing capacity for a limited fleet size, however no documentation of its benefits there were available, perhaps limiting its spread throughout other parts of the agency.

Drop-back crewing was implemented as a pilot at the Alewife Terminal during AM and PM peak periods. The Pilot instructions can be summarized as follows:

(1) At the start of peak period service, two extra operators were positioned at Alewife, one for each branch.

(2) Every operator to layover at Alewife during peak periods was instructed to drop-back to the next departure for their respective branch.

The primary motivations for the Drop-back Pilot are discussed below. The pilot was motivated by the need to increase the amount of schedule recovery achieved at the terminal, improve dispatch precision, and assess the effectiveness of the operating procedure for the future 3-minute headway objective.

4.1.1 Terminal Schedule Recovery

Under current operating procedures, the Alewife terminal fails to significantly reduce dispatch headway variability as compared to the variability of northbound headways arriving to the terminal area, as discussed in Section 2.3. This is explained by operators not having enough slack to significantly shorten their terminal platform time in response to delays. Insufficient recovery time produces negative outcomes for both operators and passengers; operators aren't able to recover from northbound delays to start their southbound trip on-time, extending their shift, while southbound passengers experience some large headways. Drop-backs are expected to result in significant improvements to schedule recovery by de-coupling the operators' layover time from the minimum turnaround time.

4.1.2 Improve Dispatch Precision

Earlier headway dispatching experiments conducted with the MBTA contended with issues of compliance with recommended dispatching times, especially when the recommended dispatch occurs prior than the scheduled dispatch, or with minimal layover time (Fabian, 2017; Maltzan, 2015). These issues are equally present in heavy rail dispatching as compared to the bus and light rail dispatch pilots since the frequency is much higher and thus tolerances for delay are reduced. The next section on the Davis Holding Pilot, will explore the negative impacts of small initial deviations from the scheduled headway. The Drop-back Pilot, aims to reduce the variability of response time, especially in unfavorable operator conditions, in order to improve dispatching precision for current and future conditions and to complement further dispatching pilot experiments.

4.1.3 Future Capacity Objective

A significant reduction in scheduled terminal platform time, from the current 6-minutes to 4-minutes is required to achieve the future 3-minute headway objective. Platform times of less than 4-minutes are seen only about 3% of the time today, although an estimated 7% of peak

layovers require this level of dwell reduction to return to schedule following disruptions. In the future, operators would be required to achieve even lower platform times, around 1-2-minutes to achieve an on-time departure following northbound delays. It is clear that changes in terminal procedures will be required in order to achieve these reductions in platform time necessary for future capacity objectives, while still allowing operators sufficient break at the terminal. In part, this pilot will be a demonstration of the capacity for future layover time reductions possible using drop-back crewing.

4.2 Davis Station Holding Pilot - Introduction and Motivation

Headway-based holding is commonly used in high-frequency transit services to improve headway regularity in response to exogenous and inherently unpredictable factors such as road congestion, unexpected passenger demand, and incidents. Improved headway regularity reduces wait times and crowding downstream at the expense of delay to in-vehicle passengers.

In this case, target headway-based holding strategy was implemented at Davis Southbound to counteract routine bunching caused by differences in allowable speed for the two Alewife Terminal departure movements, detailed in Section 2.3.1. The pilot is differentiated from the Alewife Drop-back Pilot, which also aimed to improve southbound headway regularity in that the focus is placed on reducing *minor, routine* headway variations, rather than reducing the frequency of long headways, which is an objective of the Alewife Drop-back Pilot. The pilot instructions can be summarized as follows:

(1) All trains at Davis Southbound are held to a minimum target headway of 3:15 (AM), or 3:00 (PM) ("holding target").
(2) If the headway is greater than the holding target upon train arrival, trains are released immediately.
(3) Headways are measured using a stopwatch from wheel-start of the previous train until the departure instruction for the current train.

The holding target was chosen in order to reduce downstream bottleneck delays without impeding future Alewife dispatches, which would risk reducing terminal capacity and causing northbound delays. The difference in target headway for AM and PM represents the slightly higher scheduled frequency in the PM and the associated likelihood for closely spaced trains.

The Davis Pilot was proposed to counteract speed differential between platforms, address the dispatching need which could not be met at Alewife, prevent downstream congestion, and improve the passenger experience. These motivations will be explained in further detail below.

4.2.1 Variation in Headway Caused by Platform Differences

As detailed in Section 2.3.1, the design and condition of the crossover switch just south of the Alewife terminal cause berthed P1 trains to depart the terminal with a speed limit of 10 mph,

as compared to a speed limit of 25 mph for berthed P2 trains. The speed code differential was shown to result in routine, and predictable southbound bunching. In high-frequency services like the Red Line, even small initial variations in headway can cause large deteriorations in service quality, impacting passenger wait times, producing uneven train loads, and causing increased train congestion. Holding trains with short headways at Davis was devised to reduce the downstream impact of this running time differential.

4.2.2 Barrier to Implementation at Point of Dispatch

In the short term, alternatives to improve consistency of Alewife dispatches included dispatching uneven headways at Alewife (anticipating the running time differential), or enforcing a 10 mph speed limit on both tracks. These alternatives were discarded in favor of downstream holding due to their contribution to increased terminal processing time/delays. In the long term, crossover repairs to allow all trains to travel at the same speed departing the terminal is the preferred solution. Davis was selected as the downstream holding point because of its immediate proximity to the terminal, and thus minimal impact to in-vehicle passengers.

4.2.3 Downstream Bottlenecks

As discussed in Section 1.6.3, there are a number of system headway bottlenecks which prevent trains from maintaining low headways, around 3:00, through certain stations. This means, bunched dispatches from Alewife not only provide little passenger benefit, but they also experience congestion delays downstream, negating any potential benefits to the in-vehicle passengers. Therefore, by reducing the magnitude of bunching closer to the point of dispatch, they will provide benefit to more passengers in the form of reduced wait time and crowding, while maintaining similar overall running times.

4.2.4 Passenger Impacts

Regularizing headways at Davis Southbound serves to improve the southbound traveler experience by improving wait times, travel times and crowding. In the AM, 76% of passengers traveling through the peak load point at Central originate in the first four stations after Alewife, showing that evening out headways can result in significant improvements to crowding. In the PM peak, only 5% of southbound passengers originate at Alewife, resulting in little disbenefit, while any resulting improvements in wait times will benefit the large majority. Southbound trunk passengers should also benefit from faster travel time as a result of less train congestion in the stops after holding.

4.3 Andrew Station Holding Pilot - Introduction and Motivation

Similar to the Davis Holding Pilot, the third and final pilot, Andrew Station Northbound Holding implemented a target headway-based holding strategy to reduce bunching and improve

headway regularization following the northbound merge. The main difference from the Davis Pilot is the use of a variable headway target. The instructions can be summarized as follows:

(1) All trains at Andrew Northbound are held to a minimum target headway of 3:00 ("holding target").
(2) The holding target is increased to 4:00 ("long hold"), if the next-train arrival clock predicts the following headway is at least 6-minutes at the time the stopwatch reaches 3:00, or upon train arrival, whichever is later.
(3) In all other cases, trains are released immediately.
(4) Headways are measured using a stopwatch from wheel-start of the previous train until the departure instruction.

The holding strategy was implemented in the PM Peak only with consideration for passenger demand patterns; in the AM peak, branch passengers significantly outnumber trunk passengers in the northbound direction, so the holding benefit for the trunk headway regularity would not outweigh the cost of holding to in-vehicle passengers. The Andrew Holding pilot is motivated by a current lack of real-time coordination of branch headways, quality of service for northbound passengers, and downstream headway bottlenecks, as detailed below.

4.3.1 Northbound Trunk Headway Irregularity

Today, the northbound branch merge control focuses primarily on achieving the desired branch train sequencing rather than on headway regularization. In part, this is because during the PM peak, the southbound direction, especially the ordering of branch trains, is considered more critical, and so the northbound level of service receives lesser focus. However, northbound headway control can still be executed without impacting southbound service. The lack of headway control at the merge also belies a pervasive agency belief that the operating environment is too unpredictable to warrant precise headways. This is a self-fulfilling prophecy, a lack of precise dispatching leads to negative headway outcomes. Due to this lack of coordination effort, bunching and irregular headways frequency occur following the merge, negatively impacting northbound passengers.

4.3.2 Northbound Passenger Impacts

The resulting headway variability following the northbound merge significantly impacts the quality of service for northbound passengers in terms of wait times and crowding. The pilot proposes a simple but effective target headway holding strategy to regularize northbound headways without affecting the order of branch trains and minimal effect on their northbound running times. In-line headway-based holding must weigh the negative impact on in-vehicle passengers against future wait time benefits for downstream passengers. In the case of the PM Peak, northbound trunk passengers beyond JFK outnumber branch passengers approximately 8 to 1, so the anticipated wait time benefits are significant compared to the anticipated delay for branch

passengers. Looking beyond wait time, northbound passengers will further benefit as a result of reduced crowding and left-behinds as a result of more uniform headways downtown.

4.3.3 Downstream Headway Bottlenecks

The worst effects of bunching are partially avoided in high-frequency transit by the minimum feasible headway; for example, trains arriving simultaneously to the merge may achieve at minimum a 2:00 headway at Andrew. Trains merging at minimal spacing will experience delays due to more restrictive bottlenecks at downstream points. While a 2-minute spacing is possible at Andrew, the minimum northbound line headway is closer to 3:00 at bottleneck points of Park Street – Downtown Crossing, Harvard Northbound, and Alewife Terminal. These bottlenecks mean that pushing through bunched trains at Andrew does not translate into short headways in the southbound direction. Thus, holding trains to the minimum line headway at Andrew is recommended to provide more regular northbound service, and can be applied without impacting southbound headways.

4.4 Relation Between Service Interventions

The three proposed pilots provide lessons for developing operating strategies for the future Red Line 3-minute headway objective. This section will discuss how the various strategies relate and how they are relevant to current and future capacity objectives. The main themes explored by the pilots relevant to future service objectives are service regularity and time-tabling precision.

4.4.1 Service Regularity

Service regularity is closely linked to the quality of service as experienced by customers; if service is irregular, customers will experience inferior wait time and crowding levels. Generally, improving service regularity is less resource intensive than improving service frequency, since it does not require extra train units. In some cases, improving service frequency can even cause an effective reduction in capacity by causing increased train congestion (Heimburger et al., 1999). For these reasons, it is important to focus on improving service regularity in advance of a service frequency increase.

The three proposed pilots all target improvements in service regularity, and together form a persuasive argument for how small headway differences have large downstream impacts. Drop-backs and Davis Holding target improvements in southbound trunk regularity, by mitigating large, and small deviations in headways, respectively. With a variable holding rate, Andrew Northbound Holding also targets reductions in both large and small headway deviations. Each of these pilots demonstrates the value of service regularity in reducing passenger crowding, wait times, and journey times under the peak demand conditions. Together they demonstrate how small investments in service management can provide benefits equivalent to expensive service frequency improvements.

In the long-term, real-time headway-based dispatching at the terminal stations (Alewife in the North and Ashmont or Braintree in the South) is preferred to in-line holding at Andrew and Davis, because it causes less passenger delay (particularly in the AM peak when trains are heavily loaded in the inbound direction). However, in the short-term, these strategies provide valuable lessons in prioritizing service regularity.

4.4.2 Operating Precision

High precision time-tabling reflects an agency culture of "every second counts" which is all but necessary to reliably achieve such high frequencies. The majority of high-frequency metros schedule operations down to at least the nearest 15s (Canavan et al., 2019). Though rooted in the timetable process, this culture expands to other operating practices such as headway and dwell time control.

Today, the Red Line is scheduled to the nearest minute, and even large deviations from even headways (routine bunching following Alewife crossover or northbound merge) go unnoticed or unchallenged. In future near-capacity operations, precision in operating practices will become increasingly important as slack to absorb delays is diminished.

The three proposed pilots seek to improve influence agency culture for operating precision. In the case of the Davis and Andrew Holding pilots, handheld stopwatches were used as the primary headway measurement tool, introducing the notion that using station clocks (which display time only to the minute) and timetables rounded to the minute are insufficient for the task of headway management. In addition, the Drop-back Pilot also introduces the notion that to-the-second departures are necessary for achieving higher capacity and more resilient terminal operations. Together, these control strategies can move the needle in introducing an "every second counts" ethos into the agency.

4.5 Pilot Implementation Details

4.5.1 Proposed Schedule

A six-week period was designated for the three pilots, broken up into 3 phases, each two weeks in duration, between January 21st and March 1st, 2019. The service interventions scheduled to occur during each of the two week periods are shown in **Table 4-1**. The pilots were scheduled during weekday AM (6:30-9:30 AM) and PM (4:00-7:00 PM) peak periods, where applicable.

The control, or "Baseline" period was a two-week period immediately preceding the pilots. A new schedule was implemented on the MBTA Red Line in late December 2018, leaving a limited window to be used for pilot Baseline analysis under the same operating conditions. The first two weeks of operations under the new schedule were excluded due to holidays and to allow operations to adjust to the new schedule. The new schedule was not substantially different from the prior, so the minimal adjustment period is deemed sufficient.

Table 4-1: Pilot Phases Calendar

Pilot Phase	Dates	Intervention
Baseline	Jan 7th – Jan 18th	Baseline Period (AM & PM)
Phase 1	Jan 21st – Feb 1st	Drop-back (AM & PM)
Phase 2	Feb 4th – Feb 15th	Drop-back & Davis Holding (AM & PM)
Phase 3	Feb 18th – Mar 1st	Davis Holding (AM & PM) & Andrew Holding (PM Only)

4.5.2 Pilot Execution

The number of peak periods of each pilot combination executed is listed in **Table 4-2**, excluding days removed due to service disruptions, as discussed in the next section. A complete list of pilot days is included in Appendix 2.

Consistent execution of the pilots was hampered by winter weather (esp. Jan. 22, Jan. 31, Feb 12/13), holidays (Martin Luther King Jr. Day – Jan. 21, New England Patriots Super Bowl parade – Feb. 5) and general scarcity of extra operators in a resource-constrained agency. When MBTA staff were unavailable, the pilots could not be conducted. For the case of overlapping pilots in Phase 2 & 3, management was asked to prioritize staffing for the latter implemented of the two pilots.

Table 4-2: Number of Pilot Days Available for Analysis

Intervention				Time Period	
Baseline	Drop-backs	SB Holding	NB Holding	AM Peak	PM Peak
✓				8	10
	✓			5	4
	✓	✓		2	1
		✓		9	3
		✓	✓	N/A	4
			✓	N/A	3

4.5.3 Pilot Staffing

The three pilots were executed by MBTA "officials" in tandem with MIT research students. MBTA "officials" act as roving supervisors for a single station or group of stations (usually two) to respond to passenger incidents and act as a point of contact for service controllers on the ground. While researchers were present during much of the operations to clarify pilot instructions, MBTA officials were responsible for communicating the instructions to the motorpersons in all cases. In case of the holding pilots, official communicated the holding instructions verbally to the operator who in turn would make an announcement to passengers if warranted (holding time approximately 1-minute or greater).

4.5.4 Potential Bias in Pilot Execution Days

It is important to discuss a possible bias in terms of which days the pilots were executed. For example, pilots were not staffed during days with extreme weather events due to the extra demands placed on staff during these days. In addition, in at least one case the pilot was called off by supervisors during the first hour of the shift after an unrelated delay of 15-minutes downtown caused large back-ups. Conversely, during the Baseline, there is no self-selection since service is operated daily, regardless of weather or incidents. During the two-week Baseline Period there were zero days with precipitation or unseasonably cold temperatures. Additionally, aside from the two Baseline days already excluded, there appear to be no other days where large disruptions may have prompted early cancellation of the pilots. Therefore, the Baseline sample is judged to be adequately similar to the Pilot sample in terms of excluding days with extreme weather and service disruptions.

4.5.5 Removal of Baseline and Pilot Days

On the basis of large incidents, two AM peaks are excluded from the Baseline (Jan 11th, 18th) and one from the Davis Holding pilot (Feb 26th). Each of the aforementioned days included train breakdown or passenger incident delays of at least 15-minutes during the peak two-hours. These large disruptions violate many of the assumptions used in the pilot analysis, including that of no left-behinds.

For example, **Figure 4-1** shows the southbound trunk distance-time graph for January 11th, 2019, one of the aforementioned Baseline days which was removed due to an incident between Charles MGH and Park Street beginning at 8:10 AM. It is important to notice that there are many other smaller, but still substantial delays which did not trigger removal, including others in the figure below (e.g. at Porter at 8:50 AM). Due to the frequent nature of delays on the Red Line, the threshold was set to exclude as few days as possible, while still removing those with significant negative impacts.

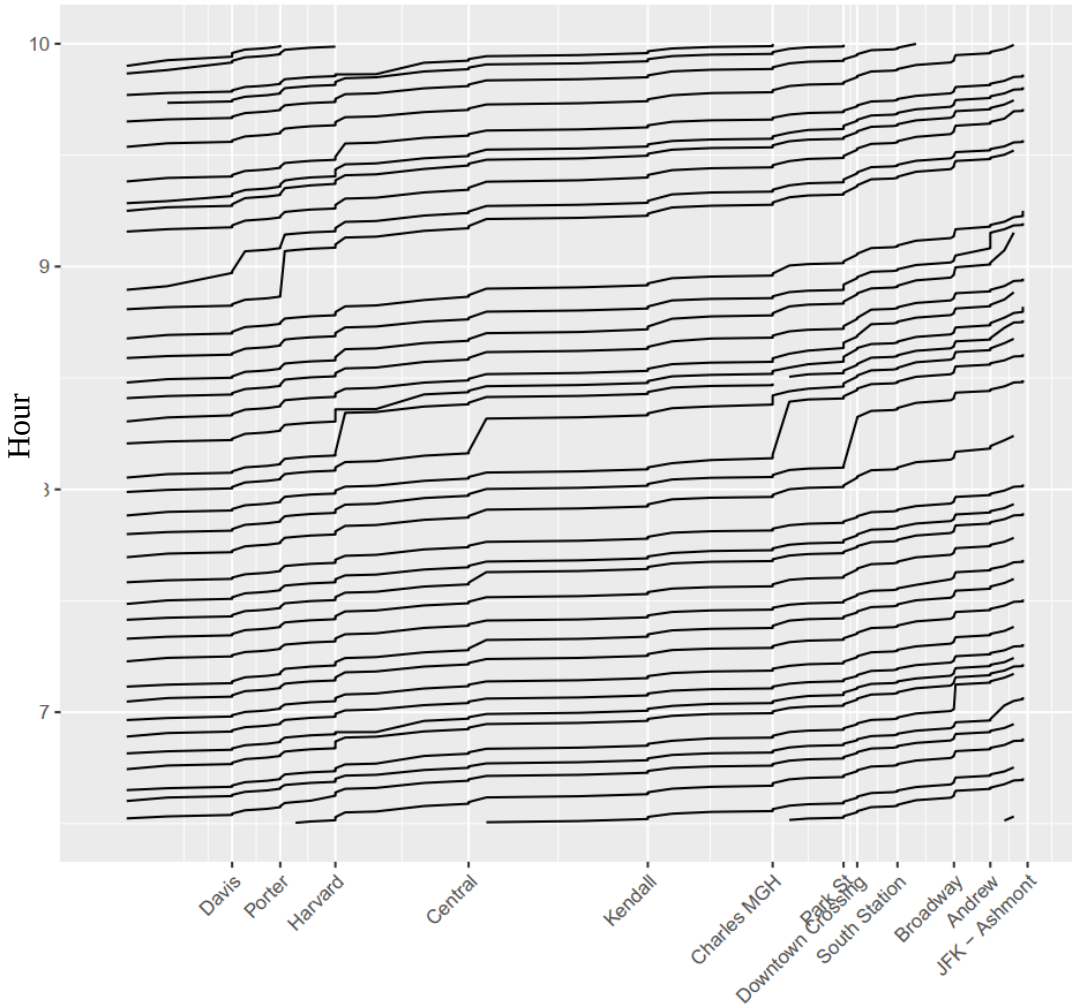

Figure 4-1: Southbound Distance-Time Graph for January 11th

The decision to remove entire peak periods rather than specific trips was made because of the lasting effect of incidents, and because it is important to measure the impact of interventions on all service, described as follows. As seen above, incidents can have long term effects on service, impacting trains which were dispatched even after the end of the incident in the form of increased train congestion. Isolating specific trips which are affected by incidents is therefore difficult. Furthermore, the pilots were intended to improve headways and running times both for the affected trains (i.e., held trains) and also for the trains with no intervention, due to changes in train congestion and passenger loading. Thus the success of the pilot should be judged on overall service impacts rather than impacts on specific trips. Removing specific trips affected by incidents can distort the overall sample, making it difficult to compare Baseline and pilot periods directly.

The first and last half hour of each Baseline and pilot peak period was also discarded from the analysis to select only periods of stable frequency and consistent pilot implementation. Thus the pilot results will consider only trains which passed through the point of interest during the middle two hours of each peak; AM (7:00-9:00 AM), PM (4:30-6:30 PM).

Due to the limited sample size, the overlapping pilots in Phases 2 and 3 do not have a sufficient number of observations, and will not be the subject of any standalone analysis. Special consideration will be taken when including data from the overlapping pilot days in their respective pilot analyses to avoid comparison of results with dissimilar initial conditions.

4.6 Principles for Analysis of Service Interventions

The following section will provide general principles for the analysis of service interventions. Providing a consistent set of principles to consider creates transparency and builds trust in the analysis. In addition, with such complex, interconnected systems, not all impacts can be discussed succinctly, meaning the most important factors must be recognized in advance of the analysis in order to draw-out representative and meaningful conclusions. These principles will guide the analysis of the aforementioned MBTA service interventions.

Considers Passenger and Operator Perspectives

It is important to consider both the passenger and operator's perspective when reviewing intervention results. Interventions which cause significant burden/complexity for the operator may not be worth the cost for minimal passenger improvements. Conversely, interventions which can provide sizable passenger benefits must be valued accordingly.

In many cases passenger and operator perspectives align. For example, running times represent both the core of the traveler experience and a major factor in service delivery costs. Crew management often presents a tension between passenger and operator perspective, since agency management must satisfy both parties, while customers are agnostic to the demands placed on crews by different operating procedures. Finding the alignment between passenger and operator perspectives is important to achieve viable, long term operating procedures with the greatest benefits.

Robust to Short Term Variations

Interventions analysis should not be overly sensitive to short term variations in passenger demand, incidents, and operating environment conditions. Service delivery should be evaluated on its robustness, in order to ensure service is designed to provide reliability to the majority of passengers regardless of trip time. Steps which can be taken to ensure robustness of statistics include use broader time periods (for example peak hour rather than peak 15-minutes), and selecting statistics which are not overly sensitive to extreme values. Of course, length of time periods should be limited in order to accurately represent passenger experience during different times of day.

Consider Magnitude of Individuals Impacts in Addition to Total Impacts

Intervention analysis should consider the magnitude of individual impacts in addition to the summation of benefits/disbenefits. Summary statistics of overall passenger impacts may mask individuals experiencing the largest magnitude of change, misrepresenting how customers perceive the travel experience. For example, Taylor & Claxton (1994) showed that customer evaluation of service attributes in the airline industry differed depending on whether they experienced a delay, even for service attributes not related to punctuality. This means that passengers experiencing delays due to train holding may have more negative views of other attributes of service quality, even beyond the added travel time, while passengers experiencing small savings may be entirely unaware of the benefits. Thus small travel time/crowding improvement for a large number of passengers may not always outweigh a large disutility for a small number of passengers. A notable exception to this in the area of transportation is in cases where a small incremental load results in significant externalities to all users. For example; each additional vehicle beyond the point of maximum capacity on the fundamental diagram of traffic flow causes a loss in system capacity overall. In these cases, strategies such as ramp-metering can improve traveler outcomes overall. This situation can arise in severely over-capacity metro systems.

Relevant to Current and Future Conditions

Analysis of service interventions should consider applicability to both current and future conditions. Interventions introduced today will be received poorly if the primary impact will not be felt until a future time. Conversely, interventions which provide only short-term benefits may receive little buy-in from management who are focused on longer-term improvements.

Some of the recommended pilots can be resolved in the future through larger capital/operational interventions, such as repairs to the Alewife Crossover or communications infrastructure for coordinated train dispatching from the southern branches. The pilots are based on the current system limitations and with consideration for the limited resources available for pilot initiatives but allow the opportunity to evaluate potential improvements if these larger interventions are embedded in the normal operations.

Transparent and Consistent Treatment of Incidents

Analysis of intervention should apply a consistent and transparent methodology for the exclusion of days/trip affected by exogenous incidents. A consistent set of observations should be used across analyses rather than selecting days which enhance specific statistics. Since not all incidents can be excluded, intervention analysis should highlight where incidents in either the Baseline or study period cause findings to differ from the expected results.

Chapter 5. Drop-back Pilot Results

A drop-back crewing pilot was implemented at the Alewife Terminal on the MBTA Red Line during the AM and PM peak periods for two weeks. The specific instructions and motivations were discussed in detail in Section 4.1. This chapter discusses the results of the Alewife Drop-back Pilot in terms of implications for current and future capacity. At the current headway, operators generally have sufficient layover time to walk the length of the platform and depart on schedule; thus improvements are primarily anticipated in extreme situations rather than the median case.

Section 5.1 examines changes in operator reaction times to the departure bell under different conditions. Section 5.2 examines the effect of drop-backs on headways departing Alewife. Section 5.3 examines improvements in northbound queuing to enter the terminal. Finally, Section 5.4 offers general discussion including the implications of the findings, related and complementary crew management techniques, compliance issues, and operator reactions to the pilot.

5.1 Operator Reaction Time

In this context, reaction time is defined as the time elapsed between the departure bell ring-off and the train pull-out from the platform. Defined in this way the reaction time may include time for the operator to check for passengers, close doors, and depart. In cases where the train arrives late or the operator is slow switching ends, reaction time may be very long, as it includes some portion of the time taken by the operator to change ends. The minimum reaction time is interpreted as a case where train doors were already closed and the operator was anticipating the departure. The procedure for determining reaction times from OCS data is described in **Appendix 1**.

Reducing the magnitude of operator reaction time is necessary in order to achieve the maximum capacity of the terminal since while the departure bell is active, any conflicting movements are restricted from using the crossover. Reducing the duration of crossover occupancy increases the available slack time between consecutive trains which can be used to absorb delays. Crossover occupancy time can be a limiting factor for terminal capacity in cases where approach and crossover speeds are limited, such as at Alewife.

Drop-backs can reduce reaction time as defined in this context by allowing more time for operators to switch ends, meaning they are more likely to be prepared for departure when the bell rings. The difference will be greatest under disrupted conditions when the operator would not normally have sufficient time to switch ends before the departure bell.

Due to scheduling intricacies and other variations in operations, it is difficult to control for the effect of drop-backs on departure headways and layover times. Analyzing reaction times is one

way to directly observe the improvement in the time for operators to switch ends. Reaction times identified from the pilot under various conditions can be used as an input to a future simulation model in order to determine terminal capacity and reliability under different crewing and dispatching procedures.

5.1.1 Reaction Time Under Regular Conditions

Under typical, non-disrupted conditions, operators have sufficient time (approximately 6-minutes) to switch ends and position themselves in or by the cab prior to the ring-off. In these cases, reaction time is primarily defined by passenger demand. Therefore, under non-disrupted conditions, limited changes to reaction times are expected under the Drop-back Pilot. However, worst-case reaction times are expected to improve significantly, as these cases may include time for the operator to switch ends, which is relieved by drop-back crewing.

During the Baseline, approximately 88% of reaction times fall within 0-60s, with a median time of 22s and a large positive skew. Median reaction times in the AM peak are 3s longer than during the PM due to the increased presence of boarding passengers at Alewife.

Overall, changes to reaction times during the Pilot closely matched the expectations outlined above, as shown in **Table 5-1**. The median reaction time showed no change in the AM and decreased by 2s in the PM. Significant improvements are seen in the worst case; the 95th percentile reaction time dropped 55s (45%) in AM peak and 43s (43%) in PM peak, contributing to fewer extreme southbound gaps. Additionally, the pilot resulted in improvements in reaction time variability; the standard deviation fell by 42% in the AM and 38% in the PM, which contributes to more precise dispatching. Even greater reductions are expected in cases in disrupted conditions, as will be explored in the following section.

Table 5-1: Changes in Overall Reaction Times (s)

Time Period	Study Period	Mean	SD	Median	95th Pctl
AM Peak (7:00-9:00 AM)	Baseline	35.7	38.8	23.0	122.2
	Drop-back	27.0	22.5	23.0	67.4
PM Peak (4:00-6:00 PM)	Baseline	31.2	31.2	20.0	101.8
	Drop-back	23.2	19.5	18.0	58.0

5.1.2 Reaction Time Under Disrupted Conditions

Under baseline terminal procedures, the duration of a train's layover time prior to the departure bell is representative of how long the operator has to walk the length of the train while still achieving a prompt departure following the ring-off. Trains which arrive on schedule, 6-minutes before the departure bell, leave the operator with ample time to switch ends and depart promptly after the bell. Conversely, trains which arrive only a few minutes before the departure bell are likely to exhibit long reaction times as the operator will still be switching ends when the

bell rings. It is in these cases where drop-back crewing is expected to make the largest impact, since the drop-back operator has ample time to walk to the opposite end of the train and prepare for departure.

A comparison of observed reaction times for different durations of layover prior to ring-off are shown in **Table 5-2** for the Baseline and pilot periods. Consistent with expectations, trains which arrived shortly prior to ring-off exhibited the largest improvements. Compared to the Baseline, the reduction in median reaction time for trains arriving only 1-2 minutes before the ring-off was 62s (43%). Although reductions in mean reaction times are seen in all categories, only improvements for trains in the 1-2 and 4-5 minute layover categories are statistically significant due to the small sample size and high variability. These results highlight how lower minimum layover times due to drop-backs aid in disrupted scenarios.

Table 5-2: Change in Reaction Time to Ring-off Bell by Duration of Dwell Prior to Ring-off

Dwell Prior to Ring-off	Mean (s)			SD (s)		N	
	Baseline	Drop-back	Change	Baseline	Drop-back	Baseline	Drop-back
1-2 min	144.6	82.3	(62.3)*	55.4	50.3	22	8
2-3 min	79.0	64.6	(14.4)	38.5	68.3	7	7
3-4 min	57.4	40.0	(17.3)	43.7	32.3	33	31
4-5 min	34.2	26.5	(7.7)*	27.7	21.8	183	98
5-6 min	31.0	26.1	(4.8)	28.9	24.1	228	175
6-7 min	24.9	24.1	(0.8)	24.7	28.0	161	124

Significance codes: * = $P \leq 0.05$, ** = $P \leq 0.01$, *** = $P \leq 0.001$

Passenger Savings due to Reaction Time Improvements:

Trains arriving 1-2 minutes before the bell accounted for just 4% of trains during the Baseline peak periods, or about 1-2 train per peak. However, past observations are not a reliable indication of the frequency of these types of extreme events, since in practice dispatchers give priority to other terminal movements of lesser priority rather than ask trains to depart momentarily after arriving. For example, dispatchers may allow a lower priority northbound train to enter platform P2 before assigning the crossover slot to a delayed P1 train scheduled for departure, recognizing that late arriving trains require additional layover time. With more experience under drop-back crewing, there is good reason to believe the dispatchers could further improve dispatching reliability by taking advantage of short reaction times under disruptions.

Additionally, while these late arriving trains make up a small fraction of dispatches overall, they have a large impact on southbound headways. Section 2.4 examined how trains with large dispatch headways are disproportionally likely to encounter high passenger volumes which can create a vicious cycle of longer dwell times, and further increases in headways. Due to this non-linear relationship, even small improvements to the worst case dispatches at Alewife enabled by drop-backs can result in substantial improvements to headways downtown.

5.1.3 Minimum Observed Layover Times

Although layover times are dependent on many external factors, the minimum observed layover times provide some anecdotal insight into the feasibility of future operations. **Table 5-3** shows the bottom 1% of observed layover times[2] from both the Baseline and pilot periods. The minimum layover time observed in the pilot was 35s lower than during the Baseline. The mean of the bottom 1% of layover times is similarly 37s lower than during the Baseline. It is also notable that the reaction time for the minimum observed layovers during the pilot is only 35s (similar to the overall Pilot mean of 27s), compared to the Baseline value of 66s. This suggest that operators were able to achieve these short pilot layovers without rushing, while the Baseline operators would have seen the bell ringing while changing ends and felt pressured to rush to the cab.

Table 5-3: Bottom 1% of Observed Layover Times[2] (s)

Study Period	Range	Mean	Mean Reaction Time
Baseline	92-134	116	66
Pilot	57-88	79	35

These minimum observed layover times suggest that at a future 3-minute headway, with a scheduled platform time of 4-minutes (*TCQSM*, 2013), there would be a maximum of about 3-minutes of schedule recovery under drop-back operations, since minimum layover times are about 1-minute. These figures suggest that a reliable 3-minute headway is not possible without drop-backs since during the Baseline operators had to rush to achieve layover times of under 2-minutes, significantly restricting the amount of schedule recovery.

5.2 Effect on Headways

5.2.1 Overall Headways

No significant changes to headways departing Alewife should be expected as a result of the pilot; under non-disrupted conditions, operators have ample time to switch ends and depart on schedule. Some improvements are expected to extreme headways, where time switching ends is a limiting factor.

Table 5-4 shows minor but not statistically significant improvements to the mean departure headways observed in both the AM and PM periods during the Drop-back Pilot. A two-sided test for differences in proportions shows that the Pilot did *not* result in a significantly lower proportion

[2] Due to the placement of track circuits used to record train movements, the layover time measured by OCS records do not correspond exactly to what an observer at the platform level would record. From a set of detailed field observations matched to OCS records, it is calculated that OCS records overstates platform times compared to what would be observed at Alewife by 70-90s, with a mean of 75s. The layover statistics presented here are therefore reduced by 75s from OCS records in order to provide values relatable to operating staff experience.

of departing headways greater than 6-minutes compared to the Baseline. This is a reasonable result; although drop-backs improve reaction times under specific circumstances, this only represent once facet of a complicated terminal system. Of course, this finding is specific to the current operating conditions where time switching ends is not a constraint in the majority of cases, and should not be extrapolated to the case of lower headways and layover times. Additionally, it is worth emphasizing that no other dispatching procedures were changed to complement the introduction of drop-backs.

Table 5-4: Alewife Southbound Departure Headways (min)

Time Period	Study Period	Median	Mean	SD	95th Pctl
AM Peak (7:00 - 9:00 AM)	Baseline	4.3	5.3	2.4	9.7
	Drop-back Pilot	4.5	5.1	2.2	9.4
PM Peak (4:00 – 6:00 PM)	Baseline	4.6	5.2	2	8.9
	Drop-back Pilot	4.2	4.8	1.8	8.2

5.2.2 Out-of-Order Trains

Lower minimum layover times due to drop-backs can also aid reducing delays due to the reordering of trains at Alewife, which occurs when northbound branch trains merge out of sequence at JFK. Reordering of branch trains is executed by breaking the normal first-in-first-out procedure; the out-of-order train is "locked in" at Alewife and the "following train" is allowed to proceed first (**Figure 5-1**). The reordering time is then defined by how fast the following train can be processed ("critical dwell"). The locked-in train is less critical as a result of the extended layover time which results from being skipped in order. This procedure typically results in a large southbound gap ("critical headway").

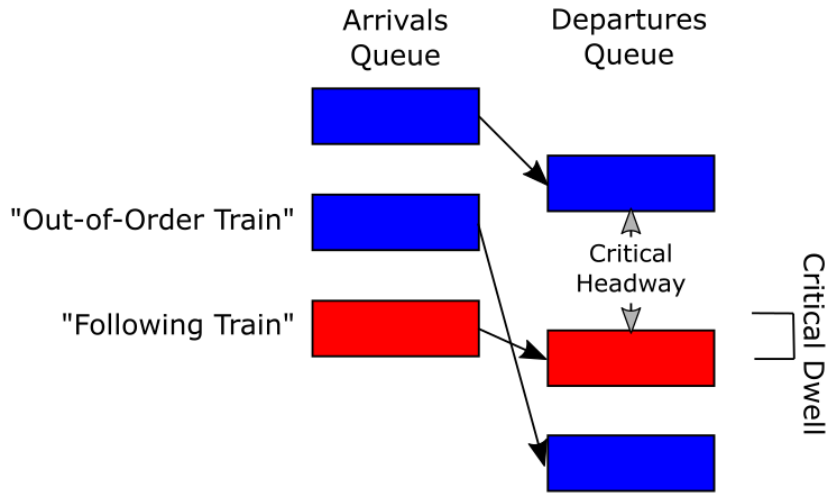

Figure 5-1: Processing Diagram for Out of Order Trains. Color denotes train branch.

During the pilot, the mean layover time[3] for the following train ("critical dwell") was reduced to 1:42, compared to 3:07 during the Baseline, a reduction of 79s (42%). The median departure headway for the following train ("critical headway") also dropped by 90s from 6:57 to 5:27, significantly reducing the impact of reordering on southbound service. With only 21 observations of reordering during the Baseline and 12 observations during the Pilot, and many factors influencing the headways of these trains, these results are intended to anecdotally rather than statistically support the power of drop-backs in disrupted circumstances. However, the results are reasonable based on the expected effect of drop-backs. Subject to further study, reduced reordering delay at Alewife due to drop-backs could contribute secondary benefits by allowing dispatchers to reorder more trains at Alewife rather than delaying trains in-line at the JFK merge.

As discussed previously, dispatching delays can become amplified over the course of the line (Section 2.4 - Headway Stability), lending importance even to small improvements in dwell reduction while reordering.

5.3 Effect on Northbound Queuing

Drop-backs are expected to reduce terminal queuing, especially in cases when the terminal is operating near or at-capacity. In this case, despite the terminal operating below theoretical capacity, significant terminal congestion is present today (as shown in Section 2.3.1), so reductions in queuing are anticipated. This section discusses the effect of the Pilot on northbound queuing in terms of running times and passenger benefits.

5.3.1 Running Times

Terminal area running times are shown in **Table 5-5** for the Baseline and pilot periods. The analysis limits are defined as departing Porter Northbound to arrival at Alewife, as explained in Section 2.3.1. During the Pilot, the mean running time through the terminal area dropped by 1.0 minutes (15%) in AM and 0.8 minutes (12%) in PM, with both changes significant at the 95th percent confidence level.

Table 5-5: Porter Northbound Departure – Alewife Arrival Running time (min)

Time Period	Study Period	Median	Mean	SD	95th Pctl
AM Peak	Baseline	6.1	6.5	2.4	11.2
(7:00 - 9:00 AM)	Drop-back Pilot	5.3	5.5	1.5	8.2
PM Peak	Baseline	6.5	6.7	1.8	9.8
(4:00 – 6:00 PM)	Drop-back Pilot	5.5	5.9	1.5	8.8

[3] Adjusted for OCS difference.

The pattern of arrival headways at Porter Northbound was examined to confirm congestion improvements were not due to incidental northbound improvements. The distribution of headways arriving at Porter Northbound are shown in **Table 5-6**. The difference in mean headway between the Baseline and Drop-back Pilot period is not significant at the 95th percent confidence interval in either the AM or PM peaks. Minimal changes of no more than 0.2 minutes are seen in other summary statistics. Finding minimal changes in headways entering the terminal area, it is concluded that reductions in terminal queuing can be attributed to improvements in processing at Alewife rather than to changes to the arrival regime.

Table 5-6: Porter Northbound Arrival Headways (min)

Time Period	Study Period	Median	Mean	SD	95th Pctl
AM Peak	Baseline	3.4	4.4	2.5	9.4
(7:00 - 9:00 AM)	Drop-back Pilot	3.4	4.4	2.4	9.2
PM Peak	Baseline	3.2	4.2	2.1	8.6
(4:00 – 6:00 PM)	Drop-back Pilot	3.3	4.2	1.9	8.4

5.3.2 Passenger Benefits

This section will discuss the estimation of passenger benefits resulting from the reduction in northbound queuing discussed above. Passenger train loads are estimated using the procedure in Section 2.5.2, and the passenger weighted mean running times are determined for Baseline and pilot trains in the terminal area. Estimation of train loads is an important step because trains arriving with high headways will be biased in the direction of large passenger volumes, although are unlikely to face congestion delays compared to bunched arrivals. The resulting mean northbound peak hour passenger savings are shown in **Table 5-7**, grouped by origin and destination station.

Overall, passenger savings are estimated as 11.2 hours for passengers traveling in the AM peak hour and 35.3 hours for passengers traveling in the PM peak hour. The magnitude of per-passenger savings is 60% higher in the AM compared to PM, but passenger demand is significantly lower, at about 20% of PM. *Considering the additional staff hours required to implement drop-backs are only about 120-minutes per hour*[4]*, these improvements in northbound travel times alone are substantial, although the primary anticipated passenger benefit of drop-back is in improved reliability southbound headways.*

[4] At 14-trains per hour with alternating branches, each operator must wait two additional headways (60/14 minutes), so staff hours are 14*2*(60/14) = 120 minutes

Table 5-7: Drop-back Pilot Mean Northbound Passenger Savings

Time Period	Origin	Destination	Hourly Passengers	Per Passenger Savings (s)	Total Savings (hrs)
AM Peak Hour (8:00-9:00 AM)	Davis	Alewife	122	-26	-0.9
	South of Porter (incl.)	Alewife	473	-61	-8.0
	South of Porter (incl.)	Davis	339	-25	-2.3
	All	All	934	-43	-11.2
PM Peak Hour (5:00-6:00 PM)	Davis	Alewife	30	-30	-0.3
	South of Porter (incl.)	Alewife	2,196	-47	-28.6
	South of Porter (incl.)	Davis	2,466	-9	-6.5
	All	All	4,692	-27	-35.3

5.4 Discussion

5.4.1 Operator Reaction Times Effect on Schedule Recovery

The expected schedule recovery achieved at different levels of delay can be calculated using a simple deterministic analysis at the current and future headway. For this analysis, it is assumed that reaction times are equal to the median observed reaction times for the corresponding duration of layover prior to ring-off (**Table 5-2**) under the given scenario. It is assumed that the previous train departs on schedule, and that the current train's departure bell rings off on-schedule despite the late arrival. Scheduled layover times of 6-minute and 4-minutes are assumed for the current and future headway, respectively.

Given the above assumptions, **Figure 5-2** shows the expected departure headways for trains arriving to the terminal with different levels of delay, between 0 to 3-minutes. Findings from the simple analysis confirm that a reliable 3-minute headway is infeasible assuming no change to the Baseline operations; for a relatively minor, 3-minute delay, Baseline reaction times would enable recovery of only 1-minute. Observed reaction times from the Drop-back Pilot are more promising; for a 3-minute delay, median Drop-back Pilot reaction times would enable 2-minutes of recovery. Given the minimum reaction times of approximately 1-minute observed under the pilot (Section 5.1), schedule recovery up to 3-minutes is possible with redefined crew procedures and an agency culture change which places a high value on on-time performance.

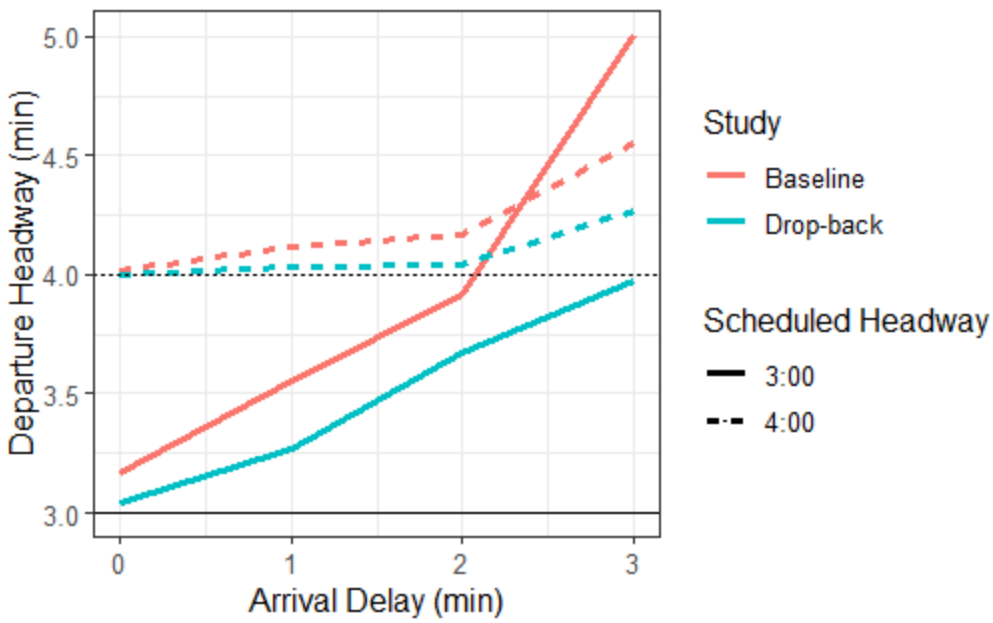

Figure 5-2: Departure Headway Under Delay Assuming Mean Reaction Times

As the maximum line capacity is approached, terminal schedule recovery becomes increasingly important, since there is less slack time available to absorb delays. When there is a delay in the system today, trains can bunch up behind the delayed train, making use of the available slack. In at-capacity operations with minimal slack-time, delays cannot be absorbed and trains may need to be pulled out of service to open up gaps. This reinforces the importance of terminal schedule recovery time at a 3-minute headway. The practical capacity of Alewife can be tested further through simulation, using reaction time distributions identified from this pilot to inform different operational scenarios.

5.4.2 Implications of Median Reaction Times

This section discusses the implication of reaction times on near-capacity operations. As identified in Section 5.1.1, the median southbound crossover slot assignment time is 180% of the minimum observed slot time, implying poor utilization of the crossover capacity. This extra crossover utilization by southbound trains results in increased queuing delays for northbound trains entering Alewife, since under normal procedures, the next arrival is assigned to the platform of the last departure, so all southbound movements conflict with the following northbound movement. One reason for the long reaction times is that operators at Alewife receive a visual/audible cue to depart only once their crossover slot is open and assigned exclusively to the departure movement.

Reducing the magnitude of the median reaction time (and thus crossover occupancy time) could have significant implications for current northbound queuing and future terminal capacity. One possible solution is to install communication infrastructure so that in addition to the departure

bell, operators receive a 30s warning before the scheduled departure, prompting them to close doors and prepare for departure.

5.4.3 Complementary Crew Management Techniques

Implementing drop-back crewing in tandem with other crew management changes can help to reduce the added resources of drop-backs, while still maintaining similar benefits. These changes which include interlining operators between the two branches, and shifting some operator relief time to Alewife are described below.

Interlining Operators

Interlining operators is one crew scheduling approach which could reduce the added resources required for drop-back crewing. Interlining operators is defined in this context as having operators switch branches during their layover at Alewife. This scheduling technique would mean operators only need to break at Alewife for one additional *trunk* headway rather than the base case of one *branch* headway (typically two trunk headways with some exceptions). In this case, a single overboard "extra" operator would be required rather than one for each branch, effectively halving the added resources compared to the baseline. Interlining, could provide ancillary scheduling benefits – scheduled half cycle times on the Braintree branch are between 11-13 minutes longer than that of the Ashmont branch, and subject to further investigation, interlining the two could allow for more efficient utilization of operator's maximum permitted shift time, with less spread time. Modern transit scheduling software, such as HASTUS, used by the MBTA, are capable of incorporating constraints such as break and finish locations into the crew scheduling problem.

Informal discussions with some operators and supervisors suggest that this could be an acceptable way to schedule shifts. Already today operators may take their breaks at a location other than their "home" terminal (for example, an operator may start their shift at Braintree, preform 2.5 round trips, and take a lunch break at Alewife), meaning alternating branches or break locations is not unfamiliar. Alternating between branches would also reduce monotony for operators. Given the potential benefits, this strategy should be made a priority for future study.

Changing break location to Alewife

Currently, operators are scheduled to take layovers approximately twice as long at the southern terminals (Ashmont and Braintree) compared to Alewife (**Table 1-3**) during peak periods. As demonstrated by the Drop-back Pilot, extending operator layover times at Alewife can help trains turn around more quickly at Alewife, increasing terminal capacity and reducing queuing delays. Subject to labor regulations, the layover time can be decreased at the southern terminals simultaneously with the increase due to drop-backs at Alewife. Assigning the longer layover time to the shared northern terminus will help to maximize capacity where most needed. However, some layover time should also be maintained at the southern terminals for the purpose of schedule recovery.

5.4.4 Related Crew Management Techniques

In addition to the complementary crew techniques described above, drop-backs provide similar benefits to other crew management techniques familiar to the MBTA, including swing-ons and two-person train operations, as discussed below.

Swing-Ons

"Swing-ons" are a crew management technique where a new operator takes control of a train from the current operator, either at an in-line relief point or at a terminal. When performed at a terminal, swing-ons function similarly to drop-backs in that terminal platform time can be drastically reduced because the operator is relieved from having to walk the length of the platform. Swing-ons are commonly used by the MBTA at Alewife during shoulder periods when operators begin or end shifts. Experience with swing-ons contributes to operator familiarity with drop-back operations. *Swing-ons may also offer one explanation for the very low layover times (90-120s) occasionally seen at Alewife during the Baseline (Section 5.1.3).* The primary differentiation from drop-backs is that while swing-ons are used mainly during shoulder periods, drop-backs are intended as a routine schedule recovery mechanism throughout peak periods.

Two-person train operations

Senior operators noted that drop-backs provide similar benefits at the terminal as having two-person train operations (TPTO), which were discontinued on the Red Line in 2012 (Tribone, 2013) in favor of single person train operations (SPTO). In the case of TPTO, operators did not have to switch ends, because roles would be reversed between the operator and conductor at each terminal. However, this technique does not address the layover problem in the case where either requires a long break, perhaps to use the restroom. Drop-backs require significantly less personnel than TPTO for similar or better improvements to terminal schedule recovery.

5.4.5 Drop-back Compliance Issues

Compared to earlier MBTA dispatching pilots by Fabian (2017) and Maltzan (2015) which contended with compliance issues, this Drop-back Pilot was largely free of such issues. As compared to the earlier pilots, one key difference was that operator layover time was only extended and never shortened compared to the Baseline. In addition, the departure instructions for the trains was governed by the same departure bell as it is during routine operations, meaning the operators noticed no difference in dispatching procedures, as in Fabian's experiment on the MBTA Green Line where a departure bell was introduced alongside the real-time dispatching intervention.

During the Drop-back Pilot, certain atypical circumstances occasionally resulted in confusion over train assignments and long reaction times. These incidents primarily arose in case of scheduled swing-ons, due to lack of clarity over whether the swing-on operator should wait an additional departure before swinging on, or the drop-back operator should drop-back an additional departure. These type of issues would be eliminated through the scheduling process. Out-of-order

trains were another issue which caused some confusion during the Drop-back Pilot. In order to correct the order while preserving the operator's branch, the operator on the out-of-order train must drop-back an extra departure (3 rather than 2), while the following operator will drop-back one fewer train (1 rather than 2). For these reasons, it is important to develop efficient communication protocols to prevent delays under these potentially confusing circumstances.

5.4.6 Operator Reactions to Drop-back Pilot

During the pilot period, drop-backs were met with reluctance by some operators because they were asked to diverge from their scheduled work assignment. This was primarily due to the fact that, for the pilot, drop-backs were implemented without changing the base operator schedules, and any operator lateness was compensated through overtime pay. Operators typically experienced increases in shift end-time of approximately 8-16 minutes (one or two branch headways), depending on if they dropped back once or twice during their shift. In some cases, operators had to take this slight shift increase out of their lunch breaks to ensure they started the second half of their shift on-time, which contributed to the frustration.

Ultimately, building drop-backs into each operator's schedule would reduce operator lateness, since all trips would be more likely to depart on schedule. Drop-backs would also mean that operators delayed in their arrival to Alewife are no longer pressured to turnaround as fast as possible, resulting in a less stressful work environment. After explaining the operator-facing benefits, operators and supervisors were generally agreeable that drop-backs would work well in the long term if written into the schedule.

5.5 Conclusion

The Alewife Drop-back Pilot results demonstrated the value of drop-backs for achieving current and future capacity objectives. Improvements were measured by the reduction in operator reaction times under pressing circumstances; for trains with arriving to Alewife just 60-120s prior to the ring-off, the reaction time dropped by an average of 62s. The minimum observed layover times (bottom 1%) were also found to be 30-60s shorter during the Pilot, and occurred without extended reaction times, an important result for schedule recovery under a future 3-minute frequency. These large improvements were evident during branch reordering; the median gap following reordering dropped from 6:57 to 5:27, significantly reducing the impact on southbound service. Northbound queuing to enter the terminal also decreased substantially, with estimated savings 35-passenger hours (or 27s per affected passenger) per PM peak hour alone. At an added cost of just *2* staff hours per hour of drop-back operations, these passengers savings are substantial. These findings support the value of drop-backs both as a tool for enhancing service reliability today and to enable higher terminal capacity in the future.

Chapter 6. Davis Southbound Holding Pilot Results

The Davis Station Southbound Holding Pilot implemented a target headway-based holding strategy to reduce variations in initial southbound headways caused by the Alewife Terminal crossover SGR. The details of the intervention were presented in Section 4.2. This section will explore the result of the Pilot. Section 6.1 details the performance impacts at Davis Station. Section 6.2 discusses the impact on southbound trunk line performance in terms of headway variability, running times, and bottleneck occupancy time. Lastly, Section 6.3 discusses passenger impacts of the Pilot, including changes in crowding, wait times, and journey times. Some general discussion on the holding pilot staffing, operator reactions, and compliance will be included in Chapter 7.

6.1 Performance Impacts at Davis

In this section, impacts of the holding on performance at Davis will be explored in terms of the ability for holding to reduce initial headway differences between the platforms and the ability of officials to accurately implement the holding regime.

6.1.1 Holding Duration and Frequency

During the pilot, staff were asked to focus on the task of headway management rather than record keeping, in order to remove distractions and improve adherence to the principal task of controlling headways. Thus, prior to analysis of pilot results, held trains must be identified post-hoc from the automatically recorded track circuit data. A set of heuristics was developed for this purpose, based on the pilot instructions and in-field observation, as detailed in Appendix 3.

Following holding identification, the holding time is calculated for held trains as the difference between the median Baseline station occupancy and the actual station occupancy. The distribution of holding time is shown in **Figure 6-1**. Holding time was distributed between 15-120s, with similar distributions in AM and PM. Both distributions are bi-modal with peaks around 30s and 50s. The mode around 50s of holding is interpreted as the holding time necessary to bring the minimum-spaced trains (approximately 2-minute departure-departure headway without holding) to a minimum headway of 3-minutes, as instructed by the pilot. The mode around 30s may denote holding for P1 trains, which generally were not held, but may have experienced a short hold if dispatched at a minimum possible headway behind a P2 train.

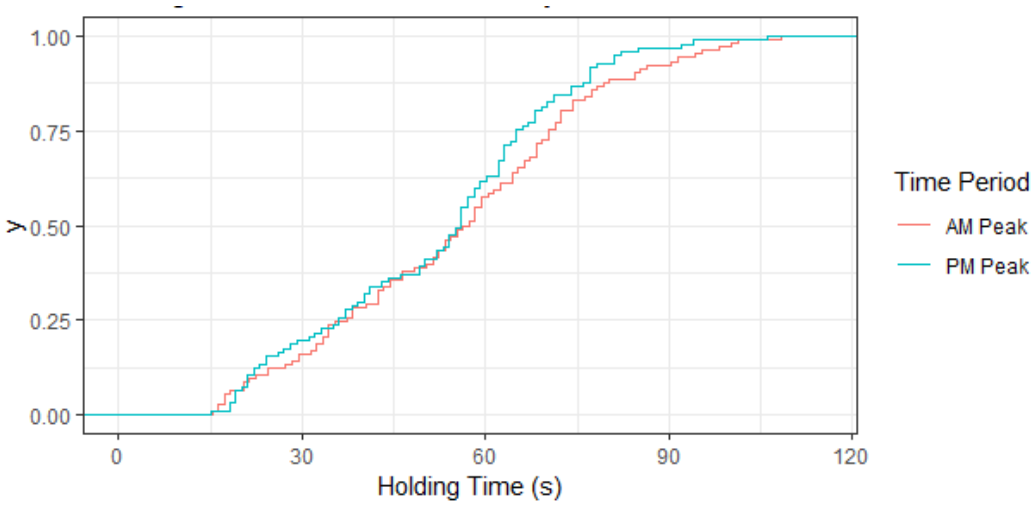

Figure 6-1: Holding Times – Cumulative Distribution

6.1.2 Holding Frequency by Platform

In order to limit complexity of the pilot, holding was not restricted to trains from a single Alewife dispatch platform. However, the holding instructions were devised in such a way to target primarily P2 trains, which make up a disproportionate number of short headways at Davis. This section will explore the outcomes of holding by platform.

As anticipated, holding was distributed unevenly between platforms, with 79% of held trains originating from Platform P2. **Table 6-1** shows the proportion of held trains and mean holding times by dispatch platform. Overall, 53% of P2 trains were held compared to only 14% of P1 trains were held. The proportions were not significantly different in AM and PM. The average holding time including only held trains was 41s for P1 trains and 58s for P2 trains. Including all trains, the average holding time was just 6s for P1 trains and 31s for P2 trains. As anticipated, these figures demonstrate that P1 trains were for the most part unimpeded by holding, and that the holding achieved the intended purpose of re-centering P2 trains between P1 dispatches.

Table 6-1: Davis Holding Statistics by Platform

			Holding Time (s)	
Alewife Dispatch Platform	N	Trains Held (%)	All Trains	Held Trains
P1 Trains	242	14%	6	41
P2 Trains	249	53%	31	58
Combined	491	34%	18	54

6.1.3 Headways Departing Davis

This section will explore the distribution of headways departing Davis, with specific discussion for held trains only, by dispatch platform, and generally for trunk and branch headways.

All Trains

Overall, holding resulted in significantly more uniformity between Davis Southbound departures as shown in **Table 6-2**. The improvements are highlighted by comparing the spread between the 15th and 85th percentile headways; in other words, the middle 70% of headways. The pilot resulted in a 1.4 minute (38%) reduction in the spread between 15th to 85th percentile departure headways from Davis in the AM and a 0.9 minute (26%) reduction in the PM. During the pilot there was also a reduction of 0.2 minutes (AM) and 0.5 minutes (PM) in the median departure headway.

Table 6-2: Trunk Headway Percentiles Departing Davis (min)

Time Period	Study Period	5th Pctl	15th Pctl	Median	85th Pctl	95th Pctl
AM Peak (7-9 AM)	Baseline	2.3	2.6	4.3	6.3	8.0
	Davis Holding	2.9	3.3	4.1	5.6	6.5
PM Peak (4-6PM)	Baseline	2.2	2.5	4.2	5.6	7.0
	Davis Holding	3.0	3.2	3.7	5.5	6.5

In contrast to the significant trunk headway improvements as a result of Davis holding, branch headways are minimally effected by the holding intervention. Under regular operations with alternating branches and dispatch platforms, each branch headway is the sum of consecutive P1 plus P2 headways. **Table 6-3** contains the trunk headway percentiles after departing Davis, showing no change in the median headways for either branch, as anticipated.

Table 6-3: Branch Headway Percentiles Departing Davis during PM Peak (4-6 PM) (min)

Branch Destination	Study Period	15th Pctl	Median	85th Pctl
Braintree	Baseline	6.1	7.7	10.0
	Davis Holding	5.9	7.7	9.9
Ashmont	Baseline	7.1	8.4	11.5
	Davis Holding	6.9	8.4	11.6

Held Trains

A histogram of departure headway for held trains only is shown in **Figure 6-2**. In the AM, the mean departure headway for held trains was 3:35, with a standard deviation of 17s. In the PM, the mean departure headway for held train was 3:20, with a standard deviation of 12s. In both periods the mean departure headway for held trains is approximately 20s greater than the target headway measured by stopwatch; this is as expected based on the time it takes to close doors and reach the first track circuit downstream of the station. The range of departure times for held trains is much greater in the AM due to large number of boarding passengers at Davis during this time which impacted the operator's ability to promptly close the doors. In the PM doors were generally

closed within a few seconds of instruction. The range of departure headways shows the importance of considering environmental conditions in devising holding targets.

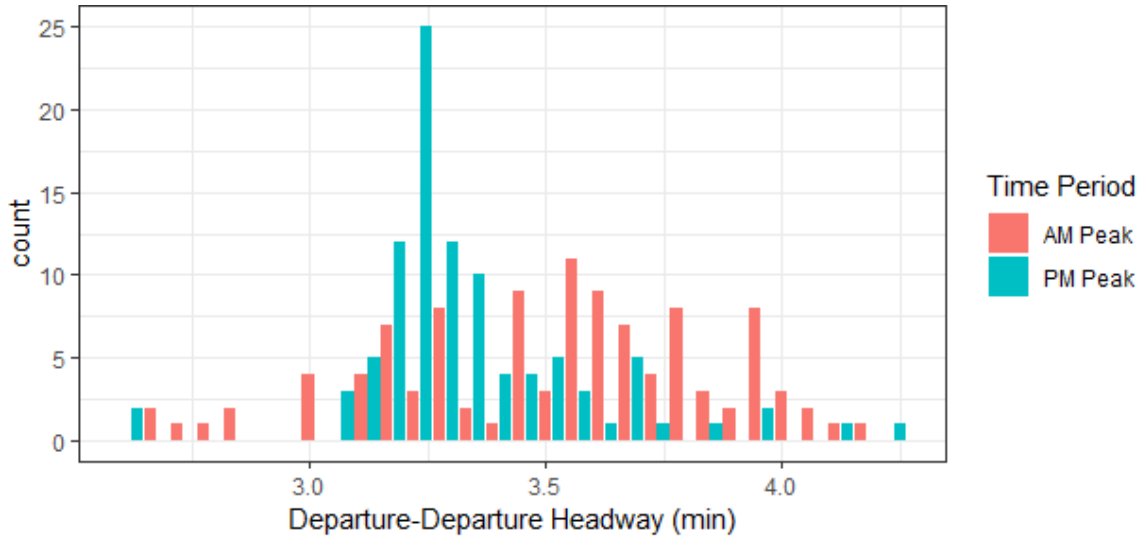

Figure 6-2: Departure Headway for Held Trains

By Platform

One of the main objectives for the pilot was to increase uniformity between Alewife P1 and P2 platform dispatches. Due to the design/condition of the crossover departing Alewife, P1 trains are limited to 10mph, while P2 trains may travel 25mph. Small variations in initial headways caused by the speed difference can have large impacts on the performance of the trains as shown in Chapter 2.

Figure 6-3 compares the fraction of trains within 1-minute of the scheduled 4-minute headway by time of day and dispatch platform between the Baseline and pilot periods. It demonstrates how holding short headways, primarily from P2, simultaneously reduces large P1 headways, and increases the overall fraction of headways departing within 1-minute of scheduled. During the AM peak, the fraction of headways within 1-minute of scheduled increased from 46% during the Baseline to 68% after holding. This was driven by a decrease in short headways less than 3-minutes (24% in Baseline vs. 6% in pilot) and a resulting simultaneous reduction in long headways greater than 5-minutes (30% in Baseline vs. 25% in pilot). It should be emphasized that the reduction of low headways did not come at the expense of raising the median headway, but rather by lowering long headways. Improvements in the PM peak are similar to those in AM. Such improvements in headway regularity would be masked by looking only at the number of headways *below* a certain threshold (i.e. the scheduled headway), emphasizing the importance in devising disaggregate performance metrics.

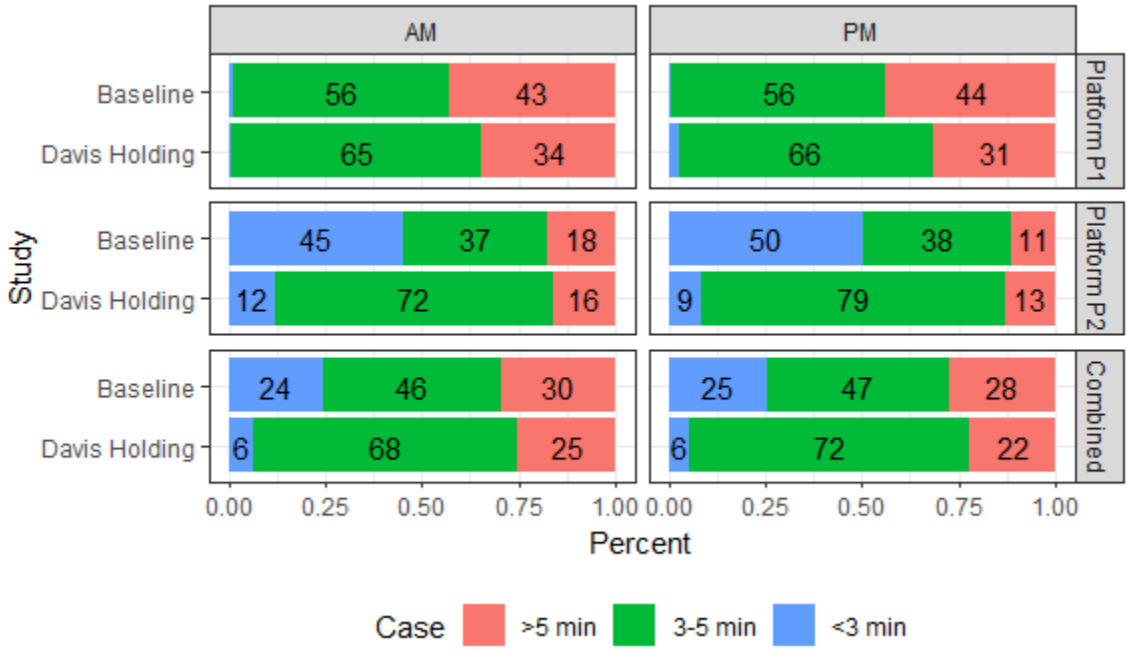

Figure 6-3: Davis Departure Headways Within 1-Minute of 4-Minute Headway (%)

6.2 Impacts on Line Performance

The previous section explored impacts of holding on headways departing Davis, finding reduced variability overall, and a reduced gap between platforms. This section will explore the impacts of Davis Holding on trunk line performance, namely headway variability, headway patterns by dispatch platform, running times, and congestion impacts to the terminal.

6.2.1 Effects On Southbound Line Coefficient of Variation

The coefficient of variation (standard deviation divided by the mean) of headways is a common measure for communicating service regularity in transit analysis. **Figure 6-4** shows the coefficient of variation (CV) of headways at each southbound trunk station. The dashed line represents the passenger weighted line-headway CV. The CV of headways is calculated for all trips dispatched from Alewife during the peak periods, regardless of what time they reached the downstream point.

In AM, a substantial, 25% reduction in passenger weighted line-CV of headway is observed. The improvements are similar throughout the line, with the reduction in CV for all southbound trunk stations between 19-29%. In PM, the CV is initially reduced by 24% at Porter, but declines throughout the line, reaching parity with the Baseline by Park Street and seeing an increase of 2-7% compared to the Baseline between Park and Andrew Southbound. Overall, the PM saw a 5% improvement in passenger weighted line-CV of headways for trunk passengers. The interpretation suggested for the smaller observed improvements in CV in the PM is that of having

lower passenger demand in the early stops (**Figure 1-2**), which reduces the divergent outcomes by platform which occur in AM, before trains reach the downtown headway bottlenecks.

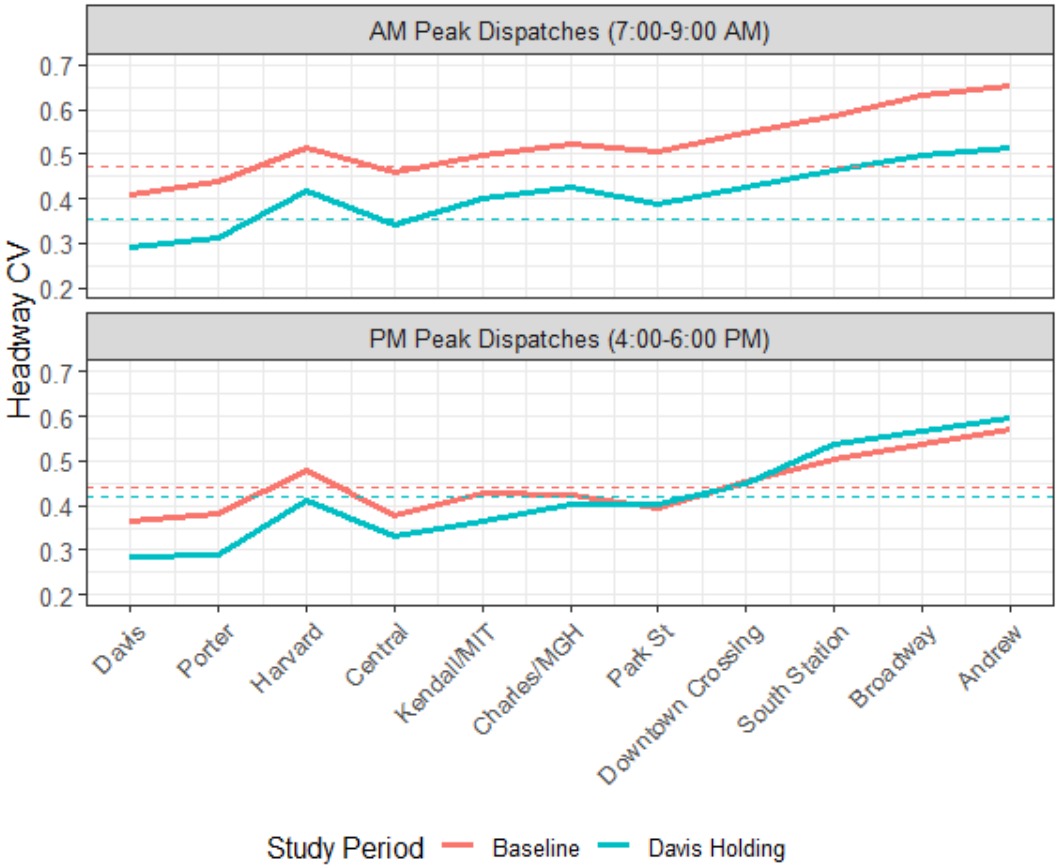

Figure 6-4: Coefficient of Variation of Headways – Peak Period Dispatches

6.2.2 Impact of Dispatch Platform On Line Performance

Minor differences in initial headway can have significant, lasting impacts on a trip's performance over the line; a long initial headway will result in long dwell times, overcrowding, and further delays, while a short initial headway will result in slow running times due to bunching and limited passenger demand. In the case of the Red Line, these initial differences are deeply engrained due to the speed differential between the two Alewife dispatch platforms.

Figure 6-5 and **Figure 6-6** below compare the headway outcomes for peak hour dispatches from Alewife during the Baseline and Davis Holding periods during the AM and PM peaks, respectively. The thick line represents median headways, while the extents of the shadow represents the 25th-75th percentile (1st and 3rd quartiles, respectively) headways. Therefore, the shadow symbolizes the headway spread for the middle 50% of all trips. Trips with an absolute headway change of more than 3-minutes over a single stop are removed from the analysis (8% of

trips in AM, 5% in PM) in order to focus on routine differences by platform rather than those caused by incidents. Only the busiest hour of each peak is included, as small initial differences will exhibit the biggest long term impact at- or near capacity. Guide lines are placed at 3 and 5-minutes on the y-axis to aid in visualizing the changes.

In the AM peak hour (**Figure 6-5**), inbound passengers demand is substantial in the first few stops, with approximately 70% of all boardings occurring by Central, causing the initial headway differences to self-reinforce in the Baseline scenario. Because P1 trains start off with a longer headway, they are more likely to experience high passenger demand and long dwell times causing them to fall further behind as indicated by the wide 50th-75th percentile distribution for P1 trains (**Figure 6-5** , Baseline). Conversely, P2 trains experience little demand between Davis and Central due to their short headway to the train in-front. Bunching is limited somewhat by headway bottlenecks downtown, but headway outcomes are locked in by the initial differences in passenger demand. In contrast to the Baseline, trains in the Davis Holding Pilot experienced similar initial headways, and thus saw near identical outcomes downtown.

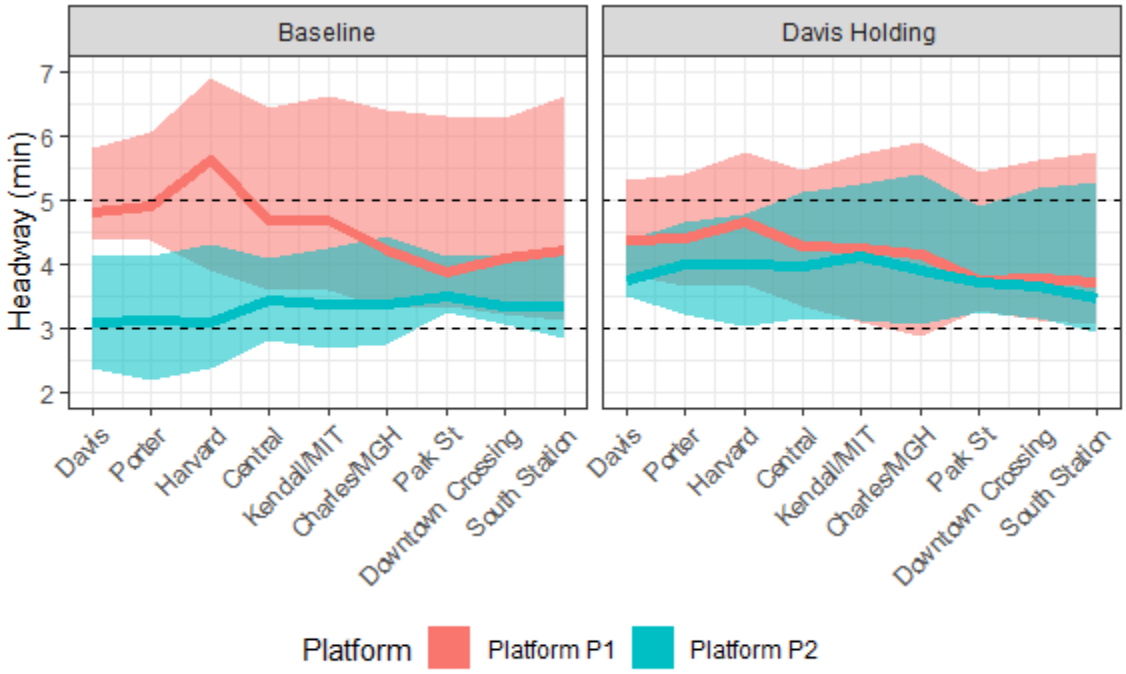

Figure 6-5: Headway Distribution by Alewife Dispatch Platform (7:30-8:30 AM Dispatches)

In the PM peak hour (**Figure 6-6**), the dispatch platform does not result in the same extreme headway discrepancies downtown as during the AM peak hour. The interpretation offered is that southbound passenger demand is much lower in the first stops (only 30% of total boardings by Central), so large initial P1 headways are less susceptible to long dwell times and delays. In the Baseline, it is observed that although trains started with distinct headway patterns, the bottlenecks at Harvard and Park create more uniform spacing, before the bulk of the passenger demand downtown. In the pilot period, initial differences are reduced, but minimal change is noted in the

resulting downtown headways. In this case the holding pilot provides the most benefit in improving trunk wait times between Davis and Harvard.

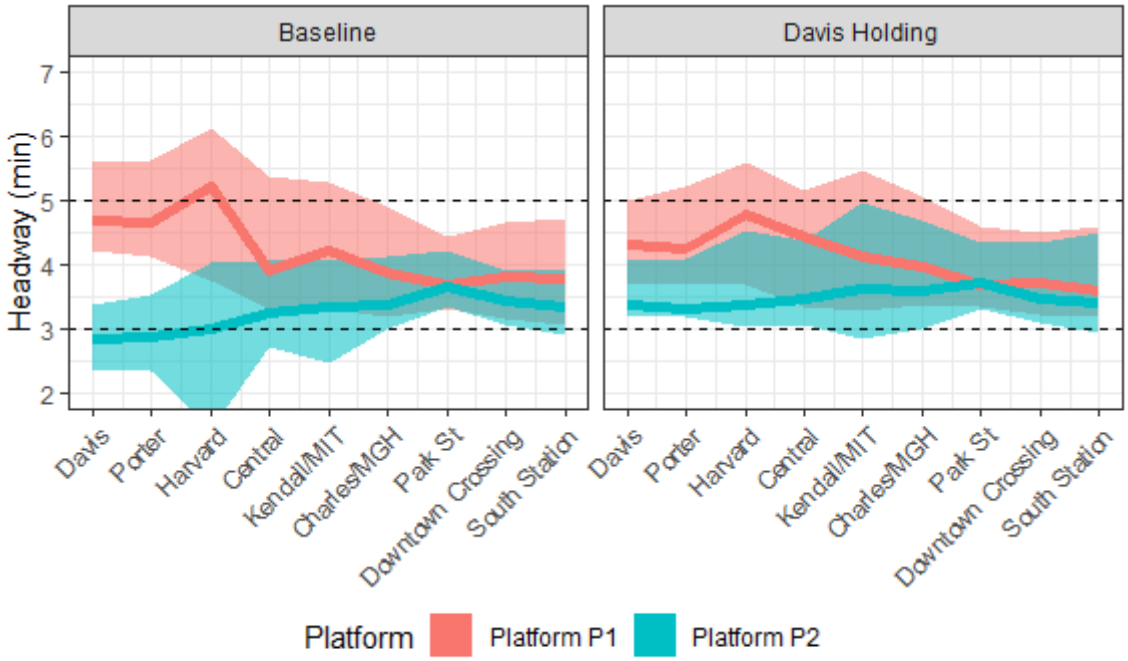

Figure 6-6: Headway Distribution by Alewife Dispatch Platform (4:30-5:30 PM Dispatches)

During Davis Holding, the initial headway differences were shown to reduce substantially, causing more similar downstream outcomes for P1 and P2 trains. In the AM, Davis Holding significantly improved ultimate outcomes by evening out the passenger demand in the first stops. In the PM, although the holding successfully reduced initial headway differences, the effect on downstream headways is less substantial due to the more diffuse demand patterns.

6.2.3 Running Times

The following sections will examine the total running time as well as the running time on individual links to identify effects of the holding pilot. By reducing the frequency of very short headways and train bunching, the holding pilot is expected to reduce overall running times from Davis.

Running times from departing Alewife to arrival at downstream trunk stations are shown in **Table 6-4**. Despite any initial delay due to holding, trains during the Pilot reached downtown in a similar running time from Alewife as compared to Baseline trips after adjusting for speed code changes between Harvard and Kendall which occurred after 02/19 (**Appendix 8**). Mean running time from Alewife to Downtown Crossing decreased by 15s in the AM Peak and increased by 15s in the PM peak. Neither difference in means is statistically significant. Running time benefits are also not apparent from Alewife to Central. Examining the 85th percentile statistics, the largest changes are seen in the PM, where there was no change at Central, yet an increase of 0.7 minutes

by South Station. This large increase over just a few stops suggests that the root cause was from incidents downtown, rather than an effect of the Pilot.

Although these summary statistics display no running time improvements, contrary to what was anticipated, it is still an important result that running times from Alewife did not increase, despite any initial delay from Davis Holding, showing that minor headway corrections can be performed without any impact on operator lateness.

Table 6-4: Running Time from Departing Alewife to Arrival at Downstream Stations (min)[5]

Extents	Time Period	Study Period	Median	Mean	CV	85th Pctl
Alewife - South Station	AM	Baseline	23.1	23.3	0.13	25.4
		Davis Holding	22.7	23.1	0.12	25.4
	PM	Baseline	22.6	23.3	0.12	26.7
		Davis Holding	22.6	23.5	0.13	27.4
Alewife - Central	AM	Baseline	12.7	13.0	0.19	14.0
		Davis Holding	12.8	13.2	0.12	14.5
	PM	Baseline	11.9	12.2	0.12	13.2
		Davis Holding	12.2	12.3	0.07	13.2

6.2.4 Running Time Through Line Bottlenecks

Improving the regularity of headways departing Davis is expected to improve the running time for trains passing through other bottlenecks; especially Harvard Southbound. Because headways below approximately 3:00 cannot be maintained through Harvard with the current track and dwell time conditions, holding at Davis to regulate headways prior to this bottleneck is expected to significantly improve train congestion through Harvard. The primary benefit of performing this holding at Davis rather than naturally at Harvard is that it reduces imbalances in passenger load that would otherwise develop over the high demand stations of Davis, Porter, and Harvard.

Figure 6-7 shows the change in running time distribution through Harvard Southbound during the AM Peak Period. The PM Peak is not shown, despite similar reductions in running time, because the bottleneck is primarily a concern during the AM when passenger volumes at Harvard are higher. Running time on the approach block and the station block are included separately as well as combined to show in which component the changes occurred. Overall, station occupancy times were reduced by approximately 30s at the 90th percentile, and the fraction of observations below 180s occupancy rose from approximately 92% to 97%. These reductions are tied to the reduction in low headway arrivals to Harvard, which experience slow speed codes on the approach

[5] Adjusted for speed code changes between Harvard and Kendall after 02/19 (Appendix 8)

block. These improvements highlight how headway regularity can increase the throughput and robustness of operations at key bottleneck stations.

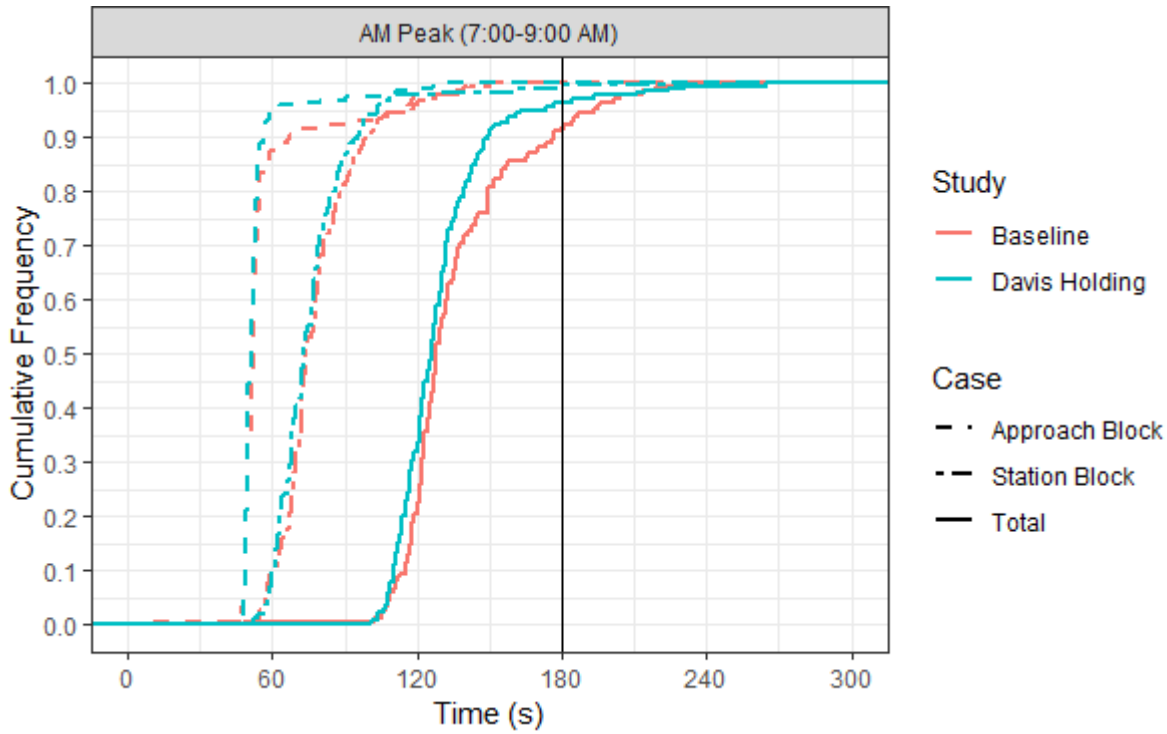

Figure 6-7: Changes in Harvard Southbound Block Occupancy Due to Davis Holding

In contrast to Harvard, other southbound bottlenecks, including Charles/MGH and Park Street exhibited only minor improvements to station occupancy times. This finding can be understood as a result of the gradual increase in headway variability over the line following holding, observed in **Figure 6-4**. This suggests that if holding is to be used to improve bottleneck capacity, the control point should be located in close proximity before the bottleneck.

6.2.5 Northbound Running Time into Alewife

One risk inherent in holding trains just after the terminal is that of impeding future dispatches, thereby reducing the terminal processing rate and increasing delay to arriving trains. In order to avoid this situation, the target headway for the holding strategy was selected as 3:00/3:15 (leading to a 3:20/3:35 headway, as seen in Section 6.1.3), which is close to the minimum headways observed departing Alewife from historical observations. Selecting a target headway close to the maximum terminal throughput minimizes the risk of knock-on delays.

The total running time from departing Porter Northbound to departing Davis Southbound ("turnaround time") is examined for any negative impacts of Davis holding on terminal throughput. The boundaries for the total turnaround time were chosen to include effects of queuing

to enter/exit the terminal, and the holding at Davis. Davis Holding days coinciding with Alewife drop-backs or Andrew Holding were excluded from the analysis due to the improvements to terminal capacity anticipated by those pilots in terms of improving processing time and reducing the frequency of bunched arrivals.

The mean and median time for each component of the total turnaround time are shown in **Table 6-5**. Overall, despite an increase in Davis Southbound dwell time due to the holding, the mean total turnaround time between departing Porter Northbound and Departing Davis Southbound dropped by 0.7-minutes in the AM and 1.6-minutes in the PM peak. The differences in means are significant at the 95th percent confidence level in both periods. Similar improvements are also seen in the median values. The increase in dwell times at Davis Southbound appears to be offset by savings in the northbound direction entering Alewife. As the pilot was designed, these findings confirm that minor holding at Davis Southbound have negligible impact on Alewife congestion and dispatching procedures under the current operating conditions.

Table 6-5: Components Alewife-Turnaround Time from Departing Porter Northbound to Departing Davis Southbound (min)

		Mean		Median	
Time Period	Component	Baseline	Davis Holding	Baseline	Davis Holding
AM Peak (7:00-9:00 AM)	1. Porter-Alewife TT	6.8	6.1	6.1	5.8
	2. Alewife Layover	6.3	6.2	5.9	5.8
	3. Alewife - Davis	1.9	2.0	2.0	2.1
	4. Davis Dwell	1.1	1.5	1.1	1.2
	5. Total	15.0	14.3	14.4	13.8
PM Peak (4:00-6:00 PM)	1. Porter-Alewife TT	6.5	5.3	6.4	4.9
	2. Alewife Layover	6.2	5.7	5.9	5.3
	3. Alewife – Davis TT	1.9	2.0	2.1	2.1
	4. Davis Dwell	1.0	1.3	0.9	1.1
	5. Total	14.6	13.0	14.3	12.6

6.3 Passenger Impacts/Improvements:

Passenger impacts of poor headway control are experienced primarily in two ways; passengers crowding and excess journey time. Passengers originating at Davis and beyond are expected to experience positive outcomes in both of these categories. Passengers originating at Alewife who are traveling short distances are the only group who are expected to be negatively affected due to extra holding time, however, longer trips from Alewife are expected to benefit from improved line headway regularity in terms of crowding and running times. Alewife passengers represent around 15-18% of southbound trunk boardings in the AM peak, and only 2-3% in the PM peak. Davis Southbound customers will benefit the most due to holding from improved wait

time reliability, while other southbound passengers will benefit from reduced overcrowding, travel time, and wait time.

6.3.1 Passenger Train Loading and Effective Utilization

Passenger train loads and effective utilization for the Baseline and pilot periods are estimated using the methodologies described in **Section 2.5.**

Results for the AM peak hour are shown in **Figure 6-8**. As expected, the Davis Holding pilot produced more uniform passenger loading, as measured by the spread between the 1st and 3rd quartile expected passenger load at Central, the peak southbound load point in the AM peak. The main driver in this difference is the reduced variability of headways between Davis and Central, where approximately half of all southbound AM boardings occur.

This significant improvement in loading diversity is attributed to the regularization of P1 and P2 headways. During the Baseline, the median effective utilization of P1 trains was 39% higher than for P2 trains; during the Pilot the difference between platforms dropped to just 8%. In terms of maximum crowding levels, train loads at Central experienced a reduction of 82 passengers (11%) at the 75th percentile, and 46 passengers (5%) at the 90th percentile.

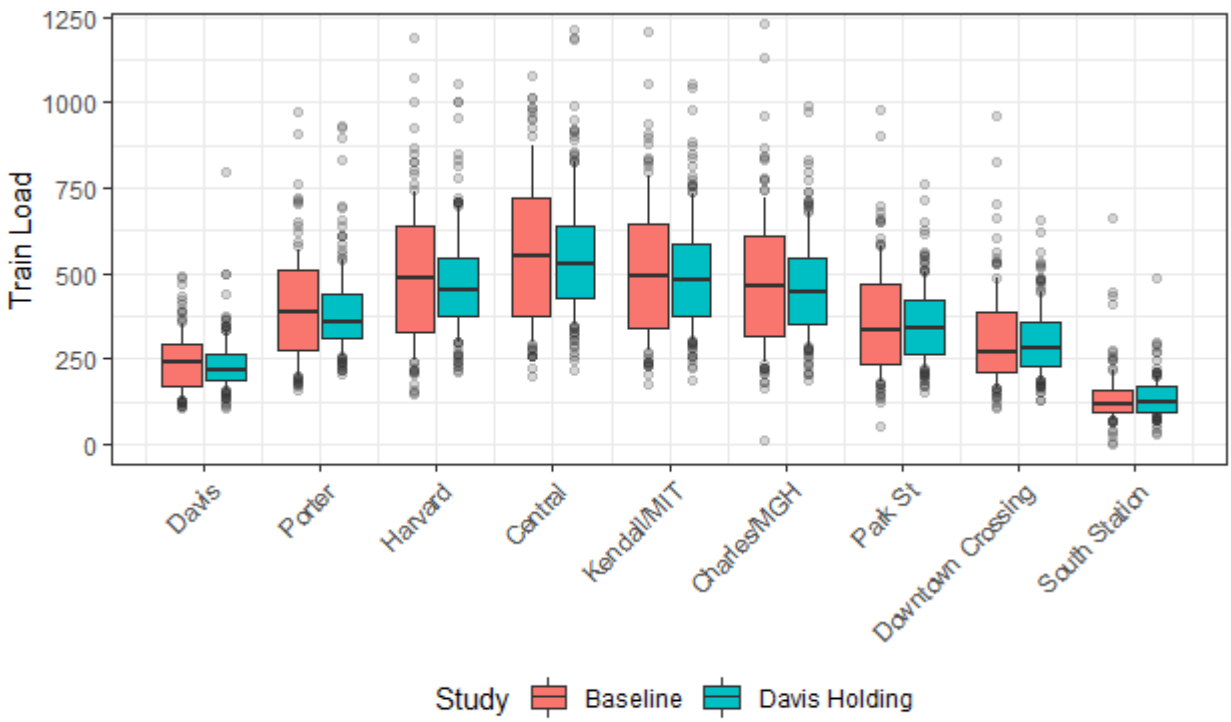

Figure 6-8: Expected Train Load by Station (Trains Departing Davis 7:45-8:45 AM)

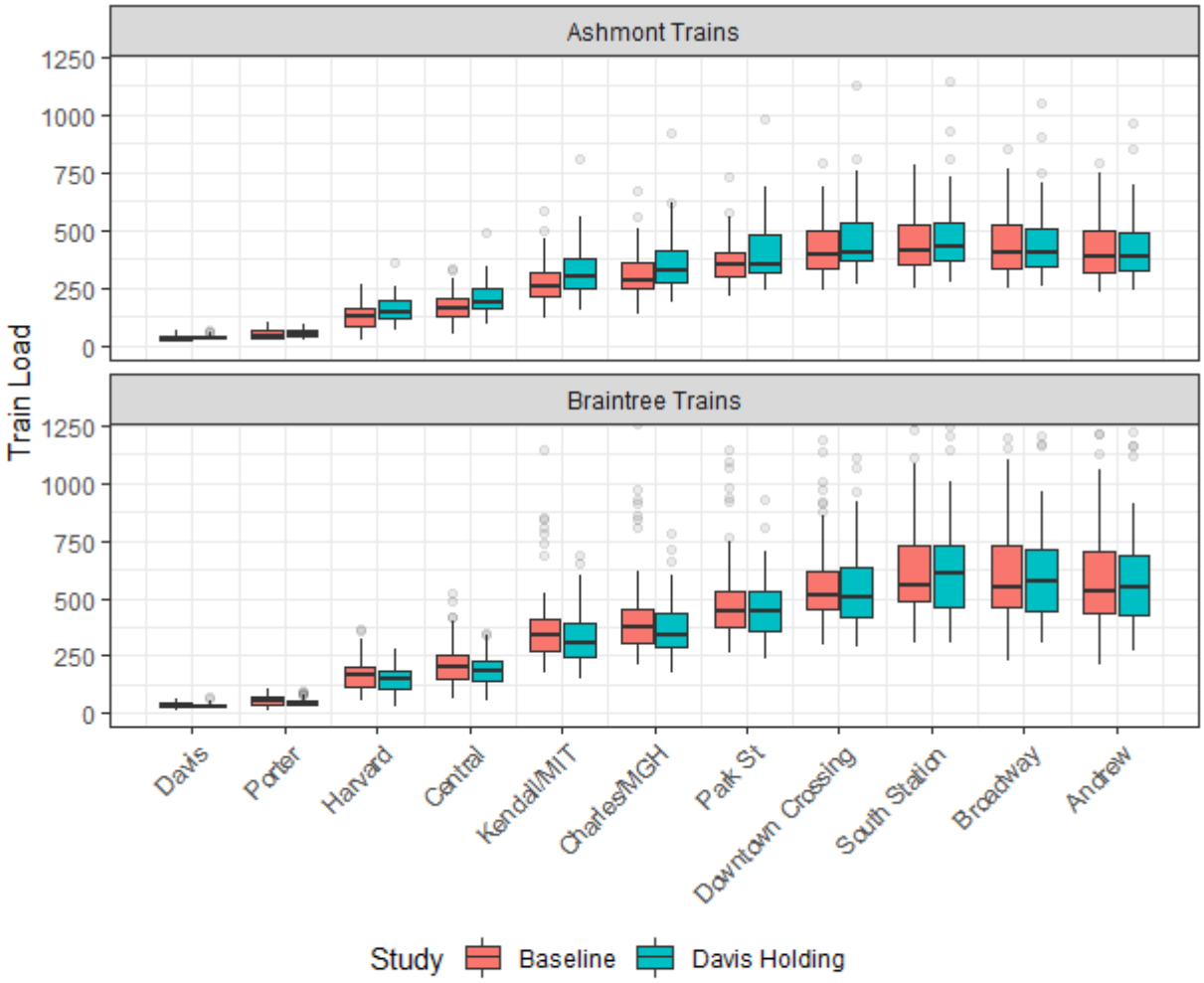

Figure 6-9: Expected Train Load by Station (Trains Departing Alewife 4:30-5:30PM)

In the PM peak, peak train loads are heavily influenced by the branch headway (the inter-arrival time between two trains of the same branch), rather than by the trunk headway. At the PM peak southbound maximum load point of South Station, approximately 80% of in-vehicle passengers are destined to the branches (compared to 8% at Central in the AM peak). As shown previously, the Davis holding pilot served mostly to regularize trunk headways, with minimal change to branch headways. Taken together, the aforementioned headway and demand patterns means the P1/P2 platform differences and Davis holding will have a much smaller impact on PM passenger peak loading compared to AM.

Figure 6-9 shows the expected train load at all southbound trunk stations during the PM peak, comparing the Baseline and holding pilot. The methodology is the same as in **Figure 6-8**, however in this case train loads are shown separately by branch due to the significant impact of branch destination on passenger loading.

Holding is observed to have had minimal impact on effective utilization, or on peak passenger loads, as measured at South Station, compared to the large reductions in peak loading diversity seen in the AM. Based on the previous discussion of headway and demand patterns in the PM, this is an anticipated result. Elsewhere along the trunk, a minor increase in passengers loads for Ashmont trains (and corresponding decrease for Braintree trains) is observed between Davis and Charles. This is not an anticipated effect of the holding pilot; this change likely denotes that more Ashmont trains operated on the P1 platform than on the P2 platform during the pilot as compared to the Baseline, however the assignment of branches to platforms occurs stochastically day to day (and hour to hour), unaffected by the holding pilot.

As implemented, the pilot procedures were intended to reduce *trunk* headway variability, which was accomplished; a future holding strategy during the PM peak could consider regulating branch headways to improve peak PM passenger loading.

6.3.2 Wait Time

A comparison between southbound trunk wait times for the Baseline versus Davis Holding Pilot is shown for each trunk station in **Figure 6-10**, (with branch passenger wait times in **Appendix 5**). The trunk line-wide passenger weighted results are also included in **Table 6-6**. The methodology for calculating average and 85th percentile wait times both at a single stop and as a passenger weighted line-wide metric was described in **Section 2.5,** and assumes a uniform passenger arrival rate.

Davis Holding resulted in large improvements to southbound trunk wait times in the AM Peak, and small but consistent improvements in the PM peak. In the AM, mean wait times improved by approximately 18s (10%) line-wide, with an improvement of similar proportion at the 85th percentile of about 36s (12%). Improvements in AM peak wait times were mostly consistent along the line, with slightly larger improvements south of downtown, although experienced by few passengers. In the PM, wait time improvements for trunk passengers where much smaller, with a line-wide mean improvement of less than 10s (5%). In this case, the wait time improvements are most significant in the first three stops, and taper off towards downtown. Trunk wait times appears to improve again after downtown, although the cause is not clear, given the lack of distinction downtown. Wait times improvements for branch passengers were much smaller, with average improvements ranging between 0-5% for the downtown stations representing the large majority of branch passenger boardings. This is explained by the earlier finding, that regulating trunk headways did little to improve branch headways, which was not the focus of this pilot.

Table 6-6: Passenger Weighted Line Wait Time Metrics

Time Period	Branch Destination	Average		85th Percentile	
		Baseline	Pilot Change (%)	Baseline	Pilot Change (%)
AM	Trunk	2.8	-0.3 (-10%)	5.0	-0.6 (-12%)
PM	Trunk	2.5	-0.1 (-5%)	4.4	-0.3 (-6%)
	Braintree	4.5	0.0 (0%)	7.7	-0.2 (-2%)
	Ashmont	5.2	-0.2 (-4%)	9.0	-0.5 (-6%)

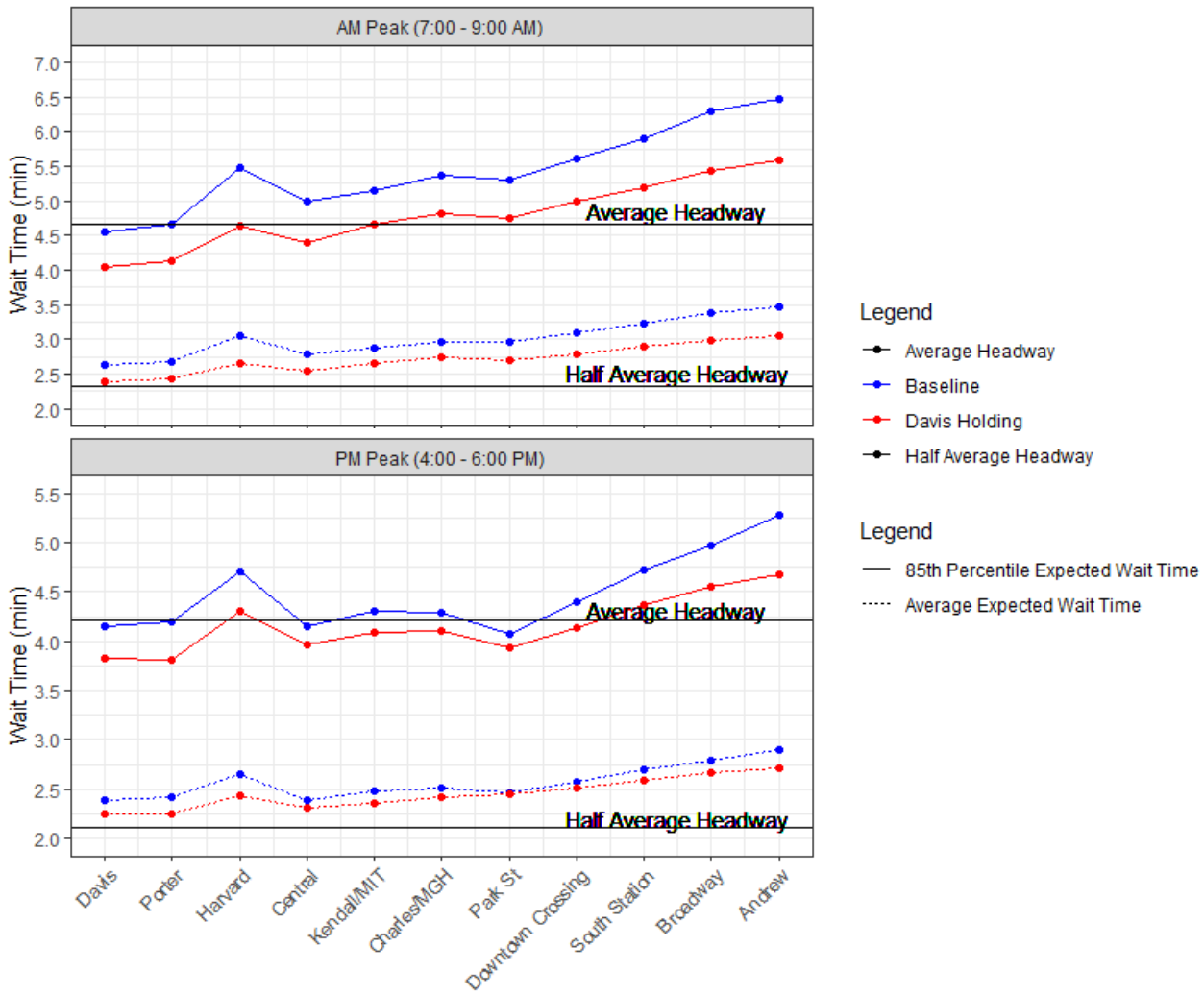

Figure 6-10: Southbound Trunk Wait Time Changes[6]

[6] The *half average headway* and *average headway* are included as reference points, representing respectively the *average* wait time and *maximum* wait time in the hypothetical case with zero headway variation at today's scheduled headway.

6.3.3 Journey Time

Journey times were calculated for all station origin-destination pairs ("OD pairs") in the southbound direction, in order to determine the impacts of the holding pilot on different passengers. Journey times are calculated using an AVL based measure (Wood, 2015), which uses only observed headways and running time between stations pairs. Wait time is calculated assuming passengers arrive at a uniform rate throughout the analysis period and board the first train to arrive with a compatible destination. The calculations represent the journey time distribution as would be experienced by a southbound passenger arriving to the station randomly during the peak hour, from arriving at the origin station to alighting at the destination station. Only the Red Line portion of the journey is included. Branch passengers are not included in the analysis on the basis that Davis Holding exhibited minimal impact on branch headways and running time beyond downtown. Journey times are corrected for minor speed codes changes which occurred during between the Baseline and pilot periods as described in Appendix 8.

The difference in mean journey times between the Baseline and pilot for trains departing Davis during the peak 2-hour period are shown in **Figure 6-11** (AM) and **Figure 6-12** (PM). A summary of the passenger weighted mean journey time changes is included in **Table 6-7**. Negative values indicate a reduction (improvement) in journey times during the Pilot period. Examining the full OD matrix in addition to summary values allows us to highlight specific passengers which experienced the most notable effects, which summary statistics would otherwise mask.

In the AM, passenger journey time savings were substantial, with estimated savings of 197 passenger hours per peak period. Passengers traveling from the northern end of the line to downtown experienced the most significant journey time improvements due to a combination of improved wait times and more consistent travel times. Only passengers traveling between Alewife and Porter experienced statistically significant increases in mean journey time; all other Alewife passengers experienced either journey time improvements, or no change.

In the PM, trunk passengers experienced a minor mean journey time increase of 2s per passenger, or 8 passenger hours per peak period. Generally, passengers seem to experience journey time benefits in early stations, with a large concentrated journey time increase between Central and Charles/MGH. This large shift in journey time improvements over relatively few stations is unexpected and appears to be caused by cascading delays caused by a handful of incidents at downtown stations during the pilot period. Without simulation analysis, it is hard to say whether the pilot contributed to these cascading delays, however, it is not anticipated, since more regular headways should contribute to more robust incident recovery.

Table 6-7: Summarized Passenger Weighted Mean Journey Time Changes[7]

Time Period	Grouping	Hourly Passengers	Passenger·Hours	Per Passenger (s)
AM Peak Period	Benefit	21,797	-197.4	-33
(7:00-9:00 AM)	Disbenefit	47	0.1	6
	All	22,815	-197.3	-31
PM Peak Hour	Benefit	2,546	-10.2	-14
(4:00-6:00 PM)	Disbenefit	4,648	18.6	14
	All	13,701	8.4	2

[7] Origin-Destination pairs without statistically significant changes in mean journey time are assumed to experience zero change for the purpose of the summary calculations.

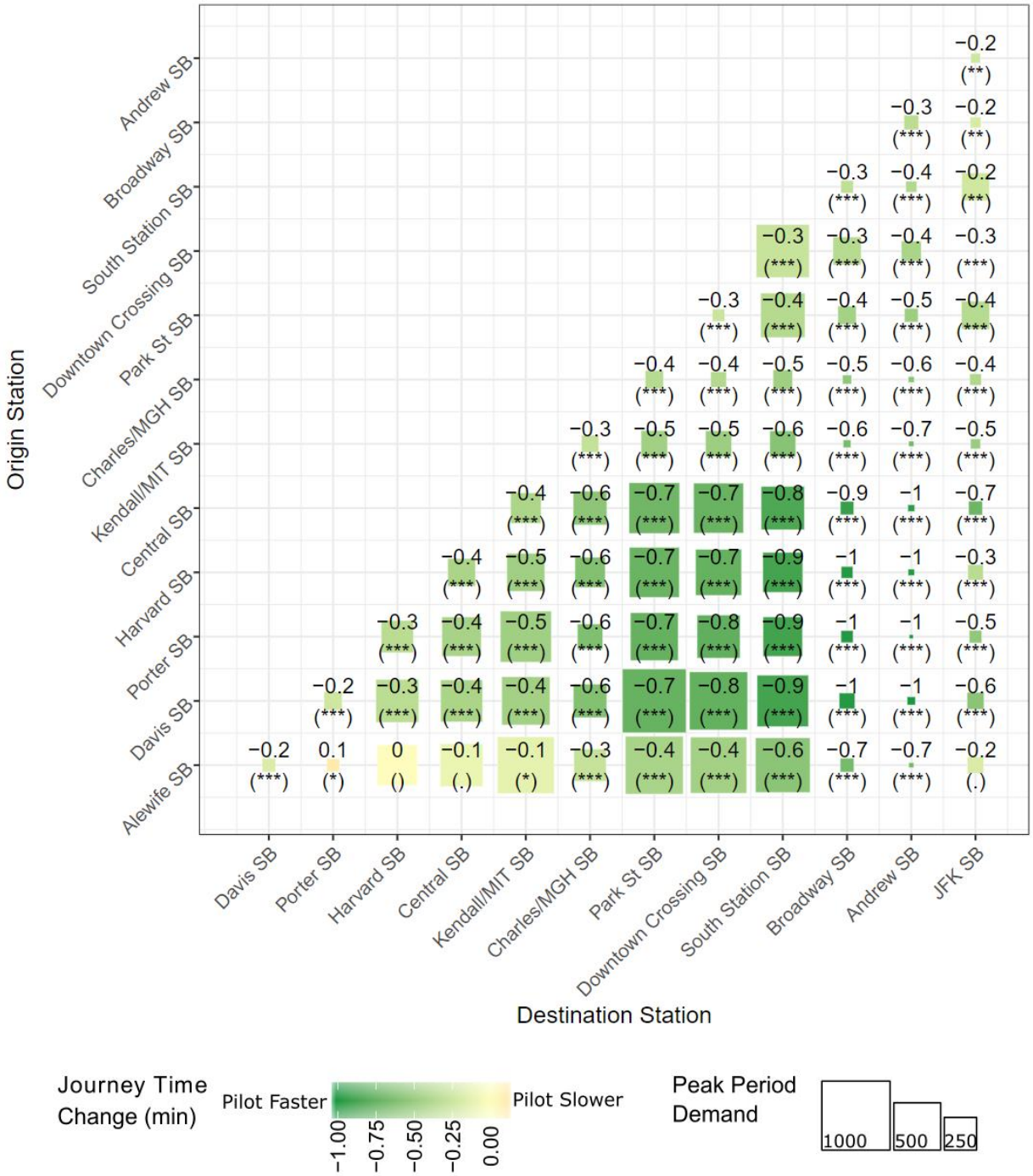

Figure 6-11: AM Peak Period Mean Journey Time Changes (min)

Significance codes: () = p ≤ 1.0, (.) = p ≤ 0.1, (*) = p ≤ 0.05, (**) = p ≤ 0.01, (***) = p ≤ 0.001

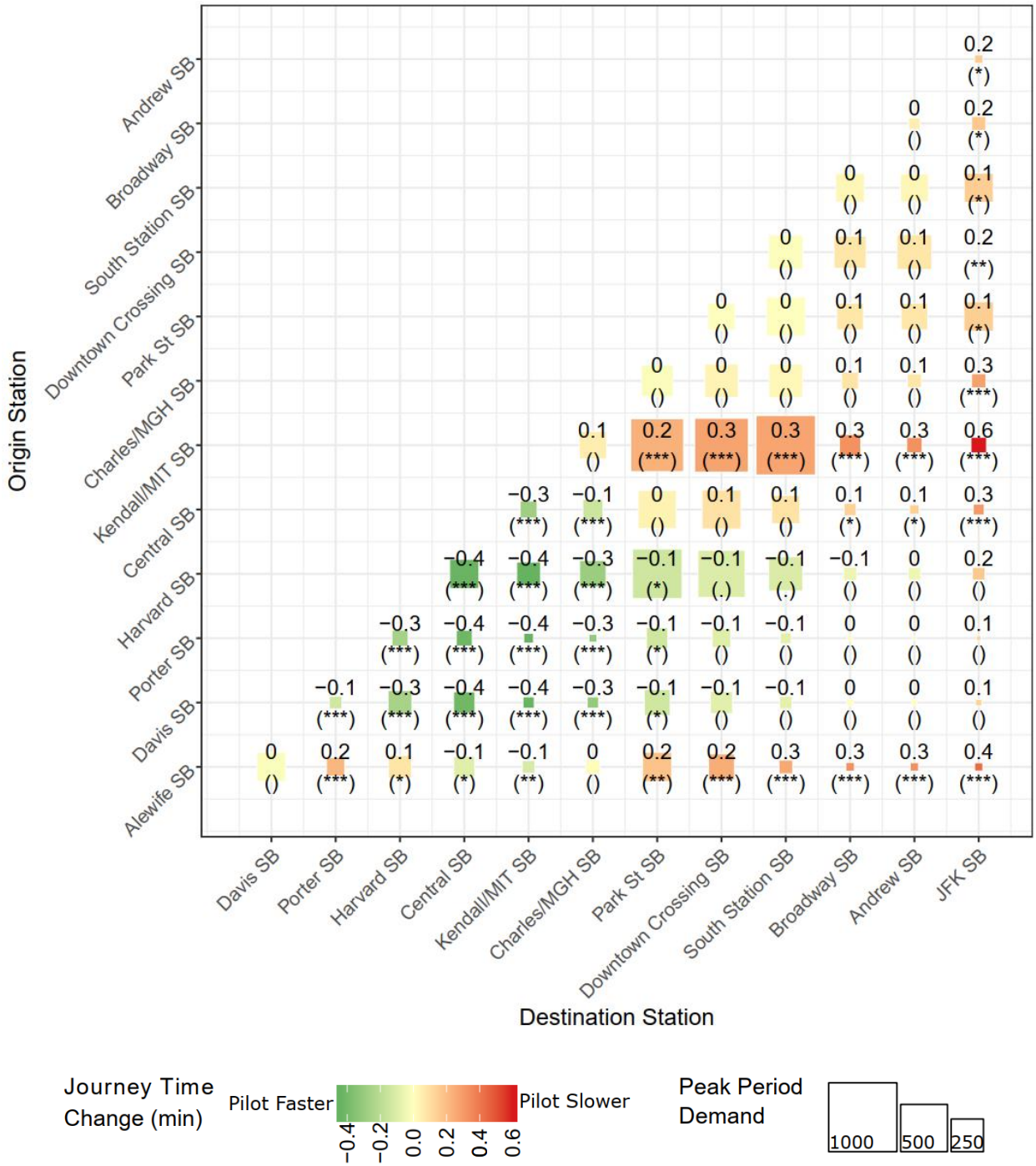

Figure 6-12: PM Peak Period Mean Journey Time Changes (min)

Significance codes: () = p ≤ 1.0, (.) = p ≤ 0.1, (*) = p ≤ 0.05, (**) = p ≤ 0.01, (***) = p ≤ 0.001

6.4 Conclusion

The Davis Holding Pilot demonstrated that significant steps to improve headway regularity and passenger experience can be made without major technological or workforce changes. In summary, trains were held to a minimum headway of 3-minutes at Davis Southbound, affecting mostly Platform 2 dispatches from Alewife which depart the terminal at a higher speed and tend to bunch up behind P1 dispatches.

In the AM, significant improvements to trunk headway regularity was observed, with a 25% line weighted headway CV improvement and 10% reduction in average wait times. Journey times improved significantly, with almost all passengers experiencing benefits due to the pilot, and an estimated 197 hours of passenger benefit per peak period. Headway regularity also contributed to a reduction in uneven passenger loading, resulting in a 20% reduction in CV passing the peak southbound load point.

In the PM, improvements were much smaller overall, attributed to the large proportion of passengers with branch destination, which were unaffected by the holding intervention. Cascading train congestion due to incidents occurring downtown also appeared to overshadow benefits from headway regularity downtown, causing overall a small increase in mean journey times of 8s per trunk passengers. The benefits to train crowding were also less notable in PM, again due to the effects of branch demand.

Overall, changes in line running times were found to be negligible after correcting for changes in speed codes which occurred during the pilot, which is an important result since even including the holding, running time did not increase. The benefits to southbound passengers were also realized without any major impacts to northbound passengers; northbound congestion approaching the terminal decreased during the pilot.

Chapter 7. Andrew Station Northbound Holding Pilot

A variable target headway-based holding strategy was implemented at Andrew Station Northbound in order to reduce northbound trunk headway variability following the branch merge. The specific instructions and motivations were discussed in detail in Section 4.3. This chapter discusses the results of the Andrew Holding Pilot in three sections; performance impacts at Andrew (7.1), performance impacts for the northbound trunk line (7.2), and passenger impacts in terms of crowding, wait times, and journey times (7.3). Finally, Section 7.4 offers some general discussion on the holding pilots implementation including compliance and staffing. Unless otherwise noted, the time frame used for the analysis of the Andrew Holding Pilot is trains passing through Andrew between 4:30-6:00 PM.

7.1 Performance Impacts at Point of Holding

This section will discuss the results of the Holding Pilot as measured at Andrew Station and in the immediately surrounding stations in terms of holding duration and frequency, resulting headways, and changes in running time.

7.1.1 Holding Duration and Frequency

Similar to the Davis Holding Pilot, a set of heuristics were developed based on the pilot procedures and in-field observations in order to identify from the train tracking data which dwell observations represent holds. The heuristics are based on the observed duration of station occupancy, and departure headway, and are discussed in detail in Appendix 3.

Following holding identification, the duration of each hold is calculated as the difference between the actual and median baseline station occupancy. The duration and frequency of holding for each target headway is shown in **Table 7-1** below. Overall, 48% of trains were identified as held. Including all trains, the mean holding time was just 32s; the median of 0s represents that under half of all trains were held. Among only trains that were held, the 34% of trains held to 3:00 had a mean holding time of 61s and the 14% of trains held to 4:00 had a mean holding time of 79s.

Table 7-1: Andrew Pilot Holding Time Statistics

Case	Fraction of All Trains (%)	Mean Hold (s)	Median Hold (s)	85th Pctl Hold (s)
Held to 3:00	34%	61	59	80
Held to 4:00	14%	79	71	116
All Trains	100%	32	0	78

The duration and frequency of holding by branch is shown in **Table 7-2**. As discussed in **Section 2.3.2**, branch merge headways can be extremely variable from day to day, resulting in the

wide variation in fraction of held trains by branch observed throughout the 7 pilot days. Between 21%-72% of Braintree trains and 44%-65% of Ashmont trains were held depending on the day. Overall, Ashmont trains were held more frequently and for a longer mean duration. It is not clear whether this is an artifact of the scheduling process, or simply a result of day to day variation as demonstrated in Section 2.3.2. During the PM peak, approximately 50% more passengers travel north of the branch merge on the Braintree branch compared to Ashmont – both volumes are limited compared to the high volumes traveling northbound from the downtown core.

Table 7-2: Holding Time by Branch

Branch	Range of Held Trains by Day	Overall Fraction of Held Trains	Range of Mean Holding Time by Day	Overall Mean Holding Time
Braintree	21% - 72%	43%	14s - 52s	28s
Ashmont	44% - 65%	56%	30s - 53s	40s

7.1.2 Headway Distribution Departing Andrew

As anticipated, the Andrew Holding Pilot caused significant improvements to headways departing Andrew in the form of less bunching and large gaps. The distribution of headways departing Andrew is shown in **Figure 7-1**, with descriptive statistics in **Table 7-3** and **Table 7-4**.

The proportion of headways below 3-minutes decreased significantly, from 30% during the Baseline to just 6.6% during the pilot, while the proportion of headways within 1-minute of the scheduled headway (4-minutes) nearly doubled from 39% to 68%. An independent samples t-test between the Baseline and pilot headway means shows no significant difference, meaning that the overall throughput was unaffected and that the holding only served to reduce variability, and did not delay departures overall.

In addition to the reduction in short headways, holding to 4-minutes ahead of a large gap significantly decreased the number of long headways. For example, the proportion of headways larger than 8-minutes (two times the scheduled headway) decreased from 4.8% to 1.4%, despite no change in the proportion of long headways at JFK (**Table 7-4**). The reduction in large headways (>8 min) at Andrew was found to be significantly lower than the proportion during the Baseline using an independent samples t-test for differences in proportions at the 95th % confidence level.

Table 7-3: Andrew Holding Departure Headway Statistics (min)

Study Period	Median	Mean	SD
Baseline	3.9	4.3	1.9
Andrew Holding	4.0	4.3	1.3

Table 7-4: Andrew and JFK Headway Proportions Before and After Intervention (%)

Station	Study Period	<3 min	3-5 min	>5 min	>8 min
Andrew Northbound	Baseline	30	39	31	4.8
	Andrew Holding	6.6	68	26	1.4
JFK Northbound	Baseline	27	38	34	5.1
	Andrew Holding	26	36	37	5.0

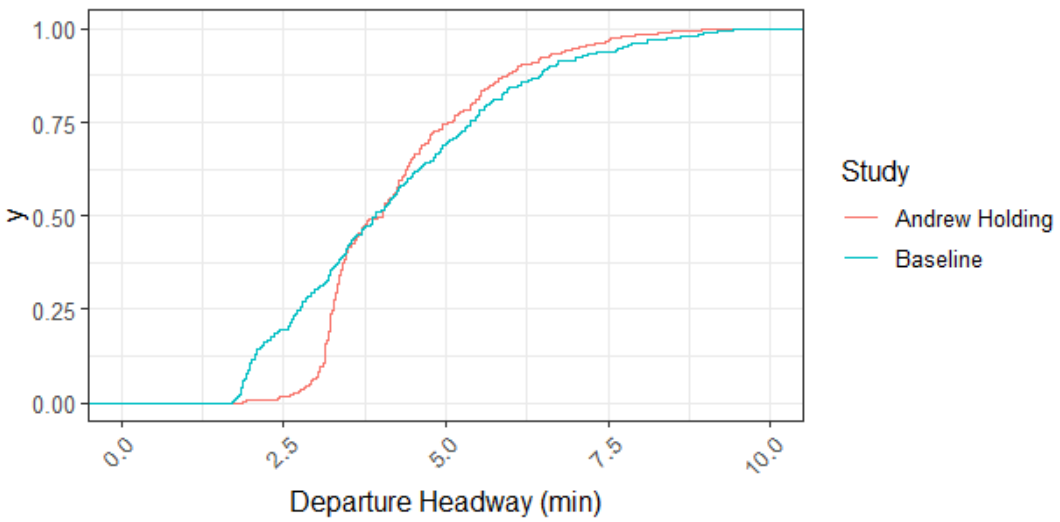

Figure 7-1: Andrew Departure Headways – Cumulative Distribution

7.1.3 Effect of Long Holds

The "long hold" target headway of 4:00 was intended to decrease the severity of the following, long headways. For trains identified as held-to-4:00, the following trains were selected, and the headways examined at JFK and at Andrew. As expected, the headway of these following trains improved significantly following the hold, with the median headways dropping from 7.5-minutes at JFK to 5.5-minutes at Andrew, while the mean decreased 1.4-minutes from 7.1-minutes to 5.7-minutes. These long holds contribute to the significant reduction in large headways greater than 8-minutes seen in the previous section.

In addition, the headway of the held-to-4:00 trains were compared to the headway of the following trains at Andrew, in order to check that trains were not "over-held", creating a longer headway in front of the held train than behind (this was one of the objectives in selecting the holding targets, as detailed in Appendix 3). It was confirmed that out of 28 trains identified as held-to-4:00, only 1 was over-held, validating the use of the next-arrival clocks for headway management, assuming proper precautions are taken in interpreting the predictions.

7.1.4 Changes in Running time - Approach to Andrew

The change in running time on blocks approaching Andrew can be used to measure any knock-on delays caused by the holding. Running times between arriving at JFK and departing Andrew Station are shown in **Table 7-5** for the Baseline and Andrew Pilot periods. These limits were chosen after finding no significant changes to travel time in the blocks approaching JFK Northbound.

Overall, trains during the pilot experienced an extra 0.6-minutes running time for a median train and 1.9 minutes for a 90th percentile train. This additional running time was concentrated at the Andrew Station Block where the holding occurred – congestion impacts between JFK-Andrew were less than 15s on average. This is an important result since it shows knock-on delays were rare.

Some delay to northbound branch passengers due to holding was deemed acceptable for this pilot, because northbound trunk passengers heavily outnumber branch passengers in the afternoon peak. In the future, the magnitude of the holding delay could be mitigated by coordinating arrivals to the merge earlier (either through real-time coordinated dispatching, or by regulating speed codes along the branches in real-time), reducing the frequency of bunched arrivals to the branch merge.

Table 7-5: Effect of Holding on JFK Arrival – Andrew Departure Running Times (min)

Analysis Extents	Study Period	Median	Mean	SD	90th Pctl
JFK Station Block	Baseline	1.0	1.1	0.3	1.3
	Andrew Holding	1.1	1.1	0.7	1.4
Departing JFK to Approaching Andrew	Baseline	2.0	2.0	0.2	2.1
	Andrew Holding	2.0	2.2	0.6	2.9
Andrew Station Block	Baseline	1.0	1.0	0.2	1.1
	Andrew Holding	1.4	1.6	0.6	2.4
Total	Baseline	4.0	4.1	0.4	4.6
	Andrew Holding	4.6	4.9	1.1	6.5

7.2 Impacts on Northbound Line Performance

This section explores impacts of the Andrew Holding Pilot on northbound trunk performance. The findings are discussed in terms of headway variability, line running times, improvements through station bottlenecks, and on the northbound running time into Alewife.

7.2.1 Effects On Northbound Line Headway Variability

An anticipated effect of the headway control at Andrew was that of reduced line headway variability, to the benefit of northbound trunk passengers. **Figure 7-2** shows the coefficient of

variation (CV) for northbound trunk headways during the Baseline and pilot periods, showing significant reductions in variability. Overall, the line passenger-weighted CV of headways between Andrew and Davis improved by 20%, from 0.50 to 0.40. The largest improvements occur departing Andrew, with a 37% reduction in CV. Large improvements are maintained through the busy downtown stations, with a 20-25% reduction in CV between South Station and Park Street, tapering off towards Alewife.

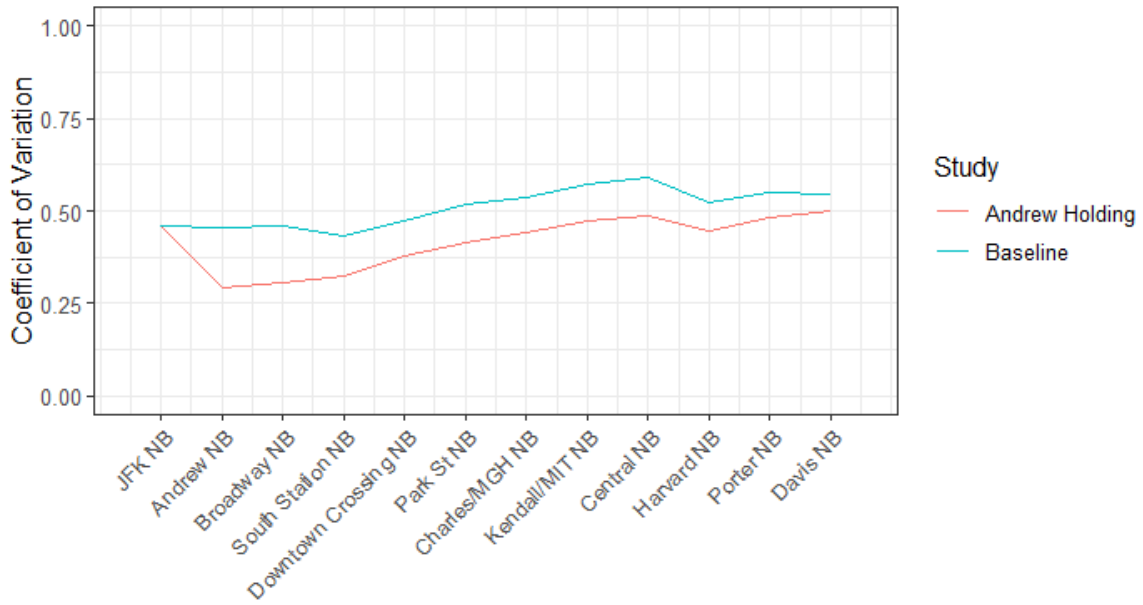

Figure 7-2: Northbound Headway Coefficient of Variation

Figure 7-3 shows the relationship between CV of headways at JFK compared to downstream stations of Andrew, South Station, and Park Street for the Baseline and Andrew Pilot. Each point represents the CV of headways for trains passing through JFK within a 30-minute period, compared to the CV of the same trains at the respective downstream station. This method of comparison allows us to understand the effect of holding at various levels of initial headway variability. The guideline in the figure, with a slope of 1 and intercept of 0, represents the hypothetical case where the CV at JFK is equal to the CV at the downstream station. Theoretically, an upper bound for this metric would be a horizontal line with y close to 0, indicating low downstream CV under any initial conditions at JFK; however, such a strategy would likely entail significant queuing delays at the holding point.

The three stations were chosen to demonstrate the evolving effects of holding. At Andrew, the Baseline CV shows almost no change compared to JFK, mirroring the guideline, with a slope close to 1. In contrast, the Pilot observations exhibit nearly even, low CV at Andrew for any level of initial CV; this is the desired outcome of holding. At South Station, the Baseline CV is reduced significantly compared to the initial CV at JFK, suggesting a natural reduction in variability

between the merge and South Station as a result of minimum safe separation requirements. The Pilot period exhibits a significant reduction in variability for trains arriving at South Station for all levels of initial CV, with the largest improvements seen in period when headways at JFK are the most uneven. The level of headway regularity at South Station is important since this is the first northbound station with heavy demand; thus improving regularity at this point will contribute largely to evening out northbound passenger loads. Finally, Park Street demonstrates significant deterioration of even headways just four stops after holding. Although the benefits of holding are still evident, the gap between the Baseline and pilot periods is much more minor here.

Together, these findings indicate the significant benefit of the Andrew Holding Pilot on limiting the worst case outcomes of the branch merge headway irregularity for northbound downtown passengers. The finding of rapid deterioration of even headways suggests that the holding should be performed as close as possible to the stations with heavy passenger demand, in order to provide maximum benefit at these key stations. In this case, holding at South Station rather than Andrew may provide increased benefits.

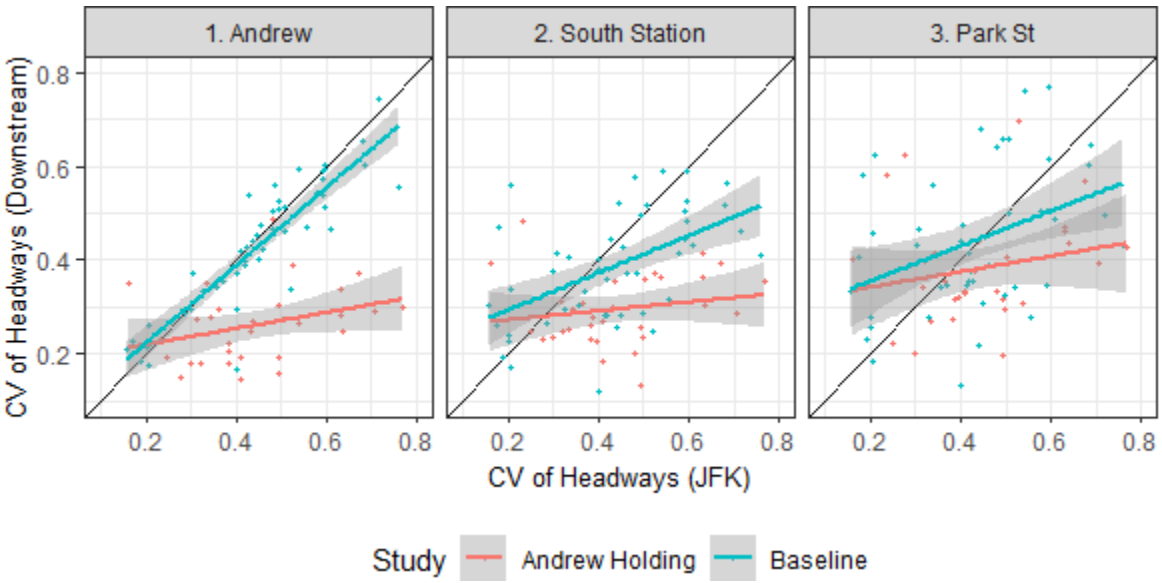

Figure 7-3: Comparison of Coefficient of Variation of Headways at JFK vs South Station

7.2.2 Running Times

Northbound running times from departing Andrew and JFK stations until arriving at Alewife are compared in **Table 7-6** for the Baseline and Andrew Holding periods. Adjustments for speed code changes during the pilot were made using the same methodology outlined previously and detailed in **Appendix 8.** Running times improved slightly between Andrew and Alewife during the pilot, although the difference in means was not significant at the 95th confidence level. Between JFK and Alewife, the mean running times increased by 0.6-minutes during the

pilot, with the differences in means found to be statistically significant at the 95th confidence level. This suggests that the delays between JFK and Andrew due to holding did not fully cancel out with equal savings in reduced congestion downstream.

Table 7-6: Northbound Trunk Running Time (min)

Analysis Extents	Study Period	Median	Mean	SD.	85th Pctl
Andrew - Alewife	Baseline	28.7	29.1	2.3	31.4
	Andrew Holding	28.4	28.7	2.1	30.7
JFK - Alewife	Baseline	33.0	33.2	2.5	35.6
	Andrew Holding	33.5	33.8	2.5	36.1

Table 7-7 compares the running time from departing Andrew to arriving at Alewife during the Baseline and Andrew Pilot periods for small and large arrival headways at JFK. Trips with headways under 2-minutes at JFK were likely subjects of holding at Andrew, while trips with headways greater than 7-minutes at JFK likely benefited from holding of the previous train at Andrew. Holding is expected to have the largest impact on these two groups.

As expected, the running time for trips with headways under 2-minutes at JFK improved significantly, with a statistically significant improvement in the mean running time of 1.8-minutes. This shows that although holding did not lower overall train running times (**Table 7-6**), it did create smoother travel for held trips after departing Andrew. The running time for trains with long initial headways, greater than 7-minutes, at JFK are also shown below; here the improvement in mean running times is not statistically significant, although the standard deviation and 85th percentile decreased substantially, owing to less extreme outcomes from the long initial headways.

Table 7-7: Andrew to Alewife Running Times for Short and Long Initial Headways (min)

Headway at JFK	Study Period	Median	Mean	SD	85th Pctl	N
< 2 min	Baseline	29.7	29.5	1.8	31.1	32
	Andrew Holding	27.9	28.2	1.8	29.8	23
> 7 min	Baseline	28.7	29.4	3.1	31.4	29
	Andrew Holding	29.1	28.8	2.2	30.5	22

7.2.3 Running Time Through Line Bottlenecks

Due to faulty circuits at some of the downtown stations, the running time through downtown station bottlenecks could not be evaluated satisfactorily. The running time through Harvard Northbound, another line bottleneck, is evaluated (**Figure 7-4**), finding small reductions in extreme station occupancy time during the Andrew Holding Pilot. The improvements are modest compared to those seen at Harvard *Southbound* during the Davis Holding Pilot (**Figure**

6-7), as a result of the holding taking place much farther upstream, and thus the headway improvements dissipating significantly by the time trains arrive at Harvard Northbound.

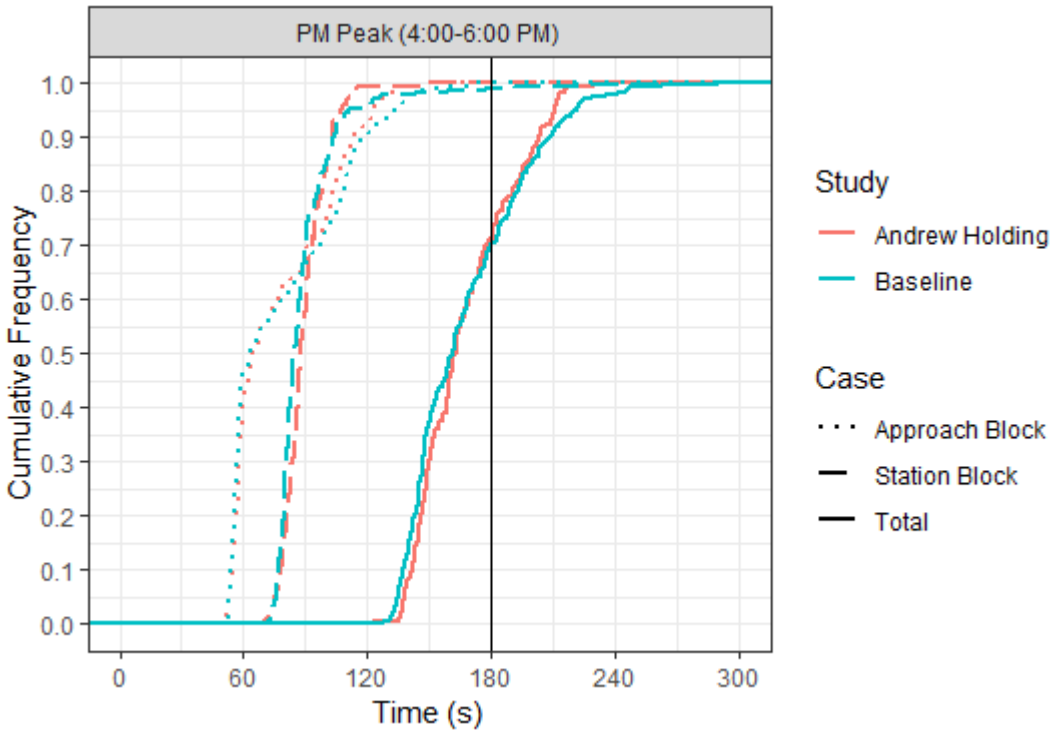

Figure 7-4: Running Time Distribution through Harvard Northbound

7.2.4 Northbound Running Time into Alewife

One anticipated outcome of the northbound holding pilot was that of reduced queuing to enter Alewife. This section will examine the impact of Andrew Holding on northbound terminal area running time, and changes in frequency of bunched arrivals to terminal area.

Observations from days with combined southbound and northbound holding will be included in the analysis based on the prior finding that southbound holding did not cause any additional congestion to northbound trains entering the terminal area.

The mean running time from departing Porter Northbound to arriving at Alewife was found to improve 15s between the Baseline and pilot periods; the difference in sample means is not statistically significant. The proportion of headways below 3-minutes was also checked to see if there were fewer cases of bunched arrivals to the terminal area. Departing Porter Northbound, the proportion of headways below 3-minutes is statistically lower during the pilot compared to the Baseline (28% compared to 38%) at the 95% confidence level. However, by departing Davis Northbound, the difference is no longer statistically significant. This indicates that northbound holding at Andrew did not significantly reduce bunched arrivals to Alewife as expected, which is attributed to queuing at Alewife.

7.3 Passenger Impacts/Improvements

This section will explore impacts of the Andrew Northbound Holding Pilot on passenger train loading, wait times, and journey times.

7.3.1 Wait Time

A comparison between northbound trunk wait times for the Baseline versus Andrew Holding Pilot periods is shown in **Figure 7-5**[8]. Average and 85th percentile expected wait time assuming a uniform passenger arrival rate are calculated using the methodology described in **Section 2.5**. Overall, Andrew Holding resulted in significant reductions in northbound trunk wait times with a line-weighted average wait time improvement of 9s (6%) and 18s (6%) at the 85th percentile level. The wait time improvements were largest at Andrew, with a 9% (14s) reduction in average and 14% (33s) reduction in the 85th percentile. Improvements tapered off towards Alewife, attributed to the bottlenecking effect of the terminal which causes increased headway regularity in both scenarios. The Pilot improvements occurred despite wait time *increases* at JFK, the station before holding. The increase in wait times at JFK is not attributed to the pilot, consistent with the earlier finding that holding delays did not affect trains until after departing JFK.

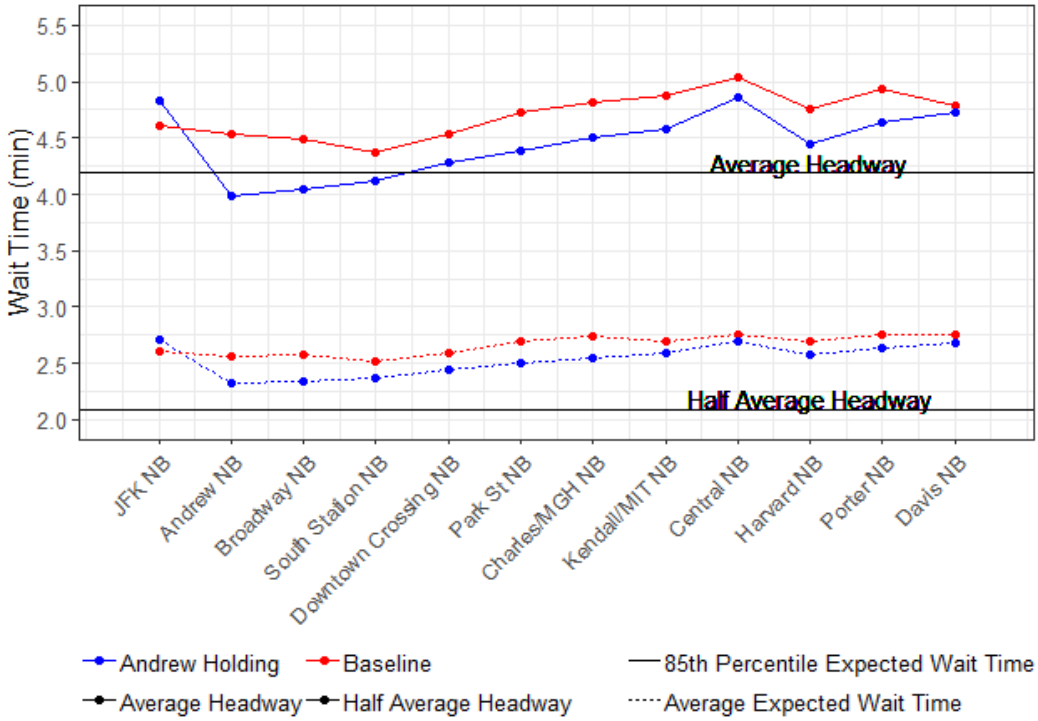

Figure 7-5: Change in Northbound Trunk Expected Wait Times

[8] The *half average headway* and *average headway* are shown as reference points, representing respectively the average wait time and maximum wait time in the hypothetical case with zero headway variation at today's frequency.

7.3.2 Journey Time

Journey time statistics are developed for the Andrew Pilot using the same methodology as described in **Section 2.5.3**, which assumes a uniform passenger arrival rate over the analysis period and no left-behinds. As before, journey times are corrected for minor speed codes changes which occurred during between the Baseline and pilot periods as described in Appendix 8. All northbound branch passengers were assumed to experience the same travel time outcomes as northbound JFK passengers, regardless of origin station. Branch and JFK passengers were also assumed to experience no changes in wait times. These assumptions follow earlier findings that holding had negligible impacts on running times prior to departing JFK, and allow us to include branch passengers in the line weighted metric without allowing incidental changes in branch wait time changes to affect the interpretation of the overall results.

Mean journey time changes for the peak period are shown for each trunk OD pair in **Figure 7-6**, with a summary of passenger weighted changes in **Table 7-8**. Overall, the peak period Pilot journey time improvements are estimated at 39.4 Passenger·Hours, or 7s per passengers. Passengers experiencing journey time benefits from the Pilot saw mean improvements of 17s, while passengers with journey time disbenefits experienced mean travel time increases of 16s. Approximately 11,400 passengers travel on OD pairs with statistically significant mean journey time improvements, compared to only 3,000 on OD pairs with significant journey time disbenefits.

Passengers experiencing the largest magnitude of benefits originate at Broadway and Andrew station. However, these two stations located just south of downtown exhibit limited demand. Passengers originating downtown and traveling north of Central represent the bulk of northbound PM peak demand, however no significant changes in journey time were found between these stations. The only statistically significant increases in mean journey tie were experienced by JFK and branch passengers traveling north of Andrew, inclusive. Destinations for branch passengers are concentrated at major transfer stations of Downtown Crossing and Park Street; the delay for branch passengers traveling to these stations is minimal, since passengers experiencing holding at Andrew experience less congestion over the next three stations.

Table 7-8: Summarized Andrew Holding Peak Period Passenger Weighted Mean Journey Time Changes[9]

Grouping	Peak Period Passengers	Per Passenger (s)	Passenger·Hours
Benefit	11,380	-17	-52.8
Disbenefit	3,052	16	13.5
All	21,466	-7	-39.4

[9] Origin-Destination pairs without statistically significant changes in mean journey time are assumed to experience zero change for the purpose of the summary calculations.

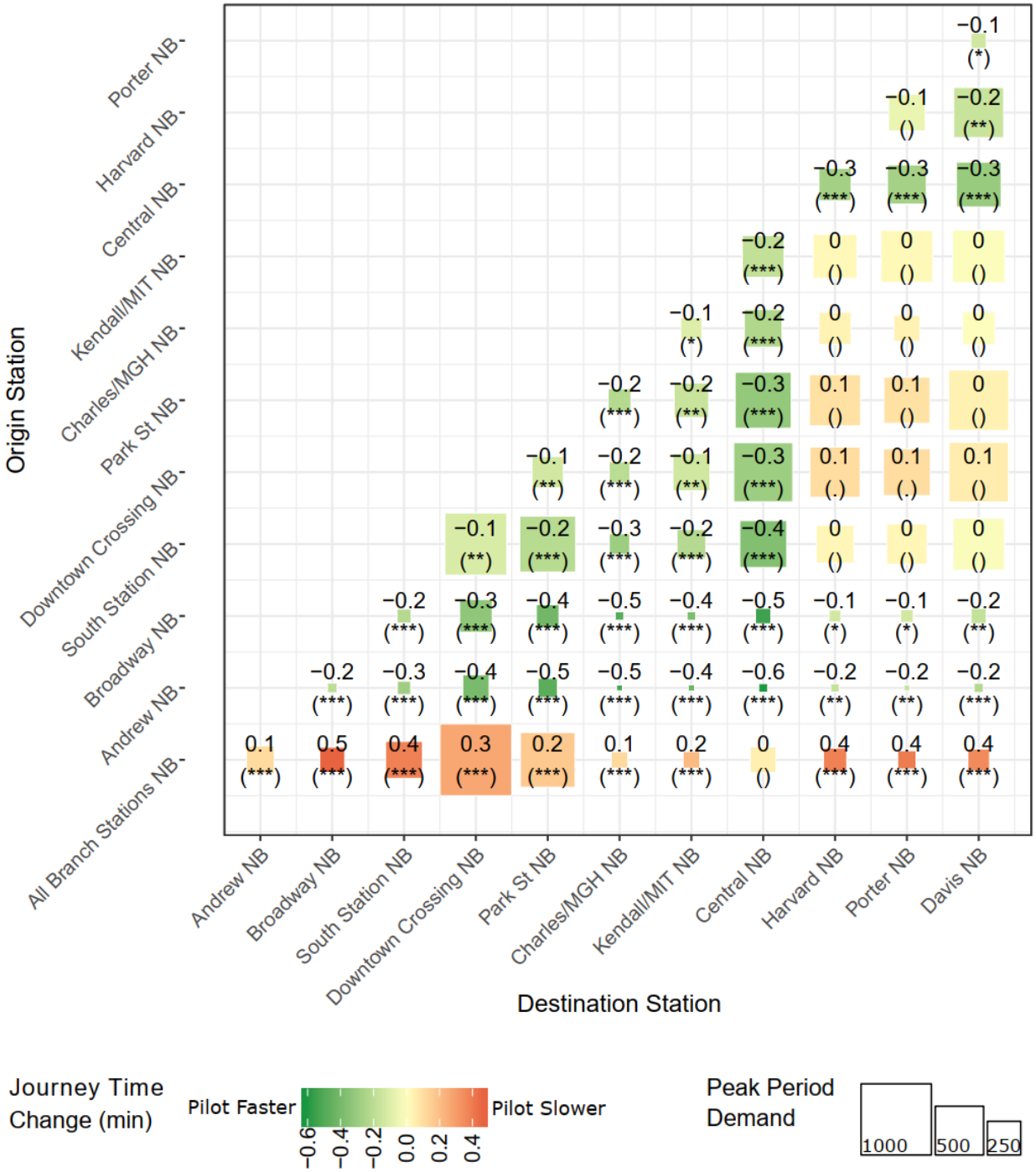

Figure 7-6: Andrew Northbound Holding Pilot - Mean Journey Time Changes by Trunk Origin-Destination Stations[10]

Significance codes: () = p ≤ 1.0, (.) = p ≤ 0.1, (*) = p ≤ 0.05, (**) = p ≤ 0.01, (***) = p ≤ 0.001

[10] Journey times for Northbound Branch stations exclude waiting time.

7.3.3 Passenger Train Loading and Effective Utilization

Expected passenger loading was determined for northbound trips using the same methodology described in Section 6.3.1, based on the observed headway at each station and historical passenger arrival rates for each OD pair. Results are shown in **Figure 7-7** for the Baseline and Andrew Pilot. The analysis was limited to the peak hour on the basis that changes to crowding are most important during this time.

As expected, a reduction in lightly loaded peak hour trains is observed, as denoted by the increase in passenger loading at the lower, middle, and upper quartile of the boxplots (25th, 50th, 75th percentiles, respectively) at all stations between Andrew and Harvard. The benefits of headway regularization are seen in the reduction in extreme crowding; the 90th percentile (upper whisker) expected passenger load after Kendall, the peak northbound load point was reduced by 112 passengers (13%). By not accounting for left-behinds, this calculation likely *underestimates the benefits of holding* on passenger comfort, overall journey time, and satisfaction, since reductions in extreme headways and crowding would also result in a reduction in left-behinds. An inventory of passenger left-behinds using archived security camera footage or wireless device probing could further quantify the benefits of headway regularity (Gonzales et al., 2018).

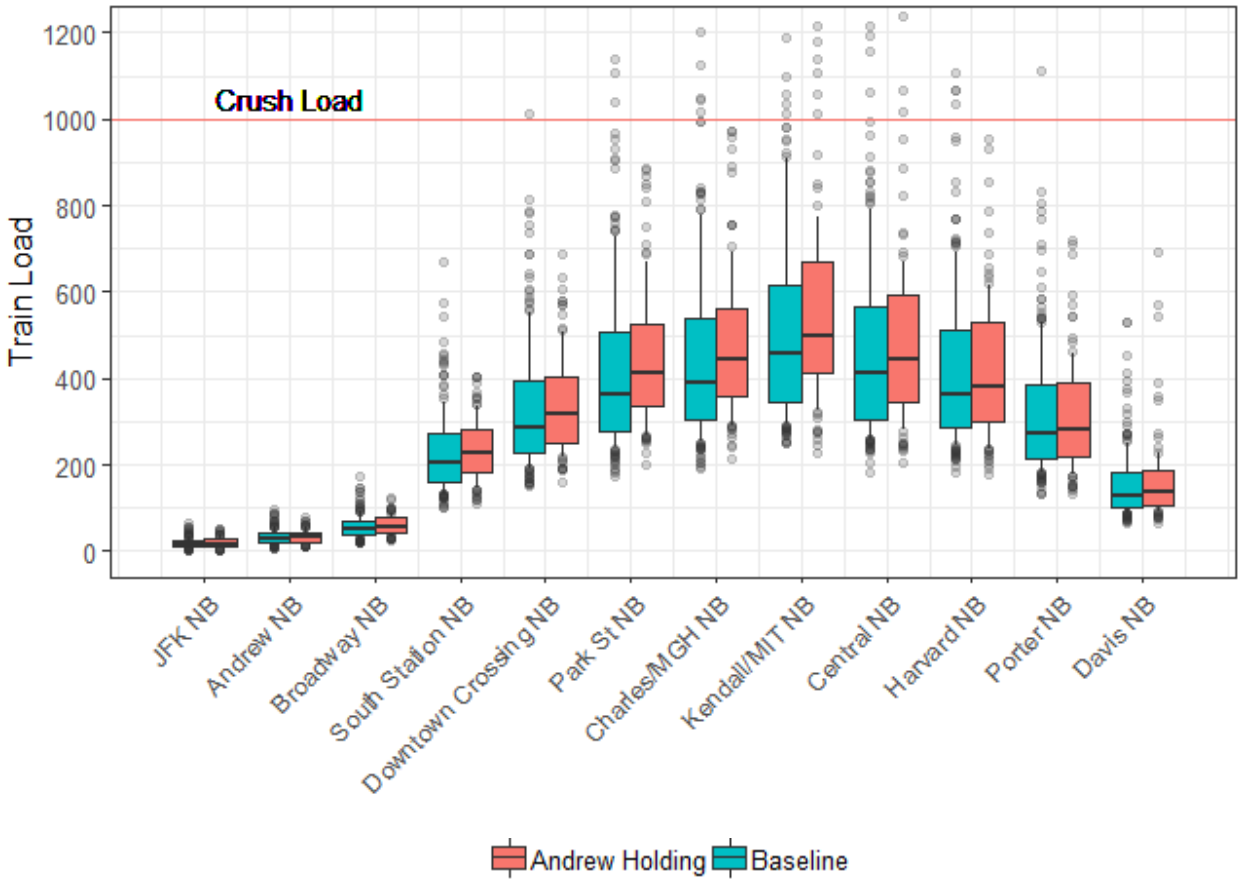

Figure 7-7: Expected Train Load (Trains Departing Andrew 5:00-6:00 PM)

Another way to communicate improvements in passenger loading is through the concept of Effective Utilization, developed in Chapter 2. The distribution of effective utilization for the Baseline and pilot periods peak hour is shown in **Table 7-9**. The main difference is found in reducing the number of under-utilized trains; the median utilization increased from 84% to 97%, demonstrating an increase in uniformity of train loads. The pilot also resulted in a decrease in high percentile utilization, seen here in the 95th percentile.

Table 7-9: Andrew Pilot Peak Hour Effective Utilization Percentiles

Study	5%	15%	25%	50%	75%	85%	95%
Baseline	0.58	0.63	0.67	0.84	1.19	1.43	1.91
Andrew Holding	0.61	0.66	0.74	0.97	1.18	1.42	1.81

7.4 Discussion

This section will discuss adherence to the holding pilots, general issues with pilot staffing, and operator reactions to the pilots discerned from informal discussion.

7.4.1 Holding Pilots Compliance Issues

As discussed extensively by Maltzan (2015) and Fabian (2017), adherence to pilot recommendations is a major factor determining the success of any intervention to improve service delivery. For the Davis and Andrew holding pilots, operator adherence to holding instructions was largely a non-issue. In devising the holding pilot instructions, it was recognized that circumstances, including passengers running for the doors, errors operating the stopwatch, and conversation between the operator and official would cause deviations from the recommended departure times for held trains. Thus headway targets were chosen conservatively, anticipating that many holds would exceed the recommendation by 10-15s. Adding this buffer time successfully prevented over-holding at the expense of some headway variability. One way to eliminate some of these delays would be to instead use automated countdown clocks rather than officials to deliver the instructions, however this should not be seen as a barrier to continuing the holding as is.

The largest deviations were seen during the AM peak holding at Davis Southbound due to the high volume of boarding passengers at this time. While passengers normally expect the doors to close on short notice, expectations differ when the train has already been holding for 1-minute; passengers were spotted entering and exiting the train, and walking along the platform to their favorite car. Operators were courteous in these cases, giving passengers several warning chimes before closing the doors. This observation highlights the importance of picking a holding target which accounts for real-world considerations including heavy passenger demand.

Another compliance issue to contend with was the adherence to the variable recommended holding time at Andrew. The two-step decision process for selecting the holding target (Section 4.3) was necessitated by the unreliability of the countdown clock; it was important to take into

account the most recent information. While students were always present during the pilot to clarify the instructions, the variable holding limit should be a focus of training if Andrew Holding is implemented in the long term.

7.4.2 Staffing

One of the primary challenges in implementing the pilots was staffing. There were a few contributing factors: (1) the unofficial nature of the pilots and understanding pilot benefits, and (2) pilot staff having other, official roles. These factors will be discussed in more detail below. In order to address these factors, the pilots would need to be scheduled officially, and management would need to be aware of the priority of the interventions relative to other tasks.

(1) Due to the "unofficial" nature of the pilots, officials did not always adhere to the pilot hours as rigidly as they might adhere to their official work assignments. This was primarily an issue during the shoulder periods, when officials viewed holding as superfluous due to the low passenger volumes at these times. In this case, it should be emphasized that holding is important not just for reducing crowding, but also for minimizing passenger wait times. Staff acceptance of the pilot procedures would also change as holding is worked in to the schedule and gains stature as a standard work assignment.

(2) Due to short-staffing, Davis Holding pilot staff were still responsible for responding to incidents at the nearby Porter Station. Twice during pilot observations, Davis Pilot staff were sent by OCC dispatchers to Porter to respond to passenger incidents. Recognizing the importance of the holding procedure and positioning officials such that staff executing the holding were largely free of other responsibilities would help to ensure the benefits of holding are delivered. For the Andrew Holding Pilot, the MBTA official was not responsible for any other stations, leading to more consistent pilot staffing.

7.4.3 Operator Reactions to Holding Pilots

Operators were largely understanding and supportive of the holding pilots. During the hold time, officials explained to them the reasons for holding, citing bunching, or a large gap for the train behind in case of the Andrew pilot. Operators are familiar with holding, although the instructions are typically received from the OCC rather than an official on the platform. One operator noted that receiving instructions from OCC over the radio can feel impersonal and frustrating, especially in case of long holds which extend the operator's shift. From this perspective it helped that the holds were always short (0-2 minutes), and that the instructions were communicated via a friendly face at the platform.

Reducing headway variability through holding is ultimately beneficial for the operator experience as well as for passengers, since it reduces the frequency of long headways and overcrowding, which contribute to stressful work conditions. In addition, it should be emphasized

to operators that minor holding up to the minimum line headway does not increase overall end to end run times due to the resulting reductions in train congestion.

7.4.4 Holding Control – General Discussion

In addition to the MBTA focused discussion, the following general observations are offered to guide future practitioners designing holding pilots in a high frequency rail environment:

(1) Holding strategy should be selected with a primary target in mind

When developing a holding strategy, a specific primary objective should be selected to guide design and post-hoc evaluation. For example, if the primary objective is to improve bottleneck throughput, then holding may be applied immediately before the bottleneck. Alternatively, if the objective is to reduce variation in passenger loading, holding may be applied just before a group of high demand stations. A primary objective is important to provide focus for the intervention in a complex decision making environment.

(2) Next arrival predictions offer a valuable resource

Next arrival clocks and real-time information available through mobile apps provides an important tool for rapid implementation of variable holding pilots, where in the past, real-time information would have been difficult to obtain. Such initiatives should be prefaced on the understanding that arrival predictions are not always accurate, and are sometimes inaccurate in predictable ways.

Prior to both holding pilots, field measurements were taken to ensure proper interpretation of the next-arrival clocks. At Davis, it was identified that predictions always assumed northbound trains would layover for a constant time at Alewife, which failed to account for schedule recovery. At Andrew, the predictions made no consideration for minimum headway, so simultaneous arrivals were commonly predicted when trains were arriving simultaneously to the branch merge. Awareness of the prediction faults allowed us to benefit from the real-time data availability.

7.5 Conclusion

The effects of the Andrew Northbound Holding Pilot were examined across many aspects of service quality and passenger experience. The largest improvements occurred at Andrew and in the stations immediately following Andrew; average wait times improved by 14s (9%) at Andrew while passenger weighted line-average wait times improved by a smaller, but still notable amount of 9s (6%) north of JFK. Overall, the pilot peak period journey time savings are estimated at 39.4 Passenger·Hours, or 7s per passengers. Customers traveling northbound from the branches experienced the only disbenefits of the pilot, in the form of additional travel time; between JFK and departing Andrew trains were delayed by a mean of about 50s, with 90th percentile delays around 110s. A portion of these delays were later recovered over successive stations in the form

or reduced train congestion; compared to the Baseline, running times from departing Andrew to arriving at Alewife improved by a mean of 1.6-minutes for trains which had an initial headway of under 2-minutes at JFK. Perhaps the most significant benefit of the pilot was of reduced train crowding at the peak load point; 90th percentile estimated train loads at Kendall/MIT Northbound fell by 13% (110 passengers) resulting from the reduction in large gaps entering the trunk.

Chapter 8. Conclusions and Recommendations

This chapter summarizes the conclusions and recommendations from this research. Section 8.1 discusses the findings and conclusions on a general level, with respect to the research objective of evaluating feasibility of a 3-minute headway on the MBTA Red Line. Section 8.2 discusses the results of the Alewife Drop-back Pilot and other recommendations to improve dispatching procedures. Section 8.3 discusses findings from the analysis of headway regularity, the results of the Andrew and Davis Holding Pilots, and other recommendations to improve headway regularity. Section 8.4 discusses results from the analysis of baseline dwell times and provides recommendations for reducing the impact of irregular and extended dwell times on capacity. Section 8.5 provides a bulleted list of all recommendations, categorized by implementation timeframe. Finally, Section 8.6 suggests areas for future research, specifically for the MBTA Red Line capacity project, and generally in the area of headway regularity and dwell time modeling.

8.1 Future Capacity Objective

The primary objective of this research was to explore the operational constraints which affect the reliable provision of a future 3-minute headway on the MBTA Red Line. From extensive investigations of current Red Line service, it is apparent that beyond the infrastructure bottlenecks already planned for extensive improvements, it is aspects of service regularity which could prevent a smooth transition to future capacity objectives. In particular, addressing current dispatch/headway and dwell time irregularities are of utmost importance to achieving reliable and efficient high frequency service on the Red Line.

Chapter 2 showed that the MBTA Red Line exhibits significant issues of unaddressed headway regularity, most significantly the initial variability due to the SGR of the Alewife crossover, and lack of real-time headway coordination between the northbound branch dispatches. At future, near-capacity operations, the margin for on-time performance will be significantly reduced, and it may not be possible to recover from such irregular initial headways without significant cascading delays or by pulling trains from service to open up slack. Chapter 3 showed the pervasiveness of extreme and irregular dwell times on the MBTA Red Line which stand to prevent the reliable provision of a 3-minute headway through many key station bottlenecks. The positive feedback loop between irregular dwell times and headways was also emphasized. While further work is required to determine precise targets for dwell time and headway regularity improvements, it is evident that significant overhauls to today's operating procedures will be required in addition to the already programmed capital investments.

To this end, this research proposed and implemented three service interventions on the Red Line with the objective of improving headway regularity, finding that even simple, low-cost strategies can deliver significant improvements in the short term. Conclusions from these interventions will be discussed in more detail in the coming sections. These interventions

demonstrated the ability for the MBTA to achieve capacity and service improvements in the short term under today's system's fiscal and technological constraints, well in advance of planned long term infrastructure improvements. Development of publicly facing intermediate reliability and frequency targets are recommended in order to signify the MBTA's commitment to the stated future service improvement. This process of continuous improvement is vital to maintain and attract ridership in the interim period, and to gradually shift the agency culture.

8.2 Terminal Operations and Dispatching Procedures

Terminal operations have a significant effect on initial headways, which in turn can define the outcome of a given trip. Chapter 2 demonstrated current inefficiencies at the Alewife Terminal, including a lack of significant recovery from northbound arrival variability, and a failure to account for headway variations caused by the southbound crossover. Irregularity in dispatch headways is a significant barrier to improving frequency since initial headway discrepancies tend to be magnified over the line and regular headways are a necessity to achieve maximum bottleneck throughput.

Chapter 5 presented the results of the Alewife Terminal Drop-back Pilot, which trialed drop-back crewing in order to demonstrate one important strategy which can be implemented as part of a larger review of terminal procedures. Based on the success of the pilot in terms of reducing northbound congestion, and improving recovery following delays, drop-back crewing is recommended for continued implementation across AM and PM peak periods, beginning with the next scheduled rating. Drop-backs are essential both in the short term to achieve fullest utilization of the limited fleet size, and in the long term to enable sufficient schedule recovery time with minimal layover.

For future implementation of drop-backs, service planning and delivery should be made aware of how the lower minimum turnaround times can benefit other operating procedures. For dispatchers, very short layover times enabled by drop-backs may allow for branch trains to be reordered primarily at Alewife rather than JFK, or to break from the traditional order of terminal movements (alternating north and southbound movements) in order to enhance schedule recovery. For schedulers, drop-back crewing could allow the shifting of operator break time to Alewife, or the interlining of operators between branches to enable more efficient shift utilization.

As observed during the pilot, it is important to include drop-backs in the official schedule in order to maintain relationships with operators by avoiding unexpected shift changes. The benefits of drop-backs from the operators' perspective should also be made explicit at the time of implementation; namely longer breaks and greater schedule reliability, in order to receive buy-in from all stakeholders. MBTA supervisors at Alewife should also explore operator incentives to ensure operators continue to depart promptly with the ring-off, despite the increase in down-time at the terminal, for example by posting public stats highlighting the top operator's departure compliance. Developing a culture of "every second matters" throughout the organization ahead of

the arrival of new Red Line vehicles is crucial for a smooth transition to higher frequencies, and the terminal is likely the most important location of all.

Beyond planned crossover repairs to allow higher and more consistent speeds departing the terminal, another important investment to improve terminal capacity would be communications infrastructure to give operators advanced warning of the departure bell. Operators could use the warning time to enter the cab, close the doors, and prepare for departure, even before the crossover is free of conflicting movements, potentially reducing crossover occupancy by up to 20-30s for every departure. Under current operations, this change could already provide substantial benefit in the form of reduced northbound queuing, while under future operations it would provide an invaluable improvement to terminal capacity and reliability.

8.3 Headway Regularity and Holding Pilots

Examination of the current state of service regularity in Chapter 2 demonstrated the necessity of operating changes to improve headway regularity, both to improve capacity today, and enable frequency improvements in the future. Chapters 6 and 7 presented the results of the Davis Southbound and Andrew Northbound Holding Pilots, respectively, which implemented headway-based holding strategies to improve headway regularity. Based on the positive results of the pilots, it is strongly recommended that a holding strategy is continued during AM Peak at Davis and during PM Peak at Davis Southbound and South Station Northbound. While the strategies as implemented have areas for improvement, a culture of continual improvement will be required in order to achieve a substantial increase in reliability ahead of the change to 3:00 headways, and these holding pilots represent a step in the that direction.

Davis Southbound is an ideal holding location to address issues with Alewife dispatching due to the potential to benefit many downstream passengers with only minor inconvenience to Alewife passengers. However, Davis should be re-evaluated as a holding location after Alewife crossover reconstruction occurs (RFQ 2018/09) which will allow P1 dispatches to travel at a higher speed over the crossover, reducing the routine headway differences currently seen at Davis. Managing headways at the terminal station is preferable to holding along the line in order to reduce the number of in-vehicle passengers delayed and maximize downstream passenger benefit.

The Andrew Northbound holding pilot showed significant benefits for northbound trunk passengers in terms of wait time, and reduced crowding. Recognizing that few passengers board at Broadway and Andrew, it is recommended that the holding point be moved to South Station. This would ensure the largest improvements are experienced at the three highest demand downtown stations, before the regularity improvements dissipate, further improving consistency of northbound train loads. Additionally, it is recommended to adjust the long gap holding strategy to target an even headway, rather than a 4-minute headway, in order to further reduce extreme wait times. To reduce the disbenefit experienced by branch passengers, it is recommended that real-time headway based dispatching is examined as a potential solution at Ashmont. Improvements to

dispatching would reduce the frequency and magnitude of in-line holding required at the merge to regularize headways. Looking further ahead, automatic train control could be used to optimize the speed profile of branch trains in real-time to achieve coordinated merge headways.

In the short term, holding tasks can be assigned to the officials already located at the designated station. Additional officials in the surrounding stations would ensure holding staff would rarely need to leave their post during peak hours as occasionally occurred during the pilot. Assigning minor holding tasks to officials rather than OCC dispatchers is recommended in order to reduce the workload of OCC and more crucially, to *disseminate the objective of on-time performance to the operating level staff*. Developing an app-based holding decision support tool (Fabian, 2017) for use by officials could further improve outcomes by recommending a variable holding time considering upstream conditions and historical passenger demand patterns.

In the longer term, it may be desirable to implement in-line headway based holding using automated ring-off bells or countdown timers, similar to the ring-off bell in place at Alewife Terminal today. The control points would be installed at pre-determined stations and the holding regime (even headway, target headway) could be set remotely by dispatchers depending on current conditions or time of day. Such automated control may be necessary to implement more complex holding strategies, and avoid compliance issues which arise with manual headway control. However, as demonstrated by the pilot, *significant gains can be achieved with simple holding strategies in the interim with currently available resources and technology prior to such investments*.

8.4 Dwell Time Control

Chapter 3 discussed the importance of reducing extreme station occupancy times in order to achieve a reliable 3:00 headway through key stations during peak hours. It was shown that as of today, there are a multiplicity of stations which exhibit extreme dwell times of 2-3 minutes, clearly limiting the reliability of a future 3-minute headway. The most constraining stations include: Harvard, Charles/MGH, Park Street and Downtown Crossing in the southbound direction, and Park Street, Harvard and Davis in the northbound direction[11]. It is notable that limiting stations include both high demand downtown stations, but also more moderate demand outlying stations, as a result of restrictive safe separation requirements and station geometry. Simulation modeling is required to determine the dwell time reductions that programmed capital improvements, including the new, uniform fleet of 4-door vehicles, and an upgraded signaling system with capacity for additional speed codes will bring, and to what extent further reductions will still be required.

[11] Downtown Crossing and South Station could not be analyzed in the northbound direction due to faulty track circuits

Looking beyond maximum dwell times, if dwell time consistency is not increased between trips, the MBTA risks maintaining the status quo of high headway irregularity and inefficient resource allocation. This is equally important in the operating environment today and in the future. Today, dwell time irregularity leads to unstable headways and bunching, which leads to an inefficient use of limited capacity. Under future, near-capacity operations, large headway gaps which open up due to irregular and extended dwell times may be impossible to recover from without significant cascading delays due to limited schedule recovery slack. A program of continuous improvement is recommended to achieve the necessary improvements to dwell time uniformity and duration, including changing staff and passenger culture towards dwell time punctuality in advance of the frequency increase.

A dedicated "Dwell Time Director" could be appointed to coordinate dwell time reduction and regularization efforts. Some of the duties would include; learning from best practices of other agencies, internal (and external) dwell time reporting to increase transparency, customer communication strategies, operator re-training, and implementation of strategies to improve passenger flow on platforms and within stations. The importance of dwell time reductions should not be understated as a barrier to frequency improvements.

In terms of reducing the impact of extreme dwell times on line capacity downtown, it is worth examining more long term solutions, including the consolidation of downtown stations Park Street and Downtown Crossing (MBTA Focus 40, 2019). Assuming the necessary customer circulation can be achieved at a single station, the strategy could reduce line cycle times and minimum headway, without significantly impacting passenger convenience due to the extremely dense stop spacing. Enforcing one-way passenger flow at Park Street is another potential solution to kick-start the large dwell time improvements necessary downtown, and could be implemented at far less cost and in a shorter timeframe than station consolidation.

8.5 Summary of Recommendations

This section serves as a summary of recommendations provided in this thesis, categorized by feasible implementation timeframe. Since no major capital expenditures are proposed, a categorization by cost is omitted.

Immediate

- Continue Davis Holding Pilot as originally designed in AM & PM Peak Periods
- Continue Northbound Holding Pilot in PM Peak Period, moving control point from Andrew to South Station, and changing long-hold target to even headway target for any train preceding a gap of at least 5-minutes

Short Term

- Include drop-backs in the next operator rating, and combine with scheduling and terminal procedural adjustments to further improve consistency of dispatches, and minimize added cost of drop-back operations
- Pilot one-way passenger flow at Park Street
- Appoint a manager to coordinate dwell time reduction efforts, including developing dwell time targets, implementing reporting & transparency measures
- Implement new headway reliability reporting and transparency measures including disaggregate analysis to raise the awareness of on-time-performance within the agency
- Develop intermediate headway and service reliability targets to serve as short term milestones on-route to achieving the long term 3-minute headway objective

Medium Term

- Implement communications infrastructure at Alewife to provide a cue to operators ahead of scheduled ring-off, in order to reduce crossover occupancy time
- Explore staff incentive programs to encourage on-time terminal departures from operators
- Develop decision support tool to coordinate northbound branch dispatches in real-time
- Develop decision support tool for in-line holding which can be deployed with officials at key stations to improve outcomes from simple holding pilots
- Implement remote, automated headway control infrastructure (countdown timer/flashing signal) at certain control points which can be controlled by OCC

8.6 Future Work

Areas for future work are divided into further study of the MBTA's Red Line capacity objectives, and areas of general research interest.

8.6.1 MBTA Future Capacity Objective

Simulation modeling is necessary to relate the headway regularity and dwell time regularity issues discussed in this research to the future Red Line conditions in terms of capacity, reliability, and resiliency to delays. Such a simulation model should be sufficiently detailed, current and future speed codes, planned future capital improvements, a realistic dwell time model, variability in operator behavior, and provisions for a mixed fleet of rolling stock.

Using a simulation tool, scenarios can be tested beyond the limits of empirical analysis which is presented here. Through scenario based simulation of future operating conditions, targets should be developed to guide headway regularity and dwell time improvement relative to today's operating practices. These targets should be an integral part of the Red Line improvement program

which currently focuses on major capital projects, but omits discussion of operating changes. Intermediate service improvement strategies could do much to placate customers wary that large capital improvements are not scheduled to bring relief for several years. Service improvement targets could also help to energize agency staff and provide a touchpoint for culture changes necessary to improve operations.

Due to the potentially limiting effect of this bottleneck on future operating conditions, great attention should be given to the headway constraint at downtown stations. A non-exhaustive list of possible solutions includes strict dwell time control, enforcing one-way passenger flow at Park Street, downtown station consolidation, major operational solutions such as skip-stop, or signaling changes to allow closer train spacing at these stations. A detailed investigation into alternatives for this issue is recommended, along with trialing of lower cost solutions such as one-way passenger flow to evaluate potential benefits.

The northbound branch merge on the Red Line presents another significant challenge in providing reliable, high frequency trunk service. In this research, a variable target headway based holding strategy was trialed following the merge to reduce the variability of trunk service; future work is required to study the efficacy of alternatives, such as real-time dispatch coordination or the feasibility of regulating speed-codes to coordinate arrival headways.

8.6.2 General Areas for Future Work

This section discusses directions for future research, not specific to the MBTA project. The main areas of interest are headway regularity and dwell time modeling with automated data.

Two concepts were developed in the exploration of headway regularity which show promise for future application. The first is that of headway stability, a concept which is well understood in practice, but lacks a formal definition in the literature. Headway stability is an important concept in communicating the effect of different dispatching headways, especially in complex systems like the Red Line with multiple branches and rolling stock. The second concept related to headway regularity introduced in this research is Effective Utilization. This metric is a promising tool to communicate to agencies the inefficient utilization of resources which arises a result of irregular headways. Further study is suggested to formalize this straightforward metric as a tool for capacity analysis and longitudinal study.

The work presented on dwell time modeling using automated data contains ample opportunity for future study. Regularly maintaining a full set of dwell times, estimated passenger movements, and passenger loading would allow agencies to pin-point the cause and reasons behind service disruptions. Application of the proposed modeling methods to a system with a tap-in and tap-out fare system is recommended to enable improved passenger to vehicle assignment estimates, and allow for further insights to the effects of crowding on dwell time and passenger behavior, on a scale not previously possible. Identifying the impact of operator behavior on dwell

times in manually controlled systems is another area for future research enabled by wide-scale dwell time monitoring.

Furthermore, the attempts to apply the well-known TCQSM (2013) methods for station bottleneck identification highlighted some gaps between the methods and state of data availability. Although automatically recorded track circuit data representing train movements are widely or perhaps universally available to service planners and schedulers today, the methods in the TCQSM still presume manual observations. Further study is recommended to confirm the use of automated track circuit data for statistical dwell time and station re-occupancy analysis, including a comparison of manual and automated data for the same context.

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Appendix 1. Reliability Buffer Time/Factor

As developed by Chan (2007), Reliability Buffer Time is the difference between 95th percentile and median journey time for a given OD pair and time period. It is method of measuring journey time reliability. Reliability Buffer Factor is the ratio of RBT to median journey time, indicating the fraction of journey time a passenger must allocate to ensure a reliable arrival time. The RBF allows reliability to be compared across OD pairs with different base journey times.

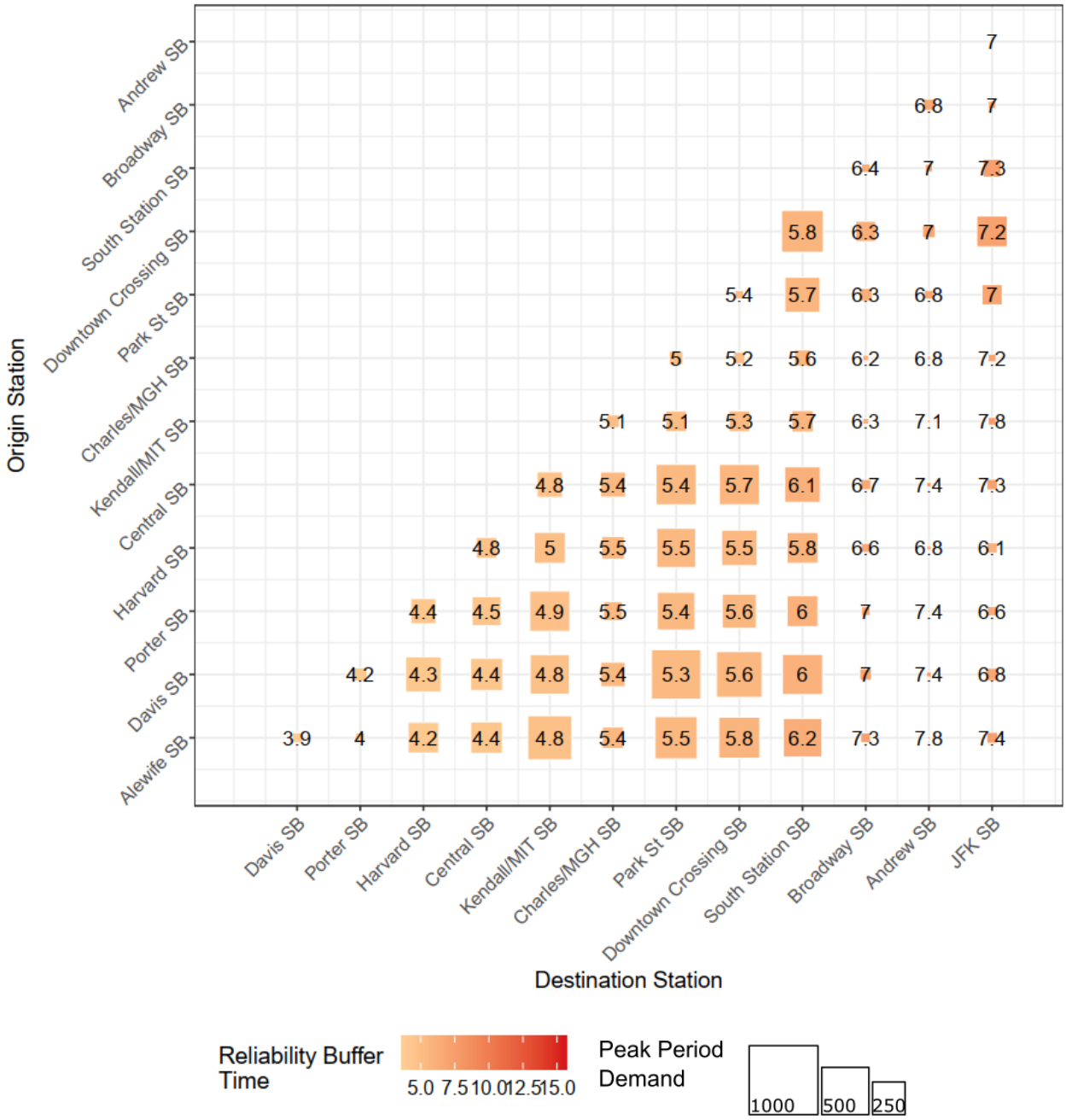


Figure 8-1: Reliability Buffer Time (AM Peak, Southbound)

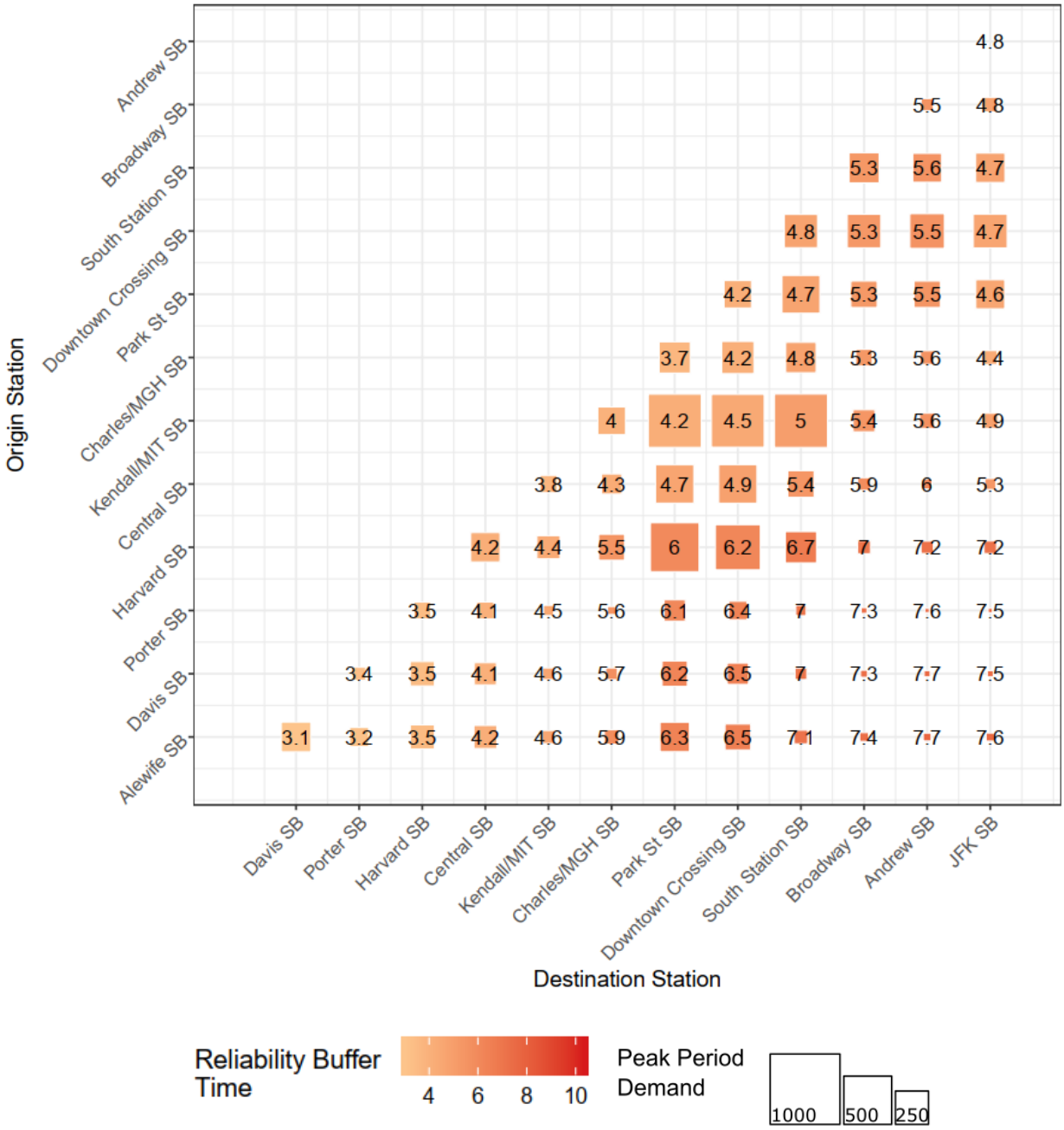


Figure 8-2: Reliability Buffer Time (PM Peak, Southbound)

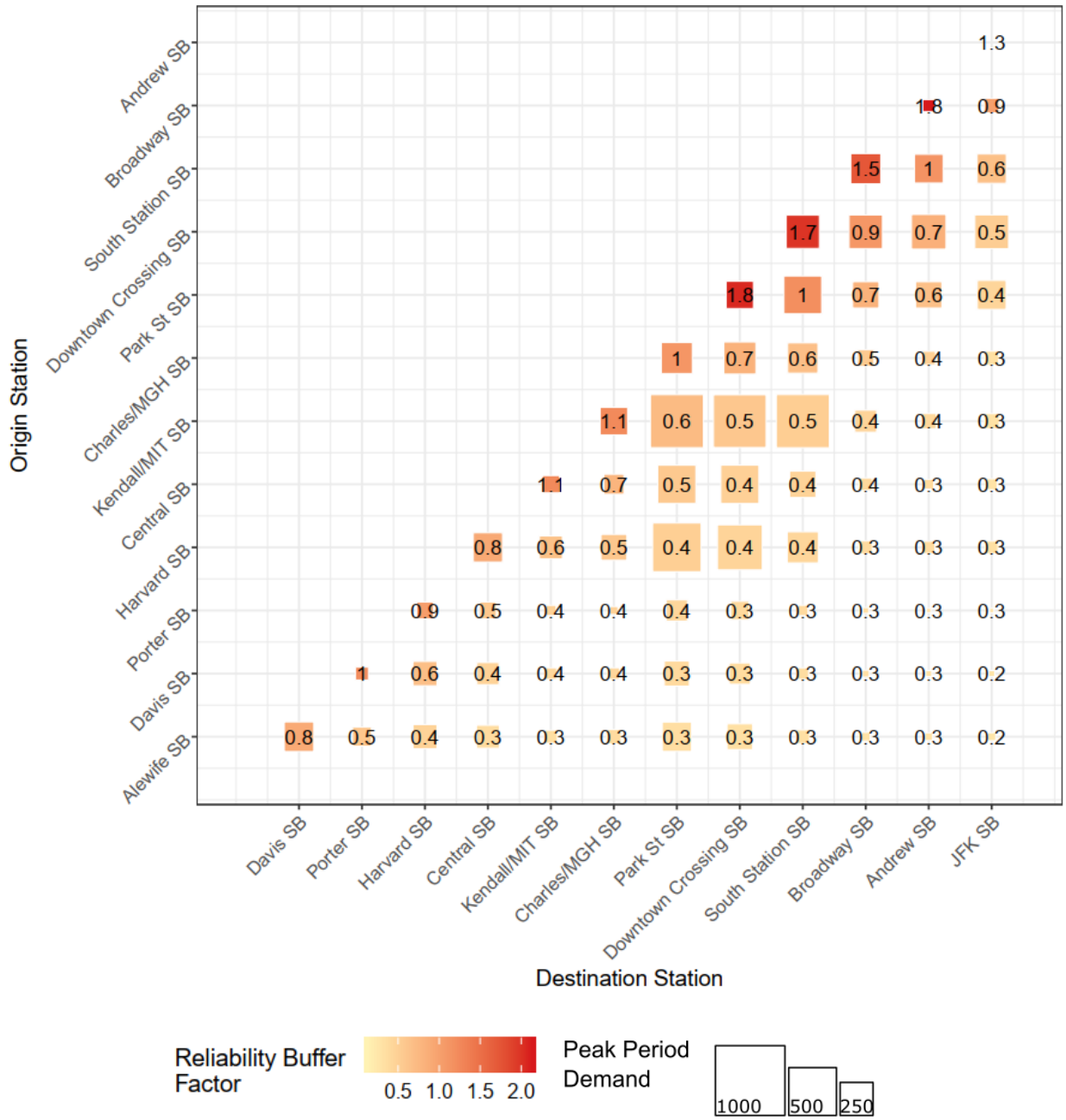


Figure 8-3: Reliability Buffer Factor (PM Peak, Southbound)

Appendix 2. Pilot Days

Days on which pilots were executed are listed in **Table 8-1**. Days on which pilots were planned but not executed are excluded for clarity.

Table 8-1: Pilot Execution Days

Date	AM Peak Pilot	PM Peak Pilot
01/07	B	B
01/08	B	B
01/09	B	B
01/10	B	B
01/11	B - Removed due to Delays	B
01/14	B	B
01/15	B	B
01/16	B	B
01/17	B	B
01/18	B - Removed due to Delays	B
01/23	DB	
01/24	DB	DB
01/25	DB	DB
01/28	DB	DB
01/29	DB	
01/30		DB
02/06	DH	DH
02/07	DB & DH	DB & DH
02/08	DB & DH	
02/11	DH	
02/14	DH	DH
02/15	DH	DH
02/19	DH	DH & AH
02/20	DH	AH
02/21	DH	AH
02/22	DH	AH
02/25	DH	AH
02/26	DH - Removed due to Delays	AH
03/01		AH

Note: B = Baseline, DB = Drop-backs, DH = Davis Holding, AH = Andrew Holding

Appendix 3. Holding Targets Selection

Andrew Holding Targets

The primary holding target of 3:00 approximates the minimum line headway in the northbound direction. Choosing a target close to the minimum line-headway ensures that queuing delays at the holding point do not exceed future savings due to reduced train interaction.

The extended holding target of 4:00 represents the scheduled trunk headway during the afternoon peak period. The criteria for initiating the extended hold was that of a large gap following the current train. A cutoff of 6-minutes predicted headway (judged when the current stopwatch has reached 3:00 since last departure, or when the train arrives, whichever is later) was chosen after field observations confirmed that a 6-minute prediction for the next train provided reasonable certainty that the train would not arrive in less than 5-minutes. The limit of 5-minutes is important because it means that a 3:00-minute train held for an additional time of 1:00 will result in a minimum of a 4:00 headway for the following train – the same as the current headway target. Over-holding trains such that their headway becomes higher than the following train is undesirable because it will cause bunching for the following train, which is the exact situation target based holding is intended to eliminate.

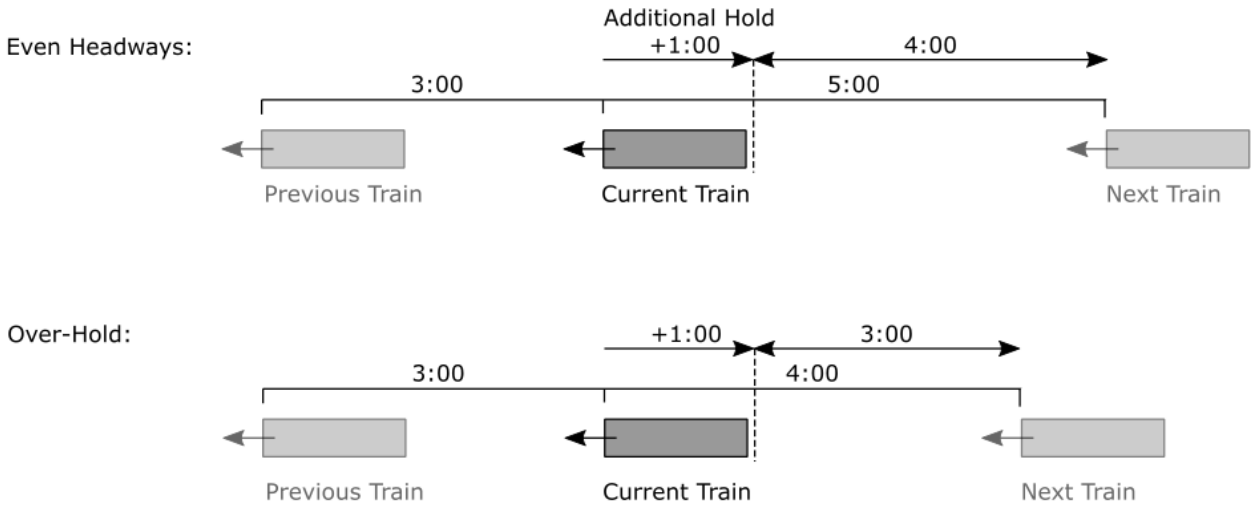


Figure 8-4: Example of Even Headway vs. Over-hold

Davis Holding Targets

A variable holding target was not possible at Davis due to the unreliability of the next arrival clocks at the station immediately downstream of the terminal. As such, a simple target based headway was selected, representing approximately the minimum southbound line headway.

Appendix 4. Determining Alewife Terminal Reaction Time from OCS Records

Reaction time is approximated using event records from the operational control system (OCS). First, each departure is matched up with a corresponding bell-ring and crossover slot. The moment when the bell rings at the platform corresponds to the moment when the crossover slot is assigned to the departing train. In the OCS records, the bell-ring event always occurs before the slot is available. In cases where the crossover is occupied by a conflicting movement at the time of the departure ring-off, there may be significant lag between the bell-ring event and the slot-open event. This frequently occurs when Automatic Train Dispatching (ATD) is in use, since the departure bells are fixed to the schedule, and remain fixed regardless of conflicting movements. Thus the departure-bell as experienced by the operators at Alewife is more accurately described as the start of the movement's crossover slot time. Finally, the slot assignment is automatically ended when the train has fully exited the crossover. Thus, the reaction time can be measured as the difference between the southbound Alewife crossover slot time assigned to the train and the minimum observed crossover slot open time for that movement for all trains during the pilot/Baseline periods. The minimum reaction time is therefore taken as zero. This interpretation of OCS event records was validated by corresponding precise field observations to the event records.

An example of the raw OCS output is provided in **Table 8-2**. Reading the events in order of occurrence, the departure bell is turned on at 7:25:09, followed 1 second later by the slot control for P2. The slot remains open for 59 seconds, until 7:26:09, and the departure bell is turned off. A northbound slot is opened, presumably to P2, following normal operating procedures. Next, the departure bell is turned on, this time followed by a southbound P1 slot after 20 seconds. This delay shows the importance of referencing both slot control and ring-off events in tandem. The slot remains open for P1 for 66s, and the departure bell is turned off. A sample of the cleaned signal events are shown in **Table 8-3**.

Table 8-2: Sample of Raw OCS Output for Alewife Signaling System

Message Sequence #	Message Group	Timestamp	Line	Message Sub-Type	Location Name[12]	On/Off
1631	DEVI	7:25:09	R	MD	ALE_bell	1
1633	DEVI	7:25:10	R	SG	ALE_S2	1
1634	DEVI	7:26:09	R	SG	ALE_S2	0
1635	DEVI	7:26:11	R	MD	ALE_bell	0
1641	DEVI	7:26:44	R	SG	ALE_N1	1
1643	DEVI	7:27:35	R	SG	ALE_N1	0
1652	DEVI	7:28:28	R	MD	ALE_bell	1
1660	DEVI	7:28:48	R	SG	ALE_S1	1
1661	DEVI	7:29:54	R	SG	ALE_S1	0
1662	DEVI	7:29:56	R	MD	ALE_bell	0

Table 8-3: Cleaned Alewife Departures from **Table 8-2**

Dispatch Platform	Bell On Time	Slot On Time	Slot Off Time	Bell Off Time	Slot Open Time (1)	Minimum Slot Open Time for Movement (2)	Reaction Time (1) - (2)
P2	7:25:09	7:25:10	7:26:09	7:26:11	59s	24s	35s
P1	7:28:28	7:28:48	7:29:54	7:29:56	66s	46s	20s

[12] ALE_bell refers to the Alewife ring-off bell
ALE_S1 and ALE_S2 refer to the southbound slot control for platforms P1 and P2, respectively.
ALE_N1 refers to the northbound slot control. There is no differentiation between P1/P2 routing.

Appendix 6. Identification of Held Trains

This appendix details the set of heuristics used in order to identify which trains were held during the Davis and Andrew Holding Pilots. The heuristics were developed with reference to the Pilot instructions (Chapter 4) and from field observations during the pilots, and applied to automatically collected train tracking data.

Davis Holding

The set of heuristics used to identify which trains were held during the Davis Holding Pilot are as follows:

(1) Station-block occupancy time at least 15s greater than median Baseline occupancy time

In practice, operators were not asked to hold if the holding time would be less than approximately 15s. Thus a threshold of 15s greater than the median Baseline station occupancy time was used to identify held trains, with separate medians for the AM and PM peaks. During the Baseline Period, only 6% of trains in AM and 7% of trains in PM would exceed the median period block occupancy time by 15s and also meet the headway criteria for holding identification, showing that a 15s threshold for holding identification will produce few false positives.

(2) Departure-departure headway less than 4:15

A high upper limit for departure-departure headway was used for holding identification to account for delays which occur after the departure instruction was given, such as passengers running to catch the train. The upper limit also takes into account that departure headways from train-tracking data are measured after the train passes a track circuit downstream of the station, and not from the time of wheel-start as the officials were asked to measure. An upper limit is also necessary to exclude cases where trains were held for reasons other than those intended in the pilot, for example instruction from OCC due to incidents up or downstream.

(3) Departure-arrival time less than 2:30

The final criteria limits holding identification to trains which arrived at short headways. In practice, officials would not ask operators to hold if holding time would be minimal (15s or less). Trains with a departure-arrival time of 2:30 would naturally meet the recommended departure-departure headway without holding.

The results of the holding identification process for the Davis Pilot are shown in **Figure 8-5**. The departure-arrival and departure-departure headways are shown on the x and y axes, respectively for each dwell observation during the pilot period. The difference between the two is the block occupancy time, which includes dwell time and pull-in/pull-out time. Observations bounded by the aforementioned set of criteria are identified as holding.

Deserving of special mention are a number of outliers in the AM in the upper left quadrant, which meet the long block occupancy time and low departure-arrival headway criteria, but exceed the departure-departure headway criteria. A number of these which were observed by researchers in person represent instances where the OCC ordered trains to hold at Davis due to downstream incidents; for example, a medical emergency at Porter. The remainder are assumed to be of similar nature and are omitted from the holding identification as this type of holding criteria is distinct from the pilot instructions.

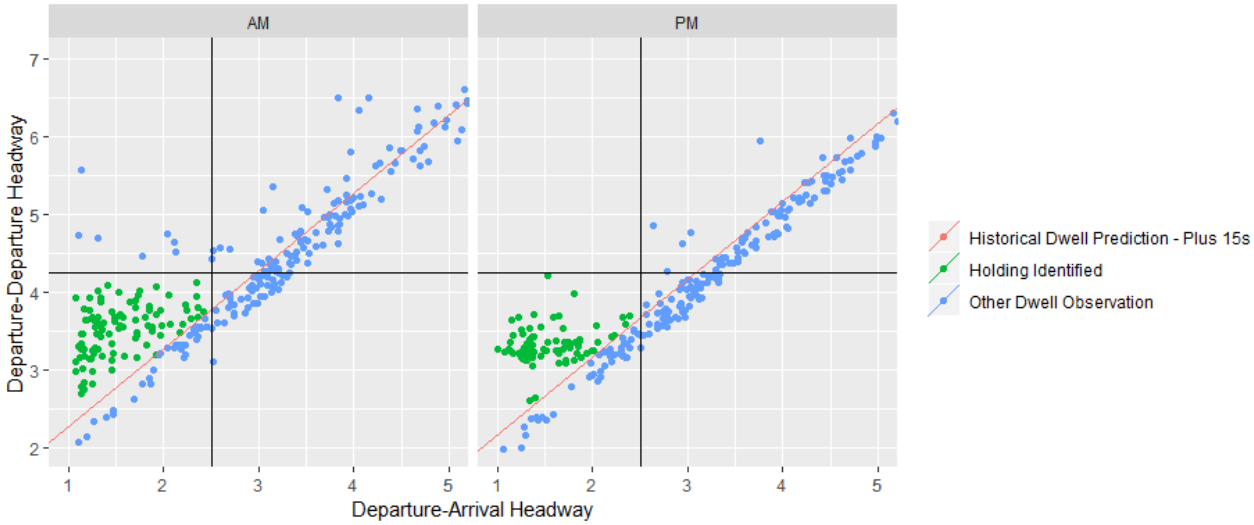


Figure 8-5: Davis Pilot Holding Identification

Andrew Holding

A set of heuristics was applied to identify which trains were held during the Andrew Holding Pilot. The main difference from the Davis Holding identification procedure is the introduction of the variable holding target. The heuristics used as follows:

(1) Station-block occupancy time at least 15s greater than the median Baseline station-block occupancy time

In practice, operators were not asked to hold if the holding time would be less than approximately 15s. Thus a threshold of 15s greater than the median baseline station occupancy time was used to identify held trains. During the Baseline, only 4% of observations with departure-departure headways less than 5-minutes exceeded this threshold, showing that a 15s threshold for holding identification will produce few false positives.

(2) For identification of 3:00 target headway holds; departure-departure headway less than 4-minutes.

A maximum of 4:00 departure-departure headway is used to differentiate between the two holding targets. In rare cases, a train held to 3:00 may exceed the target headway by 1-minute and be falsely identified due to OCC dispatch orders relating to downstream incidents.

(3) For identification of 4:00 target headway holds; departure-departure headway between 4 to 5-minutes.

A minimum of 4:00 departure-departure headway is deemed acceptable for differentiating between the two holding targets. From field observations, held trains regularly exceed the target headway by at least 10-15s, but rarely fail to meet it (under 4:00).

The results of the holding identification process are shown in **Figure 8-6**. Guidelines are shown, corresponding to each aforementioned heuristic. The dwell observations which are identified as trains held to 3:00 are observed to have a very tight distribution; this represents that many trains in this group arrived with a minimum possible spacing to Andrew, between 1-2 minutes after the previous train had departed. Trains held to 4:00 have a much sparser distribution, representing both the infrequent nature of this type of holding (14% of all trains), and the wider range of possible arrival headways. A small number of un-held trains are observed with departure-departure headways less than 3:00. These represent instances during the peak hours where staff were not able to preforming holding as instructed, either due to stopwatch error or conflicting instructions from OCC.

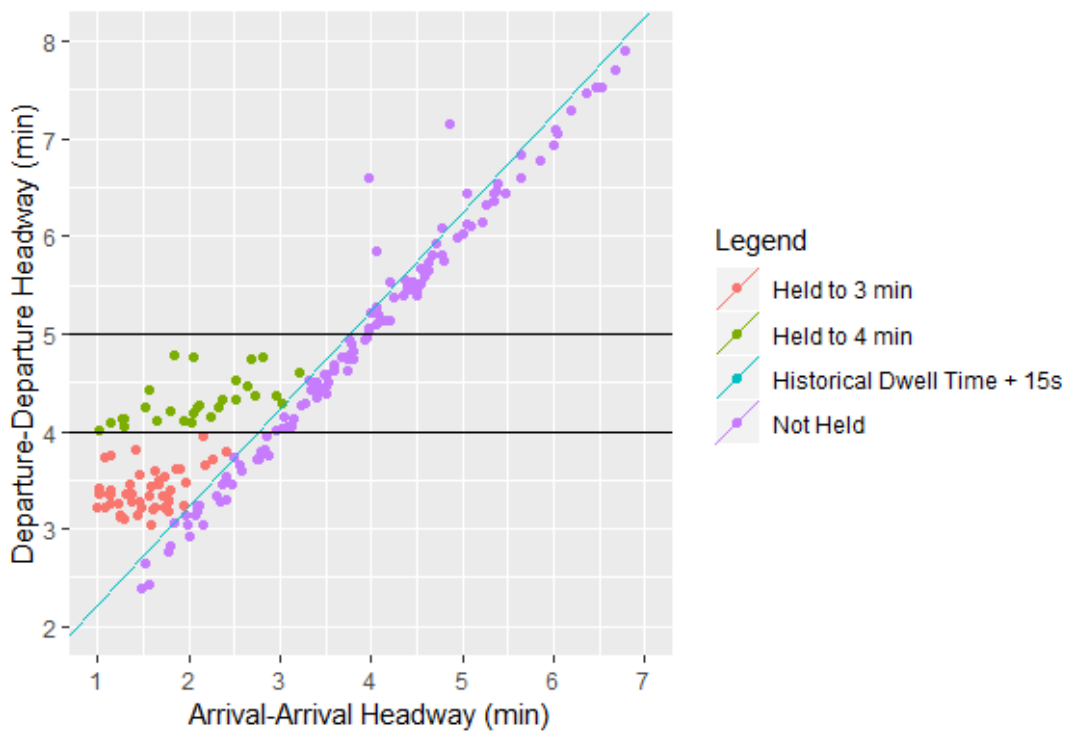
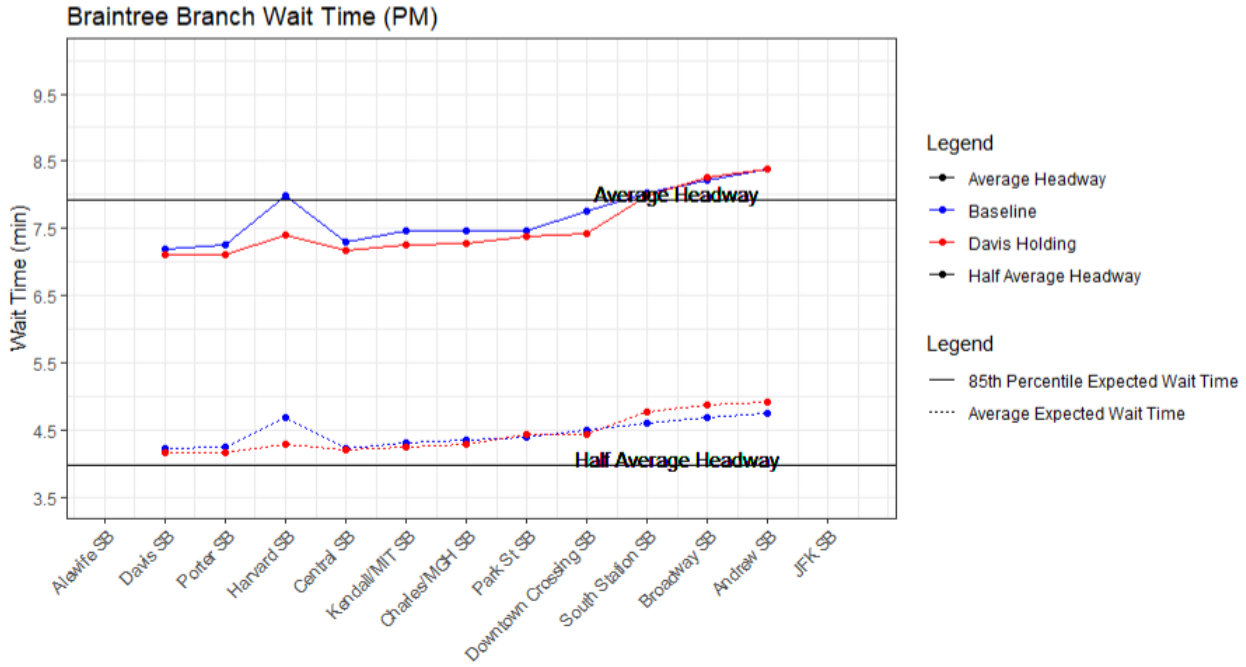
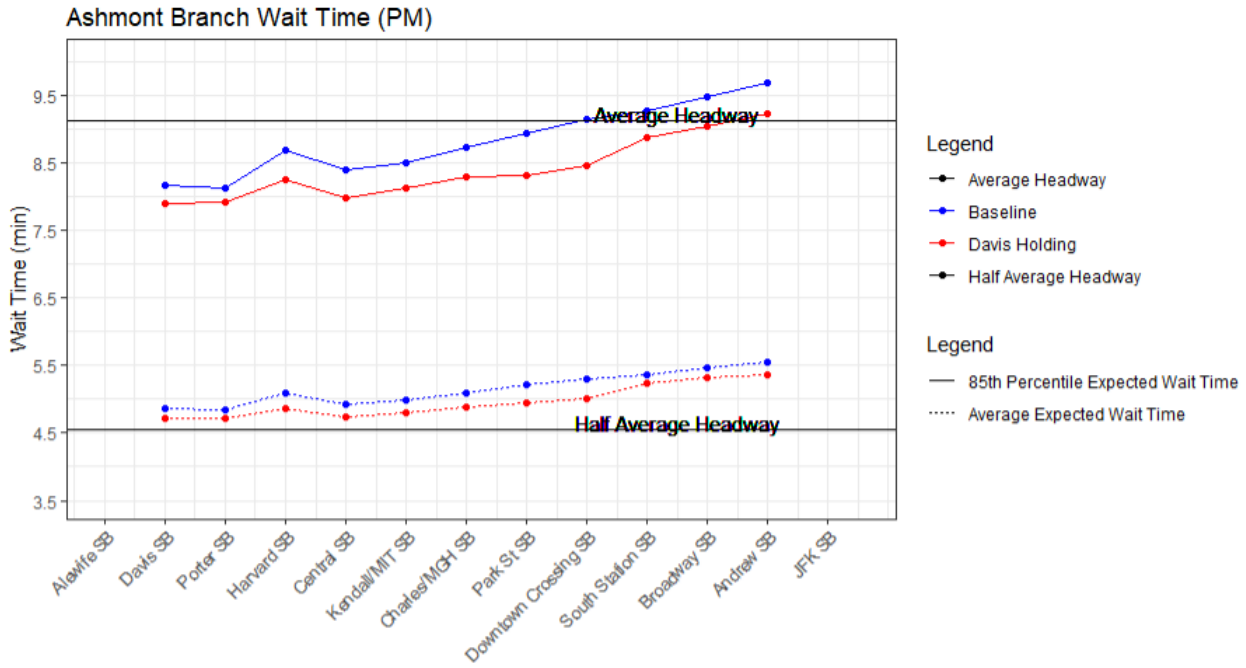


Figure 8-6: Andrew Pilot Holding Identification

Appendix 7. Davis Holding – Impacts on Branch Passenger Wait Times

Changes in wait times for passengers with branch destinations are shown below for the Baseline versus Davis Holding Pilot periods. Although branch headways were not a main focus of the pilot, which sought to regularize trunk headways, minor improvements are nevertheless seen to branch headways, attributed an overall improvement in headway stability.





Appendix 8. Running Time Adjustments due to Speed Code Changes

A number of speed code changes between the Baseline and pilot periods caused difficulty in discerning the impacts of the three interventions on line running times and journey times. Official records of speed code changes were unavailable; thus the following methodology was developed to identify any major changes.

(1) Determine affected days and station pairs
(2) Determine the magnitude and distribution of change
(3) Apply running time adjustments to pilot days.

(1) Determine affected days and speed codes

First, running time is grouped into station pairs and off-peak observations only are selected. Off-peak values only are used in order to isolate for the effect of the speed code changes, without any running time changes induced by the pilots. The median, 15th and 85th percentile running time on each link are summarized by day and any major shifts are identified visually. The 7 links with identified changes of at least 10s are shown below in **Figure 8-7**, with periods where speed codes differed from the Baseline highlighted.

(2) Determine magnitude and distribution of change

Following the identification of running time changes, the next step is determining whether the changes were constant across all trips, or if the change only affected a certain part of the distribution; for example, trips with very short headways. Running time percentiles (**Table 8-4**) and median running time by headway (**Figure 8-8**) are shown below. Across the range of percentiles and headways observed, all except for station pair "70070-70068" (Central to Harvard, Northbound) appear to have a nearly uniform impact for all trains. For the Central to Harvard speed code change, the difference appears to only apply to headways of at least 4-minutes; for closely spaced trains there was no change.

(3) Apply running time adjustments to pilot days

Finally, the running time adjustments listed in **Table 8-5** were applied as a uniform decrease to running times throughout the listed dates.

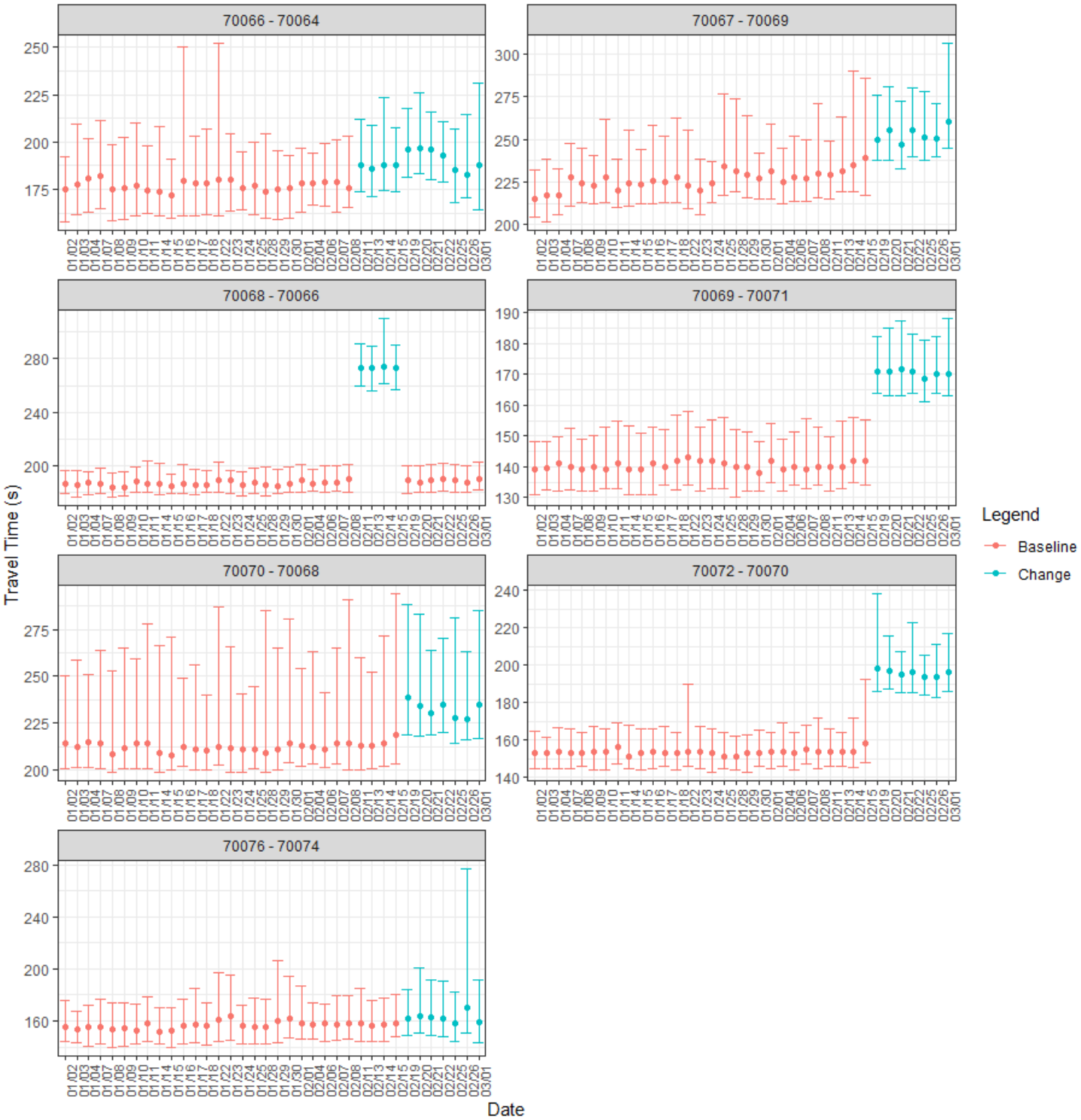


Figure 8-7: Station Pairs with Identified Running Time Changes

Table 8-4: Station Pair Running Time Percentile Changes

		Dates		Running Time Percentiles		
Station Pair	Direction	From	To	15th	Median	85th
70066 - 70064	Northbound	2/11	3/1	12	9	13
70067 - 70069	Southbound	2/19	3/1	25	25	24
70068 - 70066	Northbound	2/11	2/15	79	85	93
70069 - 70071	Southbound	2/19	3/1	30	28	24
70070 - 70068	Northbound	2/19	3/1	16	19	16
70072 - 70070	Northbound	2/19	3/1	40	39	36
70076 - 70074	Northbound	2/19	3/1	5	13	19

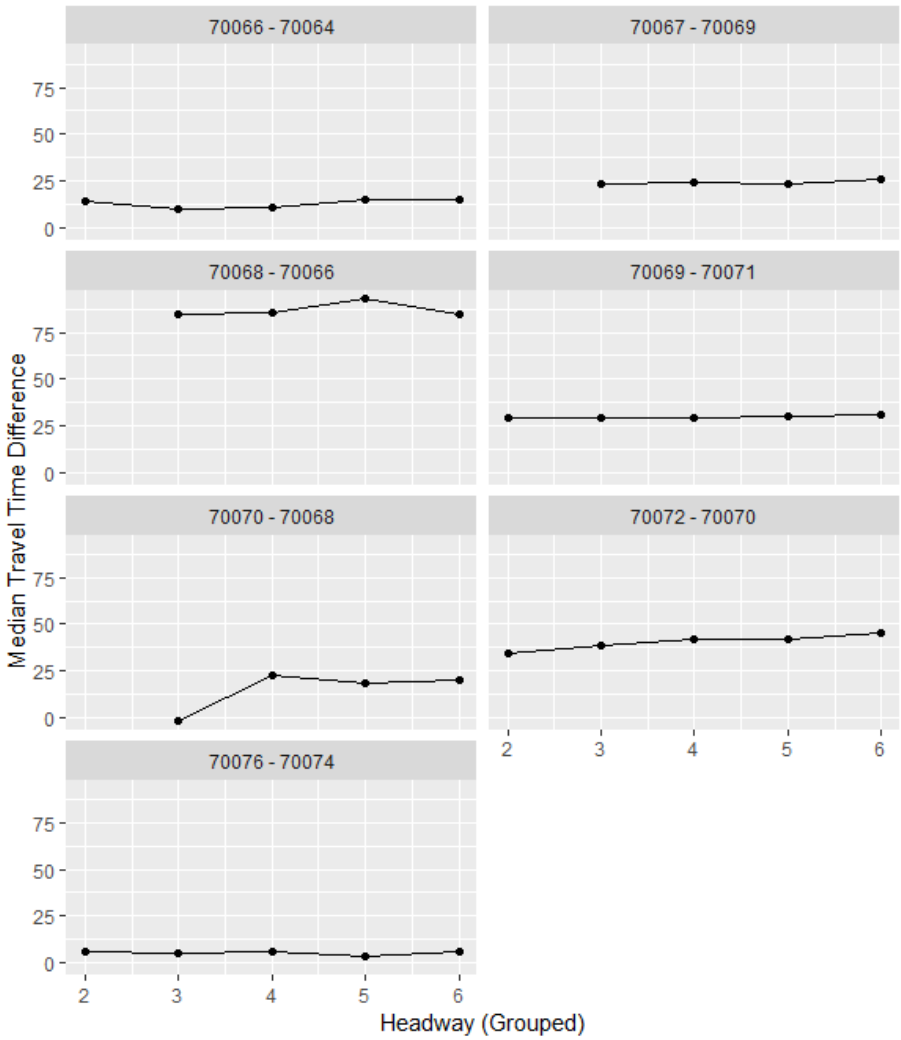


Figure 8-8: Travel Time Changes by Headway

Table 8-5: Running Time Adjustments due to Speed Code Changes

Station Pair	From Date	To Date	Running Time Increase (s)	Pilot Days Affected	Notes
70066 - 70064	2/11	3/1	9	7 NBH	
70067 - 70069	2/19	3/1	25	5 SBH	
70068 - 70066	2/11	2/15	85	N/A	
70069 - 70071	2/19	3/1	28	5 SBH	
70070 - 70068	2/19	3/1	19	7 NBH	Only applies to headways >= 4 min
70072 - 70070	2/19	3/1	39	7 NBH	
70076 - 70074	2/19	3/1	13	7 NBH	